

Affective Vulnerability and Health Correlates of Cannabis Use: Insights From Between- and Within-Person Approaches

PhD Dissertation

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Abstract

This dissertation aims to increase our understanding of affective mechanisms and health correlates associated with maintenance of cannabis use. It focused on a naturalistic sample of individuals who smoke cannabis daily or near-daily, because this group is at higher risk of cannabis-related harms, yet is underrepresented in research.

This dissertation consists of three empirical manuscripts addressing complementary aspects of the overarching aim to understand the interplay between cannabis use, affective mechanism, and health correlates. By integrating these complementary approaches, the dissertation aims to provide a comprehensive account of how between-person characteristics and within-person processes jointly contribute to the maintenance of cannabis use and the emergence of problematic cannabis use.

Manuscript 1 addresses individual vulnerability factors associated with severe patterns of cannabis use in cross-sectional data. Specifically, it examines whether difficulties in emotion regulation are associated with cannabis use severity at the between-person level, whether these associations vary as a function of gender identity, and whether coping motives operate as an intermediate process linking emotion regulation difficulties to cannabis use severity. Together, these questions address who is more vulnerable to developing harmful patterns of cannabis use.

Manuscript 2 examines within-person affective reinforcement mechanisms underlying cannabis use in daily life. Using experience sampling methods, this study investigates how momentary fluctuations in affect are temporally related to cannabis use, whether cannabis use in turn is associated with concurrent affective states, and how this varies as a function of stable between-person differences in emotion regulation difficulties. Together, these analyses address how cannabis use is dynamically initiated and reinforced in everyday contexts.

Manuscript 3 examines between-person associations of cannabis use severity and key health-related domains (in cross-sectional data), specifically sleep quality and symptoms of anxiety and depression, among individuals who use cannabis daily or near-daily. In addition, it evaluates whether these associations differ as a function of cannabis product type,

addressing heterogeneity in health-related experiences among individuals with similar levels of cannabis involvement.

Abstrakt på dansk

Formålet med denne afhandling er at øge vores forståelse af affektive mekanismer og sundhedsfaktorer, som spiller en rolle i forhold til at fastholde individer i deres cannabisbrug. Afhandlingen fokuserer på en naturalistisk stikprøve af individer, der ryger cannabis dagligt eller næsten dagligt, fordi denne gruppe har højere risiko for cannabisrelaterede skader, men på trods heraf er underrepræsenteret i forskning.

Denne afhandling består af tre empiriske manuskrifter, der adresserer komplementære aspekter af det overordnede formål om at forstå spillet mellem cannabisbrug, affektive mekanismer og sundhedsfaktorer. Ved at integrere disse komplementære tilgange, har afhandlingen til formål at bidrage med en omfattende udforskning af, hvordan karakteristika-mellem-personer og processer-indenfor-personer sammen bidrager til at fastholde individer i cannabisbrug og til at udvikle problematisk cannabisbrug.

Manuskript 1 undersøger individuelle sårbarhedsfaktorer, der er forbundet med alvorligt cannabisbrug via tværsnitsdata. Mere specifikt undersøges, hvorvidt vanskeligheder med emotionsregulering hænger sammen med sværhedsgraden af cannabisbrug (mellem-persons-niveau), hvorvidt disse sammenhænge varierer som funktion af kønsidentitet, samt hvorvidt coping-motiver fungerer som en mellemliggende proces, der forbinder vanskeligheder med emotionsregulering til sværhedsgraden af cannabisbrug. Samlet set belyser disse spørgsmål, hvem der er mere sårbare i forhold til at udvikle skadeligt brug af cannabis.

Manuskript 2 undersøger affektive forstærkningsmekanismer (inden for den enkelte person), som kan være med til at forklare cannabisbrug i dagligdagen. Studiet benytter experience sampling metoder til at undersøge, hvordan øjebliksfluktuationer i affekt er tidsmæssigt forbundet med cannabisbrug, hvorvidt cannabisbrug omvendt er forbundet med samtidige affektive tilstande, samt hvordan dette varierer som funktion af stabile forskelle i emotionsreguleringsvanskeligheder (mellem personer). Samlet set belyser disse analyser, hvordan cannabisbrug dynamisk initieres og forstærkes i dagligdagskontekster.

Manuskript 3 undersøger via tværsnitdata sammenhænge mellem sværhedsgrad af cannabisbrug og vigtige sundhedsområder (mellem personer), specifikt søvnkvalitet og

symptomer på angst og depression, blandt individer, der bruger cannabis dagligt eller næsten dagligt. Herudover undersøges, hvorvidt disse sammenhænge varierer som funktion af cannabisprodukttype, hvilket adresserer heterogenitet i sundhedsrelaterede oplevelser blandt individer med lignende niveauer af cannabisinvolvering.

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Table of Contents

Preface	15
Manuscripts included in the Dissertation	17
Chapter 1: General Introduction	18
1.1 Cannabis use: Regulations and public health context	18
1.2 Cannabis use and severity of use	19
1.3 Affective processes and reinforcement mechanisms in cannabis use	21
1.4 Emotion regulation difficulties as a vulnerability factor	22
1.5. Momentary fluctuations in affect and reinforcement	24
1.6 Sources of heterogeneity in cannabis-related outcomes.	26
1.6.1 The role of gender	26
1.6.2 Sleep quality	27
1.6.3 Mental health: Symptoms of anxiety and depression	28
1.7 Cannabis product characteristics and heterogeneity of effects	29
Chapter 2: Research gaps, aims and hypotheses	31
2.1 Manuscript 1: Individual vulnerability factors	31
2.2 Manuscript 2: Affective mechanisms	32
2.3 Manuscript 3: Health-related associations	34
Chapter 3: Methodology	36
3.1 Design	36
3.2 Approvals and open science	36

3.3 Participants	36
3.4 Recruitment.....	37
3.5 Study procedure	38
3.5.1 Baseline procedure	39
3.5.2 ESM procedure	40
3.6 Measures	42
3.6.1 Baseline measures	42
3.6.2 ESM measures	45
3.6.3 Translation and cross-cultural adaptation processes.....	46
3.6.4 Overview of constructs and measures across manuscripts.....	48
3.7 Analytical strategy	48
Chapter 4: Summary of Findings.....	52
4.1 Manuscript 1.....	52
4.2 Manuscript 2.....	52
4.3 Manuscript 3.....	53
Chapter 5: Integrated discussion	55
5.1 Overview of the PhD framework	55
5.2 Vulnerabilities	55
5.3 Momentary affective mechanisms.....	56
5.4 Health correlates	59
5.5 An integrated model of cannabis use.....	60
Chapter 6: Limitations, strengths and conclusion.....	63

6.1 Limitations across studies	63
6.2 Methodological strengths across studies	65
6.3 Future directions	66
6.4 General conclusions.....	68
References	71
Appendix A (Manuscript 1).....	94
Appendix B (Manuscript 2).....	118
Appendix C (Manuscript 3).....	153
Appendix D (Co-authorship declaration)	184

List of Figures

Figure 1 Project recruitment material	37
Figure 2 Overview of the research website	38
Figure 3 Overview of the study procedure	39
Figure 4 Instructional sequence shown to participants before the ESM period	40
Figure 5 Study procedure and within-day structure of the ESM protocol	41

List of Tables

Table 1 Recruitment contexts and number of communities approached	38
Table 2 Overview of the instruments administered at baseline	40
Table 3 Overview of ESM constructs, source of instrument, and assessment schedule	42
Table 4 Overview of instruments at baseline	45
Table 5 Verbatim ESM instruction and temporal reference windows	45
Table 6 Overview of constructs, measures, and level across manuscripts	48
Table 7 Overview of the three manuscripts	49

List of Abbreviations

AUDIT	Alcohol Use Disorders Identification Test
CBD	Cannabidiol
CUD	Cannabis use disorders
CUDIT-R	Cannabis Use Disorders Identification Test - Revised
DERS-16	Difficulties in Emotion Regulation Scale
DSM-5	Diagnostic and Statistical Manual of Mental Disorders, 5th Edition
ESM	Experience Sampling Method
FTDN	Fagerström Test for Nicotine Dependence
ICD-11	International Classifications of Diseases 11 th
MMM	Marijuana Motives Measure
OSF	Open Science Framework
PANAS	Positive and Negative affect Schedule
PSQI-b	Brief Version of the Pittsburgh Sleep Quality Index (PSQI-b)
THC	Delta-9-Tetrahydrocannabinol
TLFB	Timeline Follow-Back

Preface

This dissertation emerged from a strong interest in understanding how cannabis use, its maintenance, and its progression toward Cannabis Use Disorder (CUD) unfold in everyday life. CUD represents a significant public health concern, with rising prevalence and notable associations with affective difficulties, impaired functioning, and mental health symptoms. This dissertation aims to increase our understanding of who, why, and under what circumstances individuals initiate, maintain, and/or escalate cannabis use over time.

Cannabis use constitutes a theoretically and empirically meaningful field within addiction research. Among individuals who use cannabis daily or near-daily, use is frequent and tends to be embedded in everyday routines, making it a relevant behavior for examining how affective, contextual, and motivational processes unfold in the real-world, and which mechanisms may contribute to the maintenance of use and potential development of a problematic pattern of use.

Understanding cannabis use and the development of CUD requires approaches that capture both stable individual differences and processes that unfold over time. For this reason, the dissertation combines traditional cross-sectional designs with Experience Sampling Methods (ESM). Cross-sectional methods are well suited for identifying demographic, affective, and motivational vulnerabilities that differentiate individuals in their typical patterns of use and related functioning. In contrast, ESM allows for examination of momentary affective states, contextual influences, and short-term regulatory mechanisms as they occur in daily life.

The dissertation consists of *three* empirical manuscripts addressing complementary aspects of the overarching aim to understand the interplay between cannabis use, affective processing, and health correlates. **Manuscript 1** examines associations between cannabis use severity and emotion regulation difficulties and evaluates whether gender and coping motives shape these relationships. **Manuscript 2** investigates momentary affective mechanisms underlying cannabis use in daily life and how emotion regulation difficulties shape this. **Manuscript 3** examines associations between cannabis use severity and health outcomes (sleep quality and symptoms of anxiety and depression) and whether these

associations differ by cannabis product types. These manuscripts can be read independently while together they contribute to a coherent and integrated work.

Collectively, the findings of this dissertation contribute to a more nuanced understanding of how affective, contextual, and individual factors shape cannabis use and its potential progression toward problematic patterns. By integrating evidence from both cross-sectional and daily-life assessments, the dissertation highlights how stable vulnerabilities and momentary processes jointly inform our understanding of mechanisms related to cannabis use. These insights hold relevance for theories of addiction, affect regulation, and reinforcement. More broadly, this work underscores the value of studying psychological processes as they are experienced in everyday life and contributes to ongoing efforts to advance a more precise and contextually grounded psychological science.

Manuscripts included in the Dissertation

Manuscript 1

Cannabis Use Severity and Emotion Regulation in People Who Use Cannabis near daily:
The Role of Gender and Motives

Ana Paula Gonçalves Donate, MSc, Janna Cousijn, PhD, Kristine Rømer Thomsen, PhD

Under Review in *Journal of Psychoactive Drugs*

Manuscript 2

Cannabis Use, Affect, and Emotion Regulation in Daily Life: A 3-week Experience Sampling
Method Study in People Who Smoke Cannabis Almost Daily

Authors: Ana Paula Gonçalves Donate, MSc, Janna Cousijn, PhD, Emese Kroon, PhD,
Kristine Rømer Thomsen, PhD

Under review in *Addiction*

Manuscript 3

Beyond Cannabis Use Severity: Associations of Cannabis Product Type with Sleep Quality
and Mental Health

Author: Ana Paula Gonçalves Donate, MSc, Janna Cousijn, PhD, Kristine Rømer Thomsen,
PhD

Revise and resubmit to *Psychopharmacology*

Chapter 1: General Introduction

1.1 Cannabis use: Regulations and public health context

Over the past decades, regulatory frameworks concerning cannabis use have undergone substantial changes worldwide. While policies remain heterogeneous, an increasing number of countries have adopted more permissive approaches, including decriminalization, medical access, and legalization for recreational use (European Union Drugs Agency (EUDA), 2025, United Nations Office on Drugs and Crime (UNODC), 2025). In Denmark, as of the present date, recreational use is illegal, whereas medical cannabis is permitted under specific physician-prescribed conditions (Danish Health Authority, 2023).

From a public health perspective, these shifts are relevant because they coincide with greater availability and rising levels of frequent cannabis use (EUDA, 2025). Cannabis is the most commonly used illicit substance globally, with an estimated 228 million people reporting use in 2022 (UNODC, 2024a). In Europe, approximately 24 million adults (8.4%) used cannabis in the past year, and around 1.5% use it daily or near-daily (EUDA, 2025, UNODC, 2025a).

Individuals who use cannabis daily and near-daily are at the highest risk for cannabis-related harms, including dependence, psychotic symptoms, impaired functioning, and need for treatment (EUDA, 2024, EUDA, 2025). Among young people who use cannabis regularly (e.g., weekly or daily), 33% develop cannabis dependence (Leung et al., 2020, Leung et al., 2022, Lopez-Quintero et al., 2011).

One of the main concerns related to cannabis use is the risk of developing a cannabis use disorder (CUD). CUD is one of the most common substance use disorders worldwide (UNODC, 2024b), and contributes substantially to the global burden of disease, accounting for approximately 2 million disability-adjusted life years (Degenhardt et al., 2013, EUDA, 2025). CUD accounts for 36% of entries in specialized treatment for drug use (EUDA, 2025).

Treatment for CUD current relies primarily on psychosocial approaches (EUDA, 2025; Hoch et al., 2025a). These approaches encompass a range of structured procedures which address both psychosocial and social aspects of general drug use behavior, associated with cannabis specific modules (EUDA, 2025). The format, duration and intensity are

heterogeneous, and it includes cognitive behavioral therapy, contingency management, motivational interviewing and brief interventions (Lees et al., 2021). In Europe, sixteen countries offer cannabis-specific programs but most of these programs are not evidence-based (Hoch, 2025a). In a systematic review, psychosocial interventions in comparison with minimal treatment control, seem to be able to reduce frequency of use and severity of dependence (Gates, 2016), and treatment outcomes remain modest, with sustained abstinence rates at six-month follow-up typically not achieved in more than 35% of patients, even in structured psychosocial interventions (Hoch et al., 2014, Hoch et al., 2025a, Gates et al., 2016). In terms of pharmacotherapy to treat CUD, a meta-analysis showed low or very low evidence across 37 studies included in the analysis (Spiga et al., 2025).

Together, these findings highlight the public health significance of cannabis use, and the increased risk of harm among individuals who use cannabis daily or near daily.

1.2 Cannabis use and severity of use

Cannabis use can lead to several clinical conditions, including intoxication, withdrawal, substance-induced symptoms, and CUD. CUD and cannabis dependence can be diagnosed using the structured diagnostic systems: the Statistical Manual of Mental Disorders (DSM-5) and the International Classification of Diseases (ICD-11) (Zehra et al., 2018, American Psychiatric Association, 2013). In DSM-5, CUD is conceptualized as a single disorder that replaced the DSM-IV distinction between cannabis abuse and cannabis dependence, with severity specified according to the number of criteria endorsed. In contrast, ICD-11 retains dependence as a distinct diagnosis and differentiates it from harmful patterns of cannabis use.

According to DSM-5, CUD is a categorical diagnosis defined by a collection of eleven criteria reflecting maladaptive patterns of cannabis use. These symptoms are organized into four domains: (i) Impaired control over cannabis use, (ii) social impairment and disruption of life roles, (iii) continued use despite negative consequences, and (iv) physiological features, including tolerance and withdrawal. The severity of the disorder is based on the number of symptoms endorsed, categorized as mild (2 to 3 symptoms), moderate (4 to 5 symptoms), or severe (6 or more symptoms) (American Psychiatric Association, 2013).

Many studies, including the present dissertation, use continuous indicators of cannabis use severity, rather than a diagnostic classification. Multiple questionnaires have been developed to assess cannabis-related problems and problematic cannabis involvement (Lopez-Pelayo et al., 2015). As screening tools, these questionnaires typically capture the degree of involvement with cannabis and the extent to which use is linked to functional, psychological, or behavioral problems.

To operationalize this construct, the present dissertation uses the Cannabis Use Disorders Identification Test - Revised (CUDIT-R), a validated screening instrument widely used in cannabis research (Adamson et al., 2010, Lopez-Pelayo et al., 2015). The CUDIT-R captures multiple dimensions of cannabis involvement, including frequency of use, loss of control, and cannabis-related negative consequences, with higher scores indicating greater severity (Adamson et al., 2010). In this dissertation, cannabis use severity is hence conceptualized dimensionally as reflecting both the extent of cannabis involvement and the severity of cannabis-related problems, rather than use frequency alone.

Importantly, the CUDIT-R is a screening instrument and not a substitute for a structured diagnostic interview for DSM-5 CUD. Although CUDIT-R scores have been shown to be associated with DSM-5 CUD criteria and can help identify individuals at elevated likelihood of CUD, its diagnostic utility varies across populations and cut-off values. Accordingly, in the present dissertation, the CUDIT-R is used as a dimensional indicator of problematic cannabis involvement rather than as a direct proxy for a clinically established DSM-5 CUD diagnosis. The use of a dimensional severity measure is particularly relevant for near-daily cannabis use, as it allows variation in problematic involvement to be captured across the full spectrum of use, including subclinical levels of impairment and CUD (Adamson et al., 2010).

CUD is influenced by multiple interacting factors, including social context, developmental vulnerability, psychiatric comorbidity, and environmental exposure. Within this broader framework, affective and reinforcement-related processes are relevant for understanding why cannabis use is initiated, repeated, and maintained over time (Zehra et al., 2018, Koob & Volkow, 2016). Accordingly, the present dissertation places a particular emphasis on affective processes and reinforcement mechanisms as candidate mechanisms involved in problematic cannabis use.

1.3 Affective processes and reinforcement mechanisms in cannabis use

Different theories have been proposed to explain the maintenance of cannabis use and the progression toward CUD, including learning-based, neurobiological, and motivational models (Bickel et al., 2018, Zehra et al., 2018, Robinson and Berridge, 2008). Across these theories, affective processes are viewed as central to how cannabis use becomes established and sustained over time.

Within operant learning theory, reinforcement refers to processes through which behavioral consequences increase the likelihood of the behavior's future occurrence. In the case of cannabis: cannabis use may be positively reinforced if it produces pleasurable or rewarding internal states; or negatively reinforced if it alleviates aversive internal states, such as stress, anxiety, or irritability (Baker et al., 2004, Wycoff et al., 2018, Cheetham et al., 2010). Hence, reinforcement-based models distinguish between cannabis use motivated by the pursuit of reward from use motivated by the regulation of distress (Baker et al., 2004, Khantzian, 1997, Zehra et al., 2018).

Contemporary affect-regulation models extend these principles by emphasizing that substance use can become integrated into individuals' emotion regulation repertoires. From this perspective, cannabis use serves both hedonic functions (enhancing positive affect) and homeostatic functions (restoring or normalizing negative affect). Over repeated episodes, individuals may form learned associations that couple affective states with cannabis use, contributing to recurrent affect-use patterns and short-term feedback loops (Robinson and Berridge, 2008, Baker et al., 2004).

This learning process operates, not only through experienced affective changes, but also through anticipated affective outcomes. Motivational models highlight the role of positive and negative reinforcement expectancies - beliefs that cannabis will either produce pleasure or reduce distress - which can bias attention toward affective cues, shape interpretations of internal states, and guide momentary decision-making in daily life (Cooper, 1994, Baker et al., 2004, Stacy and Wiers, 2010). In other words, expectancies interact with momentary affective states to shape whether cannabis is perceived as a relevant and effective emotion-regulation strategy at a given moment.

At a neurobiological level, positive and negative reinforcement rely on partially overlapping, but functionally distinct neural circuits. Positive reinforcement is primarily linked to dopaminergic reward-related pathways that encode hedonic value and incentive salience, whereas negative reinforcement draws on stress and threat regulation circuitry (Berridge and Robinson, 2016, Robinson and Berridge, 2008). Consistent with this, neuroimaging evidence indicates that cannabis dependence is associated with reduced CB1 receptor availability, which begins to reverse rapidly after abstinence (D'Souza et al., 2016). Persistent CB1 receptor stimulation may downregulate endocannabinoid signaling and disrupt dopaminergic and glutamatergic transmission involved in reward and motivation (Koob and Volkow, 2016, Zehra et al., 2018). Such adaptations may contribute to reduced baseline reward sensitivity and heightened stress responsivity, shifting the functional role of cannabis from the pursuit of positive toward alleviation of distress or withdrawal-related negative affect. Critically, these reinforcement processes are context dependent. Factors such as social environment, availability of alternative rewards, developmental stage, and individual differences in emotion regulation shape the extent to which positive or negative reinforcement governs cannabis use (Volkow et al., 2016, Bickel et al., 2018, Tucker et al., 2020, Casey et al., 2011, Steinberg, 2008, Baker et al., 2004). This context dependence suggests that individual differences in emotion regulation capacity may systematically influence reinforcement processes, underscoring the importance of examining emotion regulation difficulties as a vulnerability factor or risk factor for hazardous cannabis use or CUD.

1.4 Emotion regulation difficulties as a vulnerability factor

Emotion regulation refers to processes by which individuals influence the intensity, duration, or quality of their affective states. These regulatory processes may unfold, both consciously and unconsciously, in response to environmental demands and individuals' goals (Bjureberg et al., 2016, Gratz and Roemer, 2004). The ability to regulate emotions effectively has been associated with resilience to psychopathology (Stellern et al., 2023). Theoretical models distinguish between regulatory strategies and regulatory abilities (Bjureberg et al., 2016, Gratz and Roemer, 2004, Sheppes et al., 2015, Gross, 2015, Stellern et al., 2023). Emotion regulation abilities are generally more strongly related to substance use, than the strategies (Weiss et al., 2022).

Emotion regulation difficulties capture a broader impairment in understanding, accepting, and modulating emotional experiences, particularly in contexts of heightened affective arousal. These difficulties are commonly operationalized using the Difficulties in Emotion Regulation Scale (DERS). In the literature, this scale is available in different versions (e.g., DERS-36, DERS-SF, or the shortest version DERS-16). All versions assess trait-like vulnerabilities across abilities and have been widely used in substance use research (Weiss et al., 2022, Stellern et al., 2023). In this dissertation, emotion regulation difficulties are measured using DERS-16.

Emotion regulation difficulties are associated with a wide range of maladaptive behaviors, including hazardous substance use, substance use disorders, and cannabis use (Stellern et al., 2023, Garke et al., 2021). Recently, Buckner (2024) showed that higher levels of emotion dysregulation predicted fewer protective strategies and cannabis-related problems in young undergraduate students (Buckner et al., 2024). Different dimensions of emotion regulation difficulties may relate to distinct patterns of cannabis use. For example, impulse-control difficulties appear to be more strongly associated with the frequency and quantity of cannabis use (Bonn-Miller et al., 2008), whereas limited access to adaptive regulation strategies has been more closely associated with coping-motivated and hazardous patterns of use (Dvorak and Day, 2014).

From a theoretical perspective, greater emotion regulation difficulties are associated with increased reliance on substances as external regulation tools. Individuals who struggle to tolerate, accept, or downregulate distress may be more likely to use cannabis to obtain short-term relief, consistent with negative reinforcement models of substance use (Baker et al., 2004, Khantzian, 1997). In this framework, cannabis use may become disproportionately reinforcing for individuals with higher emotion regulation difficulties, not because they experience more negative affect per se, but because they have fewer effective internal means of regulating it (Dvorak and Day, 2014, Bonn-Miller et al., 2008). As a result, relief following cannabis use may be more salient, strengthening affect–use learning over time. Taken together, individual differences in emotion regulation abilities may help explain the mixed findings in previous ESM studies examining positive and negative reinforcement in cannabis use (described in section 1.5).

1.5. Momentary fluctuations in affect and reinforcement

Reinforcement models further posit that reinforcement processes operate within individuals over short time scales, such that momentary fluctuations in affect may both precede and follow cannabis use, creating immediate affective feedback cycles that sustain behavior (Shiffman, 2009a). Studying these temporal dynamics requires methodologies that capture within-person affective variation in real time, making Experience Sampling Methods (ESM) particularly well suited (Shiffman, 2009a, Trull and Ebner-Priemer, 2009, Trull and Ebner-Priemer, 2020).

Retrospective baseline reports and real-time ESM approaches capture different levels of analysis and therefore support different inferences. Retrospective measures provide global, memory-based summaries of behavior, affect, and context, and are particularly useful for identifying relatively stable between-person differences. However, it is limited in terms of assessing momentary fluctuations or establishing a temporal ordering of affective states and cannabis use in daily life. In contrast, ESM captures affect, behavior, cognition, context in real time, allowing the examination of within-person variability and short-term temporal dynamics. This distinction is relevant in the context of reinforcement and affect-regulation theories, which concern processes that unfold within individuals over time. Accordingly, between-person associations derived from cross-sectional or baseline data should not be assumed to reflect within-person mechanisms directly.

Prior ESM research has examined both affective antecedents and consequences of cannabis use in daily life. Regarding antecedents, higher momentary positive affect predicted a greater likelihood of cannabis use later in the day (Chakroun et al., 2010), and higher intentions to use cannabis at the next assessment in university students (Sznitman et al., 2022). Evidence for negative affect as an affective antecedent of cannabis use is more heterogeneous: Higher negative affect has been associated with an increased likelihood of subsequent cannabis use among adolescents and young adults (Shrier et al., 2014a). However, other studies have reported null or inverse associations, such as no predictive effect of anxiety (Tournier et al., 2003) or findings in which higher depressed mood predicted lower cannabis use (Chakroun et al., 2010).

With respect to affective consequences of cannabis use, prior use has been associated with higher positive affect and lower negative affect in university students (Sznitman et al., 2022). In a community sample, positive affect was also higher during cannabis-use moments, whereas negative affect varied by social context - increasing when participants used alone and decreasing when they used with others (Denson et al., 2023). In a clinical adolescent sample recently discharged from psychiatric inpatient care, cannabis use was likewise associated with higher positive affect and lower irritability but showed no reduction in negative affect (Brick et al., 2023).

Existing ESM studies, examining antecedents and consequences of cannabis use, have been conducted among specific and heterogeneous populations, including university students (Chakroun et al., 2010, Tournier et al., 2003, Sznitman et al., 2022), adolescents recently discharged from psychiatric inpatient care (Brick et al., 2023), youth medical outpatients (Shrier et al., 2014a), individuals motivated to quit cannabis use (Buckner et al., 2013), and a sample meeting criteria for CUD (Buckner et al., 2015). Furthermore, one study recruited adults based on tobacco and e-cigarette use who also reported using concurrent cannabis during ESM assessments (Denson et al., 2023). Notably, very few ESM studies have examined these processes in a sample of adults with near-daily cannabis use, a group likely to show substantial overlap with individuals experiencing clinically significant cannabis-related problems or meeting the criteria for CUD. Existing studies provide an important frame, but we do not know if these findings generalize to near-daily users, who are at higher risk of cannabis-related harms and in whom reinforcement processes are more habitual. Moreover, many studies examine antecedents or consequences in isolation, preventing direct evaluation of short-term affective feedback loops central to reinforcement theories.

In addition, existing ESM studies have only to a limited extent addressed the broader sources of heterogeneity discussed in the following sections. Most have focused primarily on momentary affective antecedents and consequences of cannabis use, with less attention to whether these processes vary according to gender-related factors, sleep-related processes, or differences in cannabis product characteristics. As a result, although ESM has an advanced understanding of affect–use dynamics in daily life, important questions remain regarding how these processes may differ across individuals, contexts, and forms of cannabis

exposure.

1.6 Sources of heterogeneity in cannabis-related outcomes.

1.6.1 The role of gender

Men are nearly twice as likely to report using cannabis as women, and CUD remains approximately twice as prevalent among men, who tend to initiate use earlier and report higher frequency and severity (Shao et al., 2023, UNODC, 2025b, UNODC, 2024a, Segura et al., 2025). However, recent evidence suggests that gender differences in past month use are narrowing in several countries (Hesse et al., 2020, UNODC, 2024b, Hasin et al., 2016, Segura et al., 2025, EUDA, 2025, Shao et al., 2023). Biological sex differences have been documented in routes of administration, acute intoxication effects, and withdrawal experiences, suggesting potentially distinct vulnerability profiles for males and females (Matheson et al., 2020, Rochat et al., 2024, Gex et al., 2023, Sholler et al., 2021).

Beyond differences in prevalence, patterns of use, and frequency, gender differences in cannabis use are observed in vulnerability profiles and clinical trajectories. Despite lower overall levels of use, women tend to show more rapid progression from heavy use to CUD and show higher rates of psychiatric comorbidities, especially internalizing symptoms such as anxiety and depression (Towers et al., 2023, Hesse et al., 2020). Furthermore, women also encounter more significant barriers to accessing treatment, which may further exacerbate trajectories of affect-driven and persistent use (EUDA, 2024).

Emotion regulation difficulties represent a transdiagnostic vulnerability factor that may help explain why women are more vulnerable to cannabis-related harms despite the lower levels of use. Previous studies examining emotion regulation difficulties among people who use cannabis have largely focused on sex (rather than gender) differences and used samples with low-to-moderate frequency of cannabis use. Within these samples, findings regarding sex differences point in opposite directions. For example, in a sample with past month cannabis use (11 days/month), males reported greater levels of emotion regulation difficulties than females, and associations between emotion regulation difficulties and problematic cannabis use were stronger among males than females (Cavalli and Cservenka, 2024). In

contrast, in a large university sample with similar patterns of use (9 days/month), females reported higher levels of emotion regulation difficulties than males, and these difficulties were more strongly associated with emotional distress among females and mediated the association between cannabis use and internalizing symptoms only in females (Weidberg et al., 2023). This mixed pattern is consistent with findings from a large meta-analysis showing that although emotion regulation abilities are reliably associated with substance use, evidence for sex differences is mixed and inconclusive (Weiss et al., 2022). The limited and contrasting evidence means that gender differences in emotion regulation among people who use cannabis remain unresolved, and because previous studies are largely based on samples with low-to-moderate or heterogeneous use patterns, we do not know if these processes generalize to individuals with daily or near-daily use (Weidberg et al., 2023, Cavalli and Cservenka, 2024).

1.6.2 Sleep quality

Most individuals with CUD remain untreated (EUDA, 2024), and among those seeking treatment, treatment demand is higher for adolescents and young adults (UNODC, 2025c). Among those not seeking treatment, the annual remission rate is approximately 17% (Calabria et al., 2010). CUD is closely linked to impairments in key health domains of functioning, particularly sleep and mental health, which are central correlates of cannabis use severity (Volkow et al., 2014, Babson et al., 2017, Lev-Ran et al., 2014, Feingold et al., 2016).

Sleep is fundamental for health, and sleep disturbances are highly prevalent and prospectively associated with adverse health outcomes (Lim et al., 2023, Roth et al., 2006). Many individuals report using cannabis as a sleep aid, and sleep improvement is among the most frequently endorsed motives for use (Bonn-Miller et al., 2014). However, the empirical literature points to a disconnect between perceived benefits and observed outcomes. In community samples, endorsing current cannabis use is associated with worse subjective sleep quality, and more frequent edible use is linked to poorer sleep efficiency, shorter sleep duration, and worse overall sleep quality (Winiger et al., 2021). Longitudinal evidence indicates that poor perceived sleep quality is associated with sustained cannabis use and poorer cessation outcomes over a six-month period, even after accounting for posttraumatic stress symptoms, withdrawal severity, and other substance use (Babson et al., 2013a).

Evidence regarding cannabinoid content and sleep also provides limited support for beneficial effects of concentrations of main cannabinoids, delta-9-tetrahydrocannabinol (THC) or cannabidiol (CBD), in naturalistic settings (Winiger et al., 2021). Moreover, heavy or high-potency cannabis use has been linked to long-term disturbances such as reduced Rapid Eye Movement (REM) sleep, delayed sleep onset, insomnia symptoms, and poorer sleep continuity (Edwards and Filbey, 2021).

1.6.3 Mental health: Symptoms of anxiety and depression

Cannabis use is commonly reported as a strategy to regulate negative affective states, including anxiety and depressive symptoms (Babson et al., 2013a, Babson et al., 2013b, Babson and Bonn-Miller, 2014, Babson et al., 2017, Gates et al., 2014, Bonn-Miller et al., 2007, Gex et al., 2024). Notwithstanding these motives, a larger body of evidence indicates that more frequent or problematic patterns of cannabis use are associated with greater psychological distress and elevated symptoms of anxiety and depression (Danielsson et al., 2016, Struble et al., 2019, Xue et al., 2021, Steeger et al., 2025).

High-THC products have been linked to acute anxiety, paranoia, and stronger physiological responses, and a growing body of evidence links longer term exposure and higher potency cannabis preparations with more adverse health outcomes (Colizzi and Bhattacharyya, 2017), including elevated symptoms of CUD (Freeman and Winstock, 2015, Meier, 2017, Cinnamon Bidwell et al., 2018), increased treatment admissions for cannabis problems (Freeman et al., 2018), higher risk of developing psychosis (Hjorthoj et al., 2023, Di Forti et al., 2015), increased risk of relapse among individuals with psychotic disorders (Schoeler et al., 2016), and anxiety symptoms (de Bode et al., 2025).

Sleep disturbances and anxiety are prominent withdrawal symptoms that can persist and maintain cannabis use, especially during quit attempts (Galang et al., 2015). This suggests that higher cannabis use severity may be closely tied to worse sleep and internalizing symptoms cross-sectionally, both due to self-medication attempts and the reinforcing effects of withdrawal-related discomfort.

Taken together, sleep quality and mental health (as measured here as symptoms of anxiety and depression) represent central, yet heterogeneous, health correlates of cannabis use

severity. Mixed findings across studies likely reflect differences in baseline vulnerability, frequency and patterns of use, contextual factors, and product characteristics. Examining how sleep quality and mental health relate to cannabis use severity, in near-daily cannabis use in naturalistic settings, is critical for understanding the broader health implications and maintenance processes associated with high-frequency use.

More recently, studies have emphasized that the relationship between cannabis use and sleep is likely bidirectional and context-dependent (Babson and Bonn-Miller, 2014, Babson et al., 2013b, Wong et al., 2019, Edwards and Filbey, 2021).

1.7 Cannabis product characteristics and heterogeneity of effects

Cannabis products commonly available in Europe include herbal cannabis (flower), cannabis resin (including hash), concentrate extracts, edibles, and vaporized preparations (EUDA, 2025). Cannabis contains more than 100 cannabinoids, with the most important being delta-9-tetrahydrocannabinol (THC) and cannabidiol (CBD). THC is the primary psychoactive component of cannabis and is responsible for its intoxicating effects and addictive potential. Cannabidiol (CBD), in contrast, is non-intoxicating and may offset some harmful effects of THC (Chandra et al., 2019). Accumulating evidence indicates that the absolute doses of THC and CBD, as well as their relative ratio, are key determinants of the level of harm an individual may experience (Chandra et al., 2019, Colizzi and Bhattacharyya, 2017). For example, observational and experimental studies show that high-THC products are associated with greater risk of CUD, psychotic-like experiences, and mental health problems (Curran et al., 2019, Freeman et al., 2018, Gage et al., 2016, Hines et al., 2020). Overall, high-THC products are associated with greater risk of harms such as acute intoxication episodes, emergency department presentations, and higher levels of use severity (EUDA, 2024, EUDA, 2025, Miro et al., 2025, Hoch et al., 2025b) (see also Chapter 1.5). There is more controversy regarding potential protective effects of CBD, with recent studies suggesting that CBD does not protect against the acute effects of THC, at least in doses found in typical products on the market (Englund et al., 2023, Colizzi and Bhattacharyya, 2017, Hoch et al., 2025b, Oliver et al., 2024, Chesney et al., 2025, Baltes-Flueckiger et al., 2025). In addition to plant-based cannabinoids, synthetic cannabinoids mimic the effects of THC and can be substantially more potent (Atwood et al., 2010, Castaneto et al., 2014).

Across Europe, analyses of seized samples indicate that average THC concentrations in cannabis resin have increased over the past two decades, with high-potency products becoming more prevalent in many markets (Freeman et al., 2019, EUDA, 2024, Freeman et al., 2021). Denmark exemplifies this trend: THC concentrations in seized cannabis resin increased from 8.3% in 2000 to 25.3% in 2017, representing the highest reported THC concentration in seized cannabis resin across Europe (Freeman et al., 2021, Rømer Thomsen et al., 2019). In contrast to many other European countries, where flower is the most used illegal cannabis product (EUDA 2024b), hash is by far the most used illegal cannabis product in Denmark (Danish Health Authority 2023; Callesen et al. 2024; EUDA 2024b; Frederiksen et al. 2025). Resin-heavy markets, such as in Denmark, may produce distinct risk profiles compared to markets where herbal cannabis predominates, reflecting a systematic difference in potency and patterns of use. Importantly, naturalistic studies show that cannabis users only partially adapt their smoking behavior to variation in THC concentrations, meaning that high-potency cannabis preparations will likely deliver larger doses of THC (Freeman et al., 2014, van der Pol et al., 2014).

In addition to differences in content, cannabis products vary in their routes of administration, creating meaningful variability in users' exposure to psychoactive cannabinoids (Freeman and Winstock, 2015, Curran et al., 2016, Hindocha et al., 2017). The mode of administration influences onset, intensity, and duration of intoxication (Volkow et al., 2014). Inhaled cannabis, whether smoked or vaporized, delivers THC rapidly into the bloodstream, producing a relatively immediate onset of intoxication, whereas edibles result in a delayed but prolonged effect due to slower gastrointestinal absorption and first-pass metabolism to active metabolites such as 11-hydroxy-THC. These pharmacokinetic differences are associated with distinct patterns of subjective intoxication and potential harms.

Taken together, variability in cannabinoid composition, potency, and routes of administration contributes to heterogeneous patterns of exposure and intoxication, which in turn may help explain the mixed and sometimes contradictory findings reported in the literature on cannabis use, sleep, and mental health. In this sense, it is critical to examine the differential health effects of specific cannabinoids (e.g., THC versus CBD), products and routes of administration (de Bode et al., 2025, Foo et al., 2026, Lake et al., 2025).

Chapter 2: Research gaps, aims and hypotheses

Global surveillance data indicate that daily or near-daily cannabis use is associated with social- and health-related harm than less frequent use (UNODC, 2025a, UNODC, 2024b). In epidemiological and population-based studies, more frequent and more problematic patterns of cannabis use are associated with adverse cannabis-related outcomes, including functional impairment, psychiatric comorbidity, and greater overall burden of disease (Segura et al., 2025, Hasin et al., 2016, Hjorthoj et al., 2023, Miro et al., 2025, Subbaraman and Kerr, 2015, Volkow et al., 2014, Koob and Volkow, 2016).

Despite the consistent relevance of this subclinical population, individuals who use cannabis on a daily or near-daily basis are underrepresented in research. Instead, cannabis use is typically operationalized using broad categorical indicators based on frequency (e.g., past-month use, number of use days, or ordinal frequency bands), within which high-frequency users are analytically pooled together with individuals with occasional or moderate use (Hasin et al., 2016, Cerda et al., 2020, Steeger et al., 2025, Danielsson et al., 2016, Mauro et al., 2022).

In sum, the lack of population-specific research on individuals who use cannabis daily or near-daily constrains the field's ability to identify the psychological processes that maintain high-frequency and problematic cannabis use. Examining mechanisms such as affective vulnerabilities, emotion-regulation difficulties, and health correlates within this subgroup is important to extend the existing evidence and provide more knowledge on one of the populations at highest risk of cannabis-related harms.

2.1 Manuscript 1: Individual vulnerability factors

Research Gaps

Several unresolved gaps limit our understanding of psychological vulnerability pathways in individuals at high risk for cannabis-related harms. First, prior studies have focused on low-to-moderate use frequency individuals, making it unclear whether emotion regulation difficulties relate to cannabis use severity among individuals with near-daily use patterns. Second, although prior work has linked emotion regulation difficulties to coping-motivated

cannabis use and to cannabis-related problems (Bonn-Miller et al., 2007; Bonn-Miller et al., 2008; Buckner et al., 2024), these studies did not evaluate severity of use (as measured by CUDIT-R), nor examined these associations in near-daily users. Third, findings regarding sex differences in emotion regulation are inconsistent (Weidberg et al., 2023, Cavalli and Cservenka, 2024, Weiss et al., 2022), and existing studies rarely investigate gender-related processes, despite growing evidence that gender norms, roles, and identities shape emotion regulation, coping strategy, and mental health vulnerability (Nolen-Hoeksema, 2012, Nolen-Hoeksema and Aldao, 2011, Aldao et al., 2010). Fourth, although coping motives are consistently linked to emotion regulation difficulties and hazardous patterns of cannabis use (Bonn-Miller et al., 2008, Bonn-Miller et al., 2007), no study has tested whether coping motives can help explain the link between emotion regulation difficulties and cannabis use severity in high-frequency users. Addressing these gaps may help clarify why women experience greater cannabis-related harms than men and may identify gender-specific pathways relevant for prevention and treatment.

Aims and Hypotheses

These gaps are addressed in Manuscript 1: A cross-sectional study, including 192 individuals who use cannabis near-daily, with the aim of examining: (1) Gender differences in emotion regulation difficulties, (2) associations between emotion regulation difficulties and cannabis use severity, and (3) the role of gender identity and coping motives in these associations.

We hypothesized that higher emotion regulation difficulties would be associated with greater cannabis use severity. Given mixed findings and conceptual and methodological heterogeneity in the literature regarding gender and biological sex, we did not formulate specific directional hypotheses regarding gender. Finally, we hypothesized that coping motives would mediate the link between emotion regulation difficulties and cannabis use severity.

2.2 Manuscript 2: Affective mechanisms

Research Gaps

Despite theoretical frameworks emphasizing the role of positive and negative reinforcement and emotion regulation processes in cannabis use (Khantzian, 1997, Baker et al., 2004, Bonn-Miller et al., 2007, Cooper, 1994), daily-life evidence for these mechanisms remains limited and inconsistent.

Prior ESM studies of cannabis use or intention to use have examined both affective antecedents and consequences of use (Chakroun et al., 2010, Shrier et al., 2014a, Buckner et al., 2012, Sznitman et al., 2022, Dyar and Rhew, 2026). For example, a recent study of students found that higher positive affect predicted stronger intention to use cannabis (Sznitman et al., 2022). Overall, previous ESM studies have found support for positive reinforcement in samples with students and young samples (Buckner et al., 2012, Shrier et al., 2014a, Chakroun et al., 2010, Sznitman et al., 2022), but we do not know if it generalizes to individuals with heavier cannabis use. In relation to negative affect as antecedent of cannabis use, findings are mixed. One study found that elevated negative affect preceded cannabis use (Shrier et al., 2014b) in line with negative reinforcement theories, while others found no association between negative affect and subsequent use (Tournier et al., 2003, Chakroun et al., 2010), or found the opposite pattern, with lower negative affect predicting intention to use (Sznitman et al., 2022). These studies have great heterogeneity in terms of participants and their frequency of cannabis use, suggesting that negative affect-driven use may depend on context and individual differences. This underscores the need to identify vulnerability factors - such as emotion regulation difficulties - that may shape momentary affect–use dynamics.

To my knowledge, no ESM study has simultaneously examined affective antecedents and consequences of cannabis use in a near-daily or daily user sample, despite theoretical models predicting bidirectional affect–use feedback loops. Second, no study has tested whether trait-level emotion regulation difficulties moderate affect→use and use→affect associations. Such individual differences may help explain the divergent findings across prior ESM studies. Together, these gaps limit our understanding of short-term affective mechanisms underlying near-daily cannabis use – one of the populations at highest risk of cannabis-related harms.

Aims and Hypotheses

These gaps are addressed in Manuscript 2: An ESM study, including 113 individuals who use cannabis near-daily, with the aim of examining within-person daily associations between affect (positive/negative affect) and cannabis use, and the role of emotion regulation difficulties in these associations. More specifically, we examined whether affect (positive/negative affect) influenced the subsequent likelihood of using cannabis, and whether cannabis use influenced positive/negative affect both shortly after use and at a later moment during the day; and the role of trait-level emotion regulation in these associations. The study was preregistered on the Open Science Framework before data analysis (OSF; <https://osf.io/c74ur/>).

Our hypotheses were guided primarily by negative reinforcement theory, which offers a stronger theoretical basis for negative affect-driven use among near-daily users than prior ESM findings currently do. In effect, we hypothesized that higher levels of negative affect would increase the likelihood of cannabis use, and that cannabis use would be associated with lower negative affect shortly after use. We expected these associations to be stronger in individuals with greater emotional regulation difficulties. We had no directional hypotheses for positive affect.

2.3 Manuscript 3: Health-related associations

Research Gaps

Despite extensive work on sleep and mental health including symptoms of anxiety and depression, several gaps remain. First, few studies have simultaneously examined sleep quality and anxiety/depression symptoms in relation to cannabis-use severity, and even fewer in individuals who use cannabis near-daily. Second, most existing studies rely on categorical indicators of use frequency (e.g., occasional versus frequent) or diagnostic groupings (e.g., CUD versus no-CUD), rather than dimensional severity measures. Studies frequently classify cannabis use as *non-daily*, *daily*, or *heavy* without quantifying severity using validated continuous scales such as the CUDIT-R (Conroy et al., 2016, Winiger et al., 2021, Babson et al., 2013a). This limits the ability to identify how sleep and mental health symptoms vary across levels of cannabis use severity. Third, although product characteristics systematically influence health-related outcomes - shaping intoxication intensity, sleep architecture, and

affective responses (Freeman et al., 2019; Rømer Thomsen et al., 2019; Curran et al., 2016) – no studies have tested whether cannabis product types moderate associations between cannabis-use severity and sleep or symptoms of anxiety and depression. This may be particularly salient to examine in Denmark, because the illegal cannabis market is hash-dominated (Danish Health Authority 2023; Callesen et al. 2024; EUDA 2024b; Frederiksen et al. 2025).

Aims and Hypotheses

These gaps are addressed in Manuscript 3: A cross-sectional study, including 197 individuals who use cannabis near-daily, with the aim of examining associations between cannabis use severity, sleep quality, and symptoms of anxiety/depression and the role of cannabis product type in this association.

We hypothesized that (1) greater cannabis use severity would be associated with poorer sleep quality and higher levels of anxiety/depression symptoms; (2) the use of hash products would be associated with poorer sleep quality and higher levels of anxiety/depression relative to using flower products; and (3) product type would moderate the associations between cannabis use severity, sleep quality, and symptoms of anxiety/depression, with stronger severity–outcome links among people using hash compared to those who use flower products.

Chapter 3: Methodology

This chapter describes the research design, approvals and open science, requirements for participation, recruitment strategies, and the procedures. Also, it provides an overview of the measures and analytical approaches included in the manuscripts.

3.1 Design

The research underlying this dissertation employs a quantitative, observational, and longitudinal design. The project combines baseline assessment and a 21-day Experience Sampling Method (ESM) protocol. This design has a common methodological agenda for three empirical studies that complement each other. Specifically, two studies were cross-sectional, using baseline data to address between-person questions, and one study used intensive longitudinal ESM data to examine within-person processes over time. Although the studies differed in their analytical focus and temporal resolution, they allow complementary perspectives on affective processes and health outcomes related to cannabis use.

3.2 Approvals and open science

The research underlying this dissertation was reviewed and approved by the Research Ethics Committee at Aarhus University (BSS-2023-051) and was registered with Danish Data Protection Agency and complied with its requirements. All study procedures were conducted in accordance with the principles outlined in the Declaration of Helsinki. Hence, participant anonymity and confidentiality were ensured, all participants were fully informed, participation was voluntary, and withdrawal was possible at any time without consequences. The ESM study was preregistered on the Open Science Framework before data analysis (OSF; <https://osf.io/c74ur/>).

3.3 Participants

All recruitment materials were written in Danish. Participants were recruited both online and offline (flyers and posters). Inclusion criteria were assessed through an online screening questionnaire. To participate, prospective participants were required to meet the following criteria: (1) Be aged 18-65, (2) be able to understand Danish, (3) have smoked cannabis at

least three times per week during the past month, (4) have a smartphone with a Danish number for SMS notifications, and (5) not have used any other illicit substances in the past month. We included participants who *smoke* cannabis because it is the dominant route of administration in Denmark and to exclude participants who only use pure CBD products.

3.4 Recruitment

The research was advertised with flyers and posters in several contexts to recruit a sample of individuals with a high frequency of use which was as broad and representative as possible. See Figure 1.



Figure 1 Project recruitment material

Recruitment strategies included several contexts, such as: (a) Distribution of recruitment materials at Aarhus University campuses; (b) Distribution of recruitment materials in contexts

related to cannabis, including online platforms and community spaces. In online contexts, advertisements were published only after the moderator's approval; (c) E-mail invitations sent to individuals who had participated in a national survey at Centre for Alcohol and Drug Research, smoked cannabis at the time of the survey, and indicated that were interested in being contacted for future research; (d) Distribution of recruitment materials at the Copenhagen Cannabis Exhibition in 2023 which resulted in several people signing up to a waiting list for the study; and (e) Snowball dissemination via individuals in the cannabis communities in Denmark, who were asked to share the study information with potentially eligible participants.

Table 1 lists the communities that were approached during the entire period.

Recruitment context	Communities (n)
Copenhagen Cannabis Exhibition (2023)	1
Facebook groups related to cannabis	15
Danish Reddit subpages	2
Cannabis shops in Danish cities	3
Aarhus University campuses	2
Two prior national surveys	2

Table 1 Recruitment contexts and number of communities approached

A website was created for potential participants to access relevant information about the research. The website included: (a) Overview of the study, (b) criteria for eligibility and study procedure, and (c) frequently asked questions and contact information. See Figure 2.

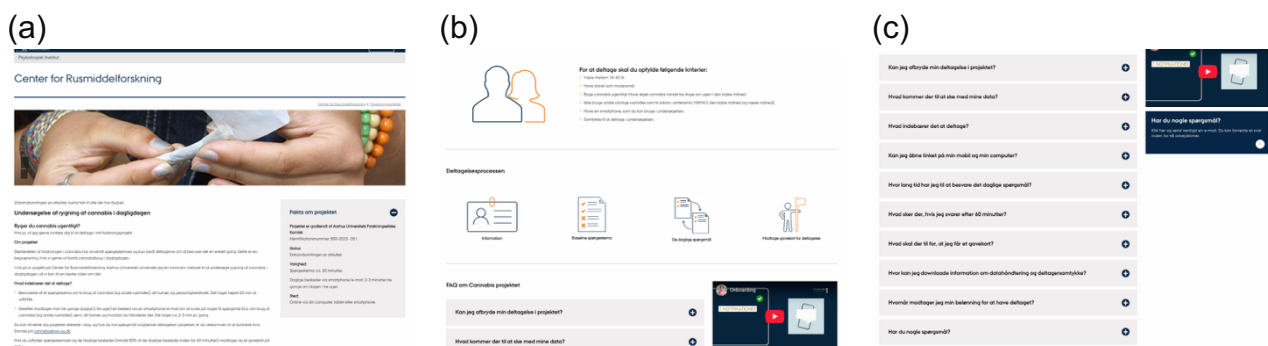


Figure 2 Overview of the research website

3.5 Study procedure

The procedure can be divided into three phases: Screening for eligibility, baseline measures (including an onboarding video), and the ESM. Figure 3 illustrates the sequential study procedure, including eligibility screening, baseline assessment, and experience sampling period. Participants who completed at least 80% of daily assessments within 60 minutes of prompt delivery received 300 kr. (approximately 40 €) as compensation for their participation. No compensation was provided otherwise.

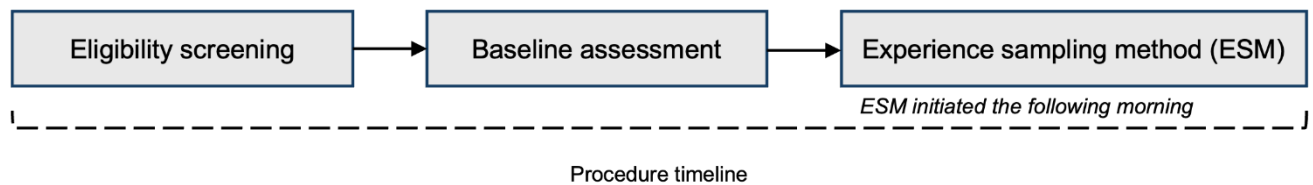


Figure 3 Overview of the study procedure

3.5.1 Baseline procedure

Data were collected from January 19, 2024 to October 10, 2024. All participants resided in Denmark, and all materials were in Danish. Research Electronic Data Capture (REDCap), hosted at Aarhus University (Harris et al., 2009) was used for data collection and management. Upon confirmation of their eligibility, participants were invited to complete the baseline online survey, which took approximately 20 minutes. Table 2 provides an overview of the instruments included in the baseline survey and their presentation sequence (instruments are explained in more detail in section 3.6.1).

	Instrument	Construct
1	Sociodemographic questionnaire	Age, gender identity, education, occupation, living situation, region of residence in Denmark.
2	Nicotine use questionnaire	Nicotine use
3	Fagerström Test for Nicotine Dependence (FTND)	Nicotine use severity
4	Alcohol use questionnaire	Alcohol use
5	Alcohol Use Disorders Identification Test (AUDIT)	Alcohol use severity
6	Cannabis use questionnaire	Cannabis use
7	Cannabis Use Disorders Identification Test–Revised (CUDIT-R)	Cannabis use severity
8	Timeline followback (TLFB)	Days using pot, skunk, and/or hash (and gram).
9	Marijuana Motives Measure (MMM)	Motives for using cannabis
10	Use of other substances	Lifetime history of other substances
11	Pittsburgh Sleep Quality Index (brief version)	Sleep quality

12	Patient health questionnaire (PHQ-4)	Anxiety and depressive symptoms
13	Difficulties in Emotion Regulation Scale (DERS-16)	Difficulties in regulating emotions

Table 2 Overview of the instruments administered at baseline

After completing the baseline survey, participants watched a video that introduced the procedures for the 21-day ESM. The video (52 seconds) was delivered via REDCap and hosted on YouTube. Each screen of the video is displayed in Figure 4. The video included: (a) Overview of the response steps, and instructions on how to (b) enable notifications, (c) open a questionnaire link, (d) answer daily questions, (e) submit responses, (f) possibility of responding using computer, phone and tablet, and (g) contact information for participant support.

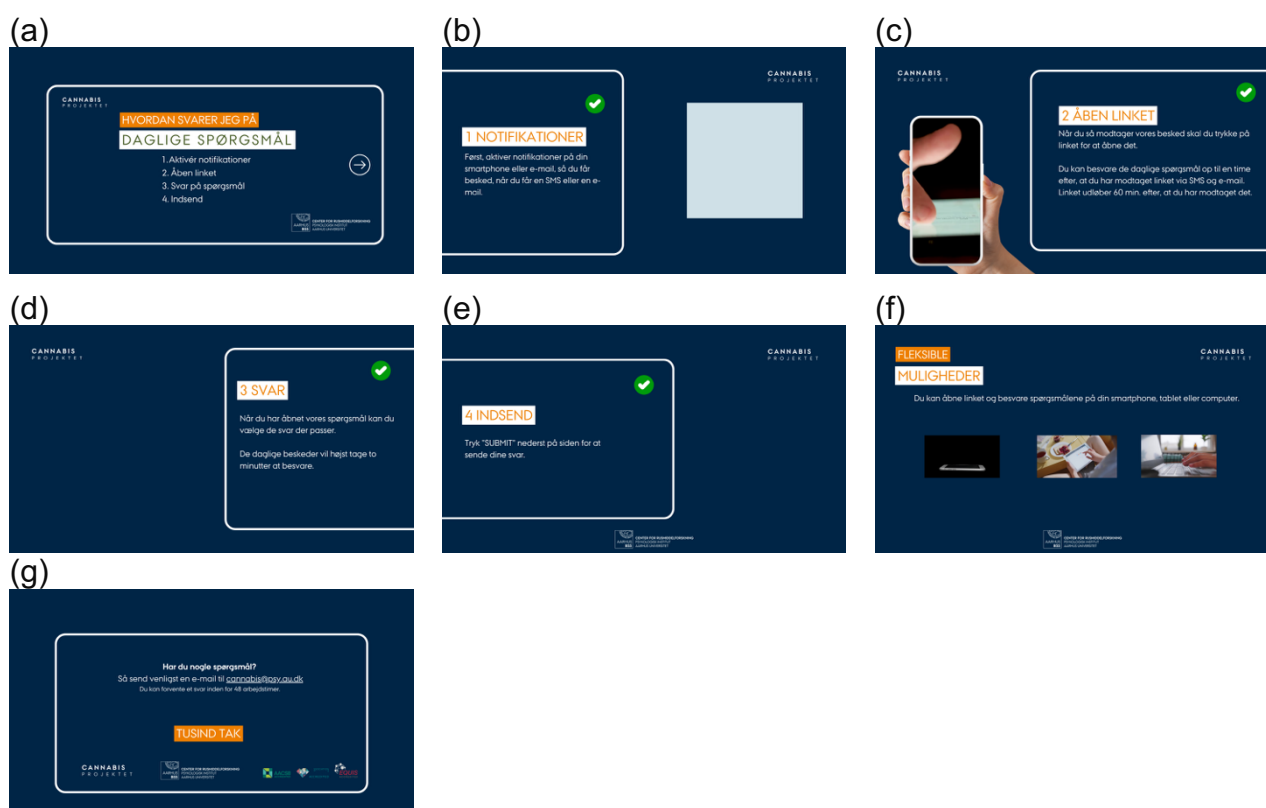


Figure 4 Instructional sequence shown to participants before the ESM period

3.5.2 ESM procedure

The ESM period began the day following baseline completion and was conducted over 21 consecutive days. Participants received three SMS-based assessments per day. The design was time-based, with assessments scheduled at semi-random times within three predefined

anchor times and daily windows: Morning (11:00), afternoon (16:00), and evening (21:00). For each window, assessments were delivered within a 20-minute time interval centered on the anchor time. Participants were informed that they needed to complete each assessment within 60 minutes. The randomization schedule and the number of prompts were identical across all participants and remained constant across days, including weekends.

See Figure 5 for more information on the study procedure: From the top to the bottom, the first part represents the sequence of the entire protocol: (a) Following baseline assessment, participants completed a 21-day ESM period, (b) each day included three semi-randomized assessments delivered within predefined morning, afternoon, and evening windows, and (c) prompts assessed momentary affect and cannabis use since the previous assessment in each time point.

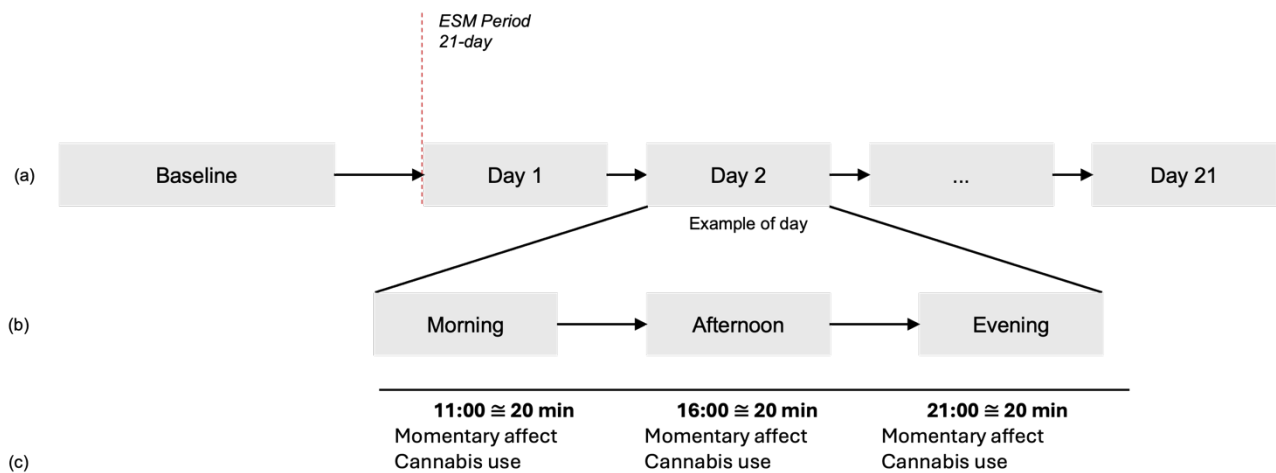


Figure 5 Study procedure and within-day structure of the ESM protocol

Experience sampling measures included assessments designed to capture momentary states and recent behaviors throughout the day. The content varied by time window to align with the temporal nature of the constructs assessed (momentary, prospective, and retrospective). An overview of the constructs and assessment timing is provided in Table 3 (these measures are explained in more detail in section 3.6.2).

All assessments included momentary affect and cannabis use since the previous assessment. This design allowed for the examination of within-day and across-day associations between affective states and cannabis use.

Construct	Instrument/ source	Time window		
		Morning	Afternoon	Evening
Momentary affect	PANAS, ESM adapted item(s)	X	X	X
Cannabis use	Custom ESM item(s)	X	X	X

Table 3 Overview of ESM constructs, source of instrument, and assessment schedule

3.6 Measures

3.6.1 Baseline measures

Sociodemographic information

In terms of demographic information, the participant was asked to indicate gender identity (“What gender do you identify yourself as?”), age, education, and current occupation.

Nicotine and alcohol use and severity of use

Use of nicotine and alcohol was assessed descriptively using a combination of lifetime use (nicotine and alcohol), age of onset, and frequency of use in the past 30 days. Severity of nicotine use was assessed by the Fagerström Test for Nicotine Dependence (FTND; Heatherton et al., 1991). Severity of alcohol use was assessed by the Alcohol Use Disorders Identification Test (AUDIT; Saunders et al., 1993). In addition, a multi-select questionnaire was used to assess lifetime use of other illegal substances such as cocaine, stimulants, or ecstasy/MDMA.

Cannabis use and severity of use

Cannabis use was assessed using a multi-select questionnaire in which participants indicated all cannabis products used in the past 30 days. Response options included hash, pot (flower), skunk, legally available cannabis products with low-THC concentrations (< 0.2 % THC), prescribed cannabis (e.g., Bediol, Bedrocan), other smoked or inhaled forms (e.g., oil used with vaping), and unknown types. Based on these responses, participants were classified into mutually exclusive cannabis product groups to account for single- and multiple-product

use. Specifically, participants were assigned to one of five groups: (1) flower only, (2) hash only, (3) multiple THC products, (4) THC-dominant concentrates, or (5) combined THC + CBD products. For each endorsed product type, participants also reported the number of use days in the past 30 days.

Past-month cannabis use frequency was also assessed using the Timeline Follow-Back (TLFB; Martin-Willett et al., 2020), specifically for pot, skunk, and hash (if selected by the participant in the single items above). In the TLFB participants also indicated the grams used (of each product; pot, skunk, hash) for each day in the last 30 days. Because the gram-level quantity estimates showed substantial imprecision, including implausible ranges and high variability, information on grams was not used in any of the analyses.

To measure severity of cannabis use, participants completed the Cannabis Use Disorders Identification Test-Revised (CUDIT-R; Adamson et al., 2010). CUDIT-R is a validated 8-item self-report measure assessing the frequency of cannabis use and cannabis-related harms over the past six months. Items are rated on a 5-point Likert scale, where higher scores indicate higher severity of cannabis use. Consistent with prior research, scores ≥ 8 indicate hazardous cannabis use and scores ≥ 12 indicate probable CUD (Adamson et al., 2010). The CUDIT-R total score was used as a continuous measure indicating cannabis use severity, reflecting both the extend of cannabis involvement and cannabis-related harms.

Cannabis use motives were assessed using the Marijuana Motives Measure (MMM; Benschop et al., 2015), a validated version with 21 items that captures five motive dimensions. Participants evaluate statements about their reasons for using cannabis using a 5-point Likert scale. The five dimensions include: (1) enhancement (e.g., because I like the feeling it gives me; four items); (2) coping (e.g., to forget my worries; four items); (3) conformity (e.g., for others to like me; two items); (4) expansion (e.g., to get to know myself better; five items); and (5) social (e.g., because it makes social gatherings more fun; six items). Dimension scores were calculated separately. In the present study, coping motives were included as a mediator based on previous evidence showing its role in linking emotion regulation difficulties to cannabis-related outcomes (Weidberg et al., 2023).

Mental health and sleep outcomes

Symptoms of anxiety and depression were assessed using the Patient Health Questionnaire-4 (PHQ-4; Kroenke et al., 2009), which is a validated 4-item measure of symptoms of anxiety and depression. The PHQ-4 includes two items for symptoms of anxiety and two symptoms for depression, scored from 0 to 4. The total scores range from 0 to 12 and higher scores indicate more severe symptoms of anxiety and depression.

Sleep quality over the past month was assessed using the brief Pittsburgh Sleep Quality Index (PSQI-b; Sancho-Domingo et al., 2021). PSQI-b is a brief, validated measure assessing the quality of sleep over the past month. Responses include self-reported times/durations or frequency rated on a 4-point Likert scale. Items are grouped and scored into five sleep components: (1) subjective sleep quality, (2) sleep latency, (3) sleep duration, (4) sleep efficiency, and (5) night disturbances. The sum of component scores (each coded 0—3) yields a total global score (PSQI-b) and higher scores indicate poorer sleep quality (Sancho-Domingo et al. 2021).

Emotion regulation difficulties

Difficulties in regulating emotions were measured using the Difficulties in Emotion Regulation Scale-16 (DERS-16; Bjureberg et al., 2016), which is a shortened version of the DERS-36 (Gratz and Roemer, 2004). DERS-16 is a validated measure that captures emotion regulation difficulties across five dimensions. Participants rate each item on a 5-point Likert scale with higher scores reflecting greater emotion regulation difficulties. The score can be interpreted as an overall sum of score or across five dimensions: (1) Lack of emotional clarity (two items); (2) inability to engage in goal-directed behaviors when distressed (three items); (3) difficulties controlling impulsive behaviors when distressed (three items); (4) nonacceptance of negative emotions (three items); and (5) limited access to emotion regulation strategies perceived as effective (five items). The total score and each dimension were calculated separately (Bjureberg et al., 2016). A Danish translation and cultural adaptation was not available, and hence this process was also part of this study (see section 3.6.3). An overview of the main constructs assessed at Baseline, and the corresponding instrument, is provided in Table 4.

Construct and instrument	Timeframe	Items	Total Range
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Nicotine dependence, FTND	Past month	6	0—10
Alcohol use severity, AUDIT	Past 6 months	10	0—40
Marijuana Motives Measure, MMM	General	21	
Cannabis use severity, CUDIT-R	Past 6 months	8	0—32
Anxiety/Depression symptoms, PHQ-4	Past 2 weeks	4	0—12
Sleep quality, PSQI-b	Past month	4	0—15
Emotion regulation, DERS-16	Trait	16	16—80

Table 4 Overview of instruments at baseline

3.6.2 ESM measures

ESM items differed in their temporal reference windows depending on the construct assessed. Momentary affect was evaluated at each prompt using a “right now” formulation. Cannabis use variables were retrospectively assessed with reference to the period since the last ESM assessment, ensuring non-overlapping exposure windows.

Construct	Temporal reference window	Verbatim ESM item prompt
Affect	Momentary (0 min)	“How do you feel right now?” (positive and negative affect)
Cannabis	Retrospective (since last assessment / same day)	“Have you used cannabis today?”
Cannabis	Retrospective (since last assessment / same day)	“When was the last time you smoked cannabis (consider when you finished smoking)?”
Cannabis	Retrospective (since last assessment / same day)	“What type of cannabis did you smoke?”

Table 5 Verbatim ESM instruction and temporal reference windows

Momentary positive and negative affect

Momentary affect was assessed at each ESM prompt, using five positive affect items (happy, satisfied, excited, relaxed, proud) and five negative affect items (anxious, regret, angry, guilt, frustration) from the short version of Positive and Negative Affect Schedule (Haney et al., 2023). Participants rated each item on a 5-point scale ranging from 1 (“very little/not at all”) to 5 (“extremely”), in response to the question “Right now, I feel...[affect]?”. The positive affect and negative affect items were chosen to capture a broad range of valence and arousal properties, following and expanding on previous studies (Sznitman et al., 2022, Wycoff et al., 2018). Sum scores were computed separately for the positive affect and negative affect, with total scores ranging from 5 to 25. Higher scores indicate greater affect intensity.

Cannabis use

Cannabis use was assessed using a branching structure. Participants first reported whether they had used cannabis today. If they responded positively, follow-up questions evaluated: the mode of consumption (smoked or edibles), the timing of the last use (using a clock), and product type (here they could indicate hash, pot, and/or skunk).

3.6.3 Translation and cross-cultural adaptation processes

Since the study was conducted in Denmark, it was decided that all instruments would be administered in Danish, despite the high level of English proficiency in the general population in Denmark. This decision was made to ensure optimal comprehension and cultural appropriateness of the measures.

The cross-cultural adaptation of psychological instruments is a complex process that requires methodological rigor to ensure equivalence between the original and adapted versions. Whenever possible, questionnaires that had already been validated and translated into Danish in prior published work were used (e.g., AUDIT, CUDIT-R, FTND, PHQ-4, MMM, and PSQI-b). However, two instruments central to the present research - the DERS-16 and the Positive and Negative Affect Schedule adapted for experience sampling methods (PANAS-ESM) - had not previously been translated or culturally adapted into Danish. For this reason, a translation and cross-cultural adaptation process was conducted for these instruments by the author of this dissertation, in collaboration with a small team of Danish experts.

Although there is no single universally agreed-upon protocol for cross-cultural adaptation, established guidelines described in the psychometric literature were followed (Sireci and Parker, 2006, Borsa et al., 2012). The adaptation process involved five essential stages: (1) Translation of the instrument from the original language into the target language, (2) synthesis of the translated versions, (3) expert evaluation of the synthesized version, (4) evaluation of the synthesized version by a small sample from the target population, and (5) back-translation.

In the first stage, independent translations were conducted by three bilingual Danish-English speakers: One psychologist, one Danish language teacher, and one individual from an

unrelated field. In the second stage, a synthesis of the translations was carried out by a committee composed of two clinical psychologists: one native Danish speaker proficient in English (main supervisor) and one bilingual Danish-English speaker with expertise in substance use research. This synthesis involved comparing the translated versions and evaluating semantic, idiomatic, conceptual, linguistic, and contextual discrepancies to reach a single consensus version. In the third stage, the synthesized version was evaluated by one bilingual Danish-English speaker who is a psychologist and experienced researcher in emotion regulation research.

In the fourth stage, the synthesized version was evaluated by a small group of participants from the target population ($N = 16$). Participants were asked to assess whether each item was (1) clear, (2) linguistically appropriate for their age, and (3) easy to understand. Responses were rated on a 5-point Likert scale ranging from 1 ("not at all") to 5 ("very clear"). Participants were also invited to suggest rewordings for any item they perceived as unclear. These ratings were used to calculate a Content Validity Coefficient (CVC, Borsa et al., 2012). A threshold of ≥ 0.80 was considered acceptable (as recommended in Sireci and Parker, 2006).

In the final stage, a consensus meeting was held to address any remaining conceptual or construct-related issues identified during participant evaluation. All modifications were incorporated following discussion and agreement, and a saturation criterion (i.e., no new emerging issues) was applied. Final decisions regarding item wording were made by the author of this thesis and main supervisor. Finally, a back-translation from Danish to English was conducted to verify consistency with the original versions (no issues occurred at this stage). Formal psychometric validation of the adapted version was beyond the scope of the present dissertation and will be addressed in future studies.

Regarding the PANAS-ESM, the adapted version was selected to capture a broad range of basic emotional states, varying in valence (positive and negative) and intensity, while avoiding redundancy across items. Momentary affect was operationalized using composite scores rather than single items, allowing for a more reliable assessment of affective states in daily life. Specifically, five positive affect items (happy, satisfied, excited, relaxed, proud) and five negative affect items (anxious, regret, angry, guilt, frustrated) were selected based on

the short ESM-adapted version of the PANAS (Haney et al., 2023). particular attention was paid to item wording to ensure suitability for repeated, momentary assessment within an experience sampling framework.

All adaptations were conducted for research purposes and followed standard academic use of the original instruments.

3.6.4 Overview of constructs and measures across manuscripts

Although each manuscript addressed distinct research questions, several constructs overlapped across manuscripts, see **Error! Reference source not found..** More detailed descriptions of instruments, scoring procedures, and psychometric properties are provided in the individual papers.

Construct	Measures	Procedure	Manuscript		
			1	2	3
Cannabis use severity	CUDIT-R	Baseline	X	X	X
Cannabis use questionnaire	Questionnaire	Baseline	X		X
Cannabis use days	TLFB	Baseline		X	
Momentary affect	PANAS	ESM		X	
Sleep quality	PSQI-b	Baseline	X		X
Emotion regulation	DERS-16	Baseline	X	X	
Marijuana Motives Measure	MMM	Baseline	X		
Alcohol use	AUDIT	Baseline	X	X	X
Nicotine use	FTND	Baseline	X	X	X

Table 6 Overview of constructs, measures, and level across manuscripts

The constructs served different analytical purposes depending on the research question, data structure, and level of inference. These differences guided the analytical strategies adopted in each manuscript.

3.7 Analytical strategy

The three empirical manuscripts employed complementary analytical strategies implemented to address distinct but complementary research questions and data structures. Analyses differed in terms of level (between-person vs. within-person), outcomes, and analytical approach, while sharing a common focus on cannabis use as a dynamic affect-regulatory process. Rather than applying identical methods across studies, the analytical strategy

reflects a coherent progression of this research agenda from individual vulnerabilities to momentary mechanisms and to health variables, thereby integrating different levels of analysis into a unified research agenda. For an overview of the manuscripts see Table 7.

Manuscript	Data structure	Level	Main outcomes	Analytic approach
1	Cross-sectional	Between-person	Cannabis use severity	Regression-based models Moderation Mediation
2	Longitudinal (ESM)	Between- and within-person	Cannabis use Momentary affect	Multilevel Models Time-lagged models
3	Cross-sectional	Between-person	Sleep quality Mental health	Linear regression models

Table 7 Overview of the three manuscripts

Manuscript 1: Main analyses

Analyses were conducted on N=192 (5 individuals with non-binary gender identity were excluded from analyses due to the low number). Ordinary least squares (OLS) regression with HC3 robust standard errors was used to test whether overall emotion regulation difficulties (DERS-16 total) and the individual subscales were associated with cannabis use severity (CUDIT-R total). Regression analysis including all DERS-16 subscales simultaneously, was performed to examine shared versus unique variance explained in cannabis use severity. Next, a moderation analysis was performed to examine whether associations between DERS-total/subscales and cannabis use severity differed between men and women. Finally, we tested a mediation model in which cannabis coping motives were specified as a mediator of the association between emotion regulation difficulties (DERS-16 total score) and cannabis use severity (CUDIT-R total score). Given the lack of gender differences in the descriptive variables, no covariates were added to any of the analyses.

Manuscript 2: Main analyses

Preprocessing

Data were exported from REDCap, followed by a series of coherence and ordering checks on the timestamps (start, end, and last cannabis use hour) to identify and correct any temporal inconsistencies. When a participant completed an evening survey the following

morning, or reported cannabis use in the morning survey, referring to consumption that occurred the previous evening, the use timestamp was adjusted to the corresponding date (i.e., previous day) to reflect the timing of the event. No major temporal inconsistencies were found (Jordan Revol et al., 2024). In addition to meeting all other eligibility criteria, participants were retained in the analysis if they completed ≥ 21 assessments and reported cannabis use at least twice per week, ensuring sufficient within-person variability for longitudinal analysis as previous studies (Sznitman et al., 2022).

Statistical modeling

Analyses were conducted on $N = 113$ (who fulfilled criteria for the study inclusion and ESM participation). To examine bidirectional relationships between cannabis use and affect, modelling both within- and between-person associations, we used multilevel linear modeling (MLMS) with random intercepts to account for within-person repeated daily assessments. Models were estimated separately for positive affect and negative affect.

Within-person predictors included affect (positive/negative affect), time until next use, and time since last use (derived from ESM surveys). Between-person predictors included cannabis use days in the last month (baseline TLFB) and emotion regulation difficulties at baseline. Within-person variables were person-mean-centered to estimate within-person associations, whereas between-person variables were grand-mean-centered (Hamaker and Muthen, 2020). Cannabis use days in the last month (baseline) were left-skewed, but as reversing and transforming the scale did not improve the fit, we maintained the non-transformed grand-mean centered version for the final models. All time variables (e.g., minutes until next use, and minutes since last use) were log-transformed after centering to reduce skewness.

Manuscript 3: Main analyses

Analyses were conducted on $N=197$ (in this paper, all 197 participants were retained, because focus was not on gender identity). We fitted ordinary least squares (OLS) regression models with HC3 robust standard errors to test whether cannabis use severity (CUDIT-R) was associated with (1) sleep quality (PSQI-b) and (2) anxiety/depression symptoms (PHQ-4). Each outcome was analyzed in a separate model. To examine the role of product type,

we first assessed whether product type was associated with cannabis use severity, sleep quality, and anxiety/depression symptoms using a one-way analysis of variance (ANOVA). Next, we tested whether product type explained additional variance beyond cannabis use severity by refitting the models with cannabis product type entered as a categorical predictor and dummy-coded using flower only as the reference category (ANCOVA). Finally, to test moderation, interaction terms between cannabis use severity and cannabis product type were added to the models. When significant, interactions were probed using simple slopes and pairwise comparisons of slopes. For descriptive purposes, adjusted marginal means for each product type were also compared using main-effects models. Further, we conducted the same analytical approach with covariates (gender, age, nicotine use, and alcohol use), and subsequently after excluding the small concentrate group.

All statistical analyses were conducted using RStudio (Posit Software, 2025). Two-sided p -values $< .05$ were considered statistically significant. Detailed descriptions of analytical procedures, model specifications, and assumptions are provided in the Methods sections of each empirical manuscript available in the respective Appendix.

Chapter 4: Summary of Findings

4.1 Manuscript 1

A total of 192 participants were included in the analyses. Most of the participants were men ($n = 127$), followed by women ($n = 65$), and a small group identified as non-binary ($n = 5$; they are not included in the analysis due to the small number of participants). The mean age was 30.89 years ($SD = 10.65$). On average, participants reported using cannabis on 26.28 days in the past 30 days ($SD = 6.00$), and the mean CUDIT-R score was 14.11 ($SD = 5.67$), indicating probable CUD (Adamson et al., 2010).

The findings showed a positive association between emotion regulation difficulties and cannabis use severity, with DERS-16 total explaining 31% of the variance in cannabis use severity. Women reported higher emotion regulation difficulties than men, but gender identity did not moderate the association between emotion regulation difficulties and cannabis use severity.

Participants with higher levels of emotion regulation difficulties were more likely to endorse cannabis use to cope with emotions, and these coping motives partially accounted for their higher cannabis use severity.

4.2 Manuscript 2

A total of 113 participants were included in the analyses. Most of the participants were men ($n = 68$), and the mean age was 31.6 ($SD = 10.4$). On average, at baseline, participants used cannabis on 27.06 days ($SD = 6.72$) in the last month, and the average CUDIT-R score was 14.36 ($SD = 5.57$), indicating probable CUD (Adamson et al., 2010). The majority had used alcohol (68.5%) and nicotine (75%) in the last month.

Regarding ESM characteristics, the analyzed data included 7119 valid observations provided by the 113 participants. On average, participants completed 85.72% of the ESM surveys ($SD = 15.69$; median = 92.1%), with completion rates varying only slightly across survey types: morning, 87% ($SD = 16.35$); afternoon, 85.54% ($SD = 16.54$); and evening, 84.61% ($SD =$

18.16). The response ranged from 36.5% to 100%, supporting the reliability of within-person analyses.

Across all ESM surveys, participants reported between 7 and 63 cannabis use events ($M = 27.1$, $SD = 14.8$; median = 24), indicating between-person variability in use frequency. Overall, cannabis use was reported in 43% of all assessments, corresponding to an average of 27 events of use during the 21-day assessment period ($SD = 14.76$). The median time since last cannabis use was 5.8 hours (IQR = 1.02 –21.03), whereas the median time until next cannabis use was 11.06 hours (IQR = 4.23 –23.08). Mean positive affect was 14.92 ($SD_{\text{between-person}} = 4.33$, $SD_{\text{within-person}} = 2.90$, $ICC = 0.691$), and mean negative affect was 6.46 ($SD_{\text{between-person}} = 2.03$, $SD_{\text{within-person}} = 1.90$, $ICC = 0.534$).

The main findings revealed two distinct but co-occurring affective reinforcement processes operating at the momentary level: Higher positive affect predicted a greater likelihood of cannabis use before the next assessment, and cannabis use predicted increased positive affect at the same assessment; but not at the next assessment. Negative affect was not related to use or vice-versa at the group level.

Furthermore, emotion regulation difficulties moderated the association between positive affect and subsequent cannabis use, such that individuals with greater difficulties had a weaker positive association between positive affect and subsequent cannabis use. Finally, emotion regulation difficulties moderated short-term effects of cannabis use on negative affect, with individuals with higher emotion regulation difficulties showing lower negative affect shortly after use compared to non-use moments.

4.3 Manuscript 3

A total of 197 participants were included in the analyses. Most participants were men ($n = 127$), followed by women ($n = 65$), and a small group identified as non-binary ($n = 5$). The mean age was 30.7 years ($SD = 10.63$). Educational attainment was distributed as follows: 28.9% lower/pre-secondary education, 28.4% upper secondary education, 16.2% vocational education, 19.8% higher education, and 6.6% other or none. On average, participants

reported using cannabis on 26.15 days in the past 30 days ($SD = 6.23$), and the mean CUDIT-R score was 14.26 ($SD = 5.99$), indicating probable CUD (Adamson et al., 2010).

In relation to the derived cannabis product categorization (mutually exclusive), most participants reported using multiple THC-dominant products ($n = 78$) or hash only ($n = 63$), followed by combined THC + CBD products ($n = 27$), flower only ($n = 26$), and THC-dominant concentrates ($n = 3$).

The findings showed that greater cannabis use severity was consistently associated with poorer sleep quality, explaining 8.1% of the variance in sleep quality and 24.5% of the variance in anxiety/depression symptoms. Cannabis product type explained additional variance in sleep quality ($\Delta R^2 = .068$) but not in anxiety/depression symptoms, but this was not the case for anxiety/depression symptoms. More specifically, individuals who reported using hash only or multiple THC-dominant products exhibited poorer sleep quality compared with those who used flower only, independently of cannabis use severity. These findings remained substantively unchanged after adjustment for covariates (gender, age, nicotine use, and alcohol use), and after excluding the small subgroup of participants reporting THC-dominant concentrates.

Chapter 5: Integrated discussion

5.1 Overview of the PhD framework

This dissertation investigates how cannabis use is related to stable individual vulnerabilities related to emotion regulation, momentary affective processes, and broader health correlates among individuals with indications of hazardous cannabis use and probable CUD. These components were studied through an integration of between-person and within-person processes. By combining cross-sectional and intensive longitudinal data from a sample that uses cannabis near-daily, the three manuscripts collectively delineate the role of these vulnerabilities, mechanisms, and health-related correlates, and how they may be linked. Together, these levels can be seen to form a coherent research framework that provides insights into how stable vulnerabilities influence momentary affective mechanisms, which in turn may accumulate into or be affected by health-related correlates.

5.2 Vulnerabilities

Manuscript 1 showed that emotion regulation difficulties are a key affective vulnerability associated with cannabis use severity among people who use cannabis near-daily, consistent with prior work (Weiss et al., 2022, Bonn-Miller et al., 2008). Individuals with emotion regulation difficulties were more likely to engage in coping-motivated use, which in turn was associated with greater cannabis use severity. This is in line with negative reinforcement models (Baker et al., 2004, Aldao et al., 2010, Aldao et al., 2016, Dvorak and Day, 2014) and affect-regulation models (Weiss et al., 2022, Khantzian, 1997), which argue that substance use can become integrated into individuals' emotion regulation repertoires as a means of normalizing or reducing negative affect.

Furthermore, women reported higher levels of emotion regulation difficulties than men. Previous studies in samples with lower frequency of use, have had mixed findings and

have focused on biological sex (Weidberg et al., 2023, Cavalli and Cservenka, 2024). This is the first study examining gender differences in emotion regulation in individuals with near-daily cannabis use, and hence more studies are needed - preferably also in samples with more gender diversity.

Although women reported higher levels of emotion regulation difficulties than men, gender identity did not moderate the association between emotion regulation difficulties and cannabis use severity. This suggests that emotion regulation difficulties operate as a general transdiagnostic psychological vulnerability across genders in this population, characterized by high frequency cannabis use and a high likelihood of meeting criteria for CUD. Together, these findings establish emotion regulation difficulties and coping-motivated use as important vulnerability factors, which can help explain who is at risk of developing more severe patterns of cannabis involvement.

5.3 Momentary affective mechanisms

Manuscript 2 examined affective processes underlying cannabis use in daily life focusing on how momentary fluctuations in affect relate to subsequent use and how these processes vary as a function of trait-like emotion regulation difficulties. The present study is the first to examine these processes in individuals with near-daily cannabis use with probable CUD, thereby extending prior findings from ESM studies in samples with less frequent use (Chakroun et al., 2010, Shrier et al., 2014, Sznitman et al., 2022, Brick et al., 2023, Denson et al., 2023), and the first to examine the role of trait-like emotion regulation difficulties.

At within-person level, higher momentary positive affect was associated with a greater likelihood of cannabis use at the next assessment. In addition, cannabis use was followed by higher positive affect at the same assessment. Together these findings indicate a bidirectional association between positive affect and cannabis use that operates over short time scales, with positive affect functioning as both antecedent and an immediate affective consequence of cannabis use. This pattern is consistent with positive reinforcement models positing that cannabis use is more likely to occur in positively valenced contexts and is maintained through immediate rewarding affective consequences (Berridge and Robinson, 2016, Robinson and Berridge,

2008). These findings also align with prior ESM studies conducted in student samples, high-risk samples and younger populations (Chakroun et al., 2010, Shrier et al., 2014, Sznitman et al., 2022, Brick et al., 2023).

From a temporal perspective, the observed increase in positive affect was limited to the assessment at which cannabis use was reported and did not persist to the subsequent assessment. This supports the interpretation that the positive affect-related effects of cannabis use are acute and short-lived, rather than sustained over longer intervals. This pattern is consistent with pharmacological accounts of cannabis effects, as well as with reinforcement-based models in which immediate hedonic responses strengthen use behavior without necessarily producing lasting changes in affective baseline (Hindocha et al., 2017, Zehra et al., 2018, Koob and Volkow, 2016).

Furthermore, the presence of positive affect both before and immediately after use suggests that cannabis use may be embedded in positively valenced contexts. Moments of elevated positive affect may reflect leisure, social interaction, or availability-related contexts in which cannabis use is more likely to occur and more rewarding. Prior ESM studies have shown that cannabis use frequently occurs in social settings and during times characterized by positive mood or reduced constraints, such as evenings and weekends (Buckner et al., 2015, Shrier et al., 2014, Buckner et al., 2012). Repeated pairing of cannabis use with such positive affective states may strengthen learned associations between positive affect-related cues and use behavior, thereby increasing the likelihood of future use during similar affective contexts (Robinson and Berridge, 2008). Contrary to our hypotheses, higher momentary negative affect was not associated with a greater likelihood of subsequent cannabis use, nor was cannabis use followed by systematic changes in negative affect at the group level. This pattern is consistent with prior ESM studies in young adult samples reporting null or inverse associations between negative affect and cannabis use (Chakroun et al., 2010, Shrier et al., 2014, Sznitman et al., 2022). Although a small number of ESM studies have reported positive associations between negative affect and subsequent cannabis use, these effects typically emerge at broader temporal windows (e.g., hours or days before use; Shrier et al., 2014) or indirectly via craving (Buckner et al., 2012). We observed relatively low overall levels of negative affect and

limited moment-to-moment variability in negative affect. This pattern is consistent with research suggesting reduced emotional variability among individuals who use cannabis regularly (Volkow et al., 2014, Koob and Volkow, 2016). This restricted negative affect variability may have limited our ability to detect negative affect–use coupling. Alternatively, this pattern may reflect methodological constraints (see Chapter 6 for further detail).

Moderation by emotion regulation difficulties

The within-person association between positive affect and subsequent cannabis use was moderated by baseline emotion regulation difficulties. While higher positive affect increased the likelihood of use at the sample level, this effect was attenuated among individuals with greater difficulties regulating emotions. This pattern suggests that momentary positive affect fluctuations function as more salient proximal cues for cannabis use among individuals with relatively lower emotion regulation difficulties. In this group, elevated positive affect may signal contexts or internal states in which cannabis use is anticipated, rewarding, or socially embedded, consistent with goal-directed positive reinforcement processes (Aldao et al., 2010, Aldao et al., 2016). In contrast, among individuals with greater emotion regulation difficulties, cannabis use appeared less contingent on momentary increases in positive affect, indicating reduced behavioral sensitivity to positive affective cues. One possible interpretation is that greater emotion regulation difficulties are associated with more habitual or automatized patterns of use, in which cannabis consumption is less tightly coupled to current affective states (Everitt and Robbins, 2016). Importantly, this attenuation does not imply an absence of positive reinforcement, but rather a reduced reliance on momentary positive affect as a decision-relevant signal.

Emotion regulation difficulties did not moderate the association between momentary negative affect and subsequent cannabis use, suggesting that vulnerability in emotion regulation did not increase the likelihood of using cannabis in response to momentary negative affect. However, emotion regulation difficulties significantly moderated the association between cannabis use and subsequent negative affect at the same assessment. Specifically, individuals with greater emotion regulation difficulties reported lower negative affect at use moments compared with non-use moments,

whereas no such difference was observed among individuals with lower emotion regulation difficulties. This finding indicates that relief-like reductions in negative affect following cannabis use emerge selectively among individuals characterized by elevated emotional vulnerability, consistent with negative reinforcement and self-medication models (Bonn-Miller et al., 2007, Baker et al., 2004, Khantzian, 1997). This selective moderation may explain the absence of group-level main effects of cannabis use on negative affect, as averaging across heterogeneous emotion regulation profiles may obscure subgroup-specific relief effects (Fried and Robinaugh, 2020, Fisher et al., 2018).

5.4 Health correlates

Manuscript 3 examined how cannabis use severity relates to key health-related domains, specifically sleep quality and symptoms of anxiety and depression, among individuals who use cannabis daily or near-daily. Across the sample, greater cannabis use severity was consistently associated with poorer sleep quality and higher levels of anxiety and depressive symptoms (Degenhardt et al., 2013, Hasin et al., 2016). This pattern aligns with a growing body of evidence linking heavier use and CUD to impairments in sleep and mental health and further suggest that severity of cannabis use captures a cumulative burden of use that extends beyond a clinical diagnose and the immediate or momentary effects of intoxication (Lev-Ran et al., 2014, Babson et al., 2017, Gates et al., 2014, Conroy et al., 2016).

More broadly this finding suggests that individuals with severe cannabis involvement tend to experience broader disturbances in emotional and behavioral functioning extending beyond the immediate consequences of use (Adamson et al., 2010, Hasin, 2013, Hasin et al., 2016). Importantly, severity-based measures such as CUDIT-R integrate multiple dimensions, including frequency of use, loss of control, and negative consequences, which may be more closely linked to health correlates than categorical diagnoses or short-term exposure indicators alone. Cannabis use severity can be conceptualized as a latent marker of cumulative dysregulation, capturing both behavioral persistence and its longer-term psychological correlates.

Cannabis product type contributed with additional explanatory variance for sleep quality. Specifically, individuals who reported using hash exclusively or in combination with other THC-containing products exhibited poorer sleep quality compared to those using flower only (Babson et al., 2017). This pattern suggests that pharmacological characteristics of commonly used products may shift average levels of sleep disturbance, likely via differences in potency or cannabinoid composition, even when they do not fundamentally alter the severity–outcome relationship.

In contrast, cannabis product type did not substantially modify the strength of associations between cannabis use severity and either sleep quality or anxiety/depression symptoms. This pattern aligns with interpretations that severity reflects a cumulative index of behavioral and psychological involvement that is more strongly tied to longer-term functioning than exposure to specific product characteristics alone (Petrilli et al., 2022). Once patterns of use reach higher levels of severity, variability in product characteristics may contribute relatively little additional explanatory power beyond the established behavioral and functional burden of use. This distinction highlights that product type and severity may operate at different levels of analysis: product characteristics influencing proximal physiological effects, and severity reflecting longer-term adaptation and impairment. In this way, Manuscript 3 provides the health-related layer of the PhD framework by showing that severity co-occurs with clinically relevant patterns of sleep and mental health disturbances, complementing the vulnerability- and mechanism-based findings of the preceding manuscripts.

5.5 An integrated model of cannabis use

Across the three manuscripts, a cannabis use emerges that links stable emotional vulnerabilities, momentary affective mechanisms, and broader health-related correlates. Together, the findings indicate that cannabis use among individuals with near-daily use is shaped by the interplay between trait-like emotion regulation difficulties, health correlates, and affective dynamics that unfold in daily life.

A central insight of this integrated framework is that affective reinforcement processes are conditional, at least in individuals with near-daily and with probable CUD. Positive

reinforcement appeared at between-person level, whereas relief from negative affect was conditional to levels of emotion regulation difficulties. This vulnerability-dependent pattern can help clarify why retrospective reports commonly emphasize relief-based motives for use (Simons et al., 2000, Bonn-Miller et al., 2007), yet previous ESM studies have found limited evidence for negative reinforcement in samples with lower levels of use (Sznitman et al., 2022, Wycoff et al., 2018, Shrier et al., 2012, Chakroun et al., 2010), which we replicated in a sample with high frequency use and probable CUD. The finding that negative reinforcement is at play selectively among emotionally vulnerable cannabis subgroups is novel and requires replication, preferably in larger samples, and with different ranges of use and severity of use.

These affective mechanisms align with the broader profile observed in Manuscript 3, in which individuals with higher cannabis use severity also showed poorer sleep quality and higher symptoms of anxiety and depression. Rather than implying temporal direction or causal pathways (which is not possible in the cross-sectional design), these patterns suggest that severity captures a constellation of behavioral and emotional difficulties that co-occur with sleep disturbance and psychological distress in this population. Although the hash only group had a differentiated average sleep quality, product type did not alter associations between severity and health indicators, suggesting that pharmacological characteristics and severity may reflect partly distinct sources of variability.

The three manuscripts situate cannabis use within contemporary transdiagnostic models emphasizing emotion regulation capacity as a central factor related to both immediate affective responses and broader patterns of functioning (Aldao et al., 2016, Weiss et al., 2022, Kuppens et al., 2010). Emotion regulation difficulties influence the contexts in which cannabis is used and the affective consequences of use, forming a bridge between trait vulnerabilities, momentary mechanisms, and health-related correlates. Clinically, these findings highlight the potential relevance of interventions targeting emotional awareness, distress tolerance, and regulatory flexibility for individuals with near-daily use. From a public health perspective, the results illustrate how short-term affective dynamics - particularly when intertwined with underlying emotional vulnerabilities - coexist with sleep and mental health disturbances in ways

that may help explain the complex clinical presentation observed in heavy cannabis users. Taken together, these findings support the coexistence of two affective reinforcement processes that are differentially expressed across individuals. Positive reinforcement processes, reflected in positive affect-related antecedents and consequences of use, appear robust at the within-person level, particularly among individuals with relatively intact emotion regulation capacities. In contrast, relief-like reductions in negative affect following cannabis use are not ubiquitous but emerge selectively among individuals with greater emotion regulation difficulties, consistent with conditional negative reinforcement processes. By demonstrating that emotion regulation difficulties shape affect–use dynamics in distinct ways for positive affect and negative affect, the present study may help clarify why prior ESM studies have reported null or inconsistent findings for negative affect. These results highlight the importance of accounting for person-level vulnerability when examining affective mechanisms of cannabis use in daily life. Although several studies report evidence suggestive of substantial individual differences — such as large intraclass correlations or divergent descriptive patterns — these differences are rarely modeled directly. The work in this dissertation addresses this gap by explicitly separating within- and between-person processes, modeling fine-grained temporal associations between affect and cannabis use, and testing theoretically motivated cross-level moderation by emotion regulation difficulties. This approach avoids averaging across opposing processes and allows for a more precise characterization of when, for whom, and under what conditions cannabis use reinforcement operates in everyday life.

Chapter 6: Limitations, strengths and conclusion

While this dissertation has provided novel contributions, it is also subject to several limitations, alongside methodological and conceptual strengths. Whereas study-specific limitations and strengths are discussed within each individual manuscript, this chapter offers broader reflections on the constraints and strengths of this dissertation. These considerations are particularly relevant for researchers aiming to investigate cannabis use through survey- and ESM-based methodologies.

6.1 Limitations across studies

Across all research procedures, data collection was conducted online. Although this mode of data collection offered several advantages, it also limited the researchers' ability to directly monitor participant engagement and adherence to study protocols.

A second limitation concerns systematic reporting biases and assessment reactivity related to the social and legal context of cannabis use. Cannabis remains a regulated substance in Denmark (Danish Health Authority, 2023), and current legal frameworks and social norms may shape how participants report cannabis use patterns and associated outcomes. Prior research in psychology and the social sciences suggests that such contexts may give rise to social desirability bias and stigma-related underreporting (Tourangeau and Yan, 2007). For example, previous studies showed a tendency to underreport adverse or impairing experiences related to cannabis use and other substances (Khalili et al., 2021, King et al., 2024, Eisele et al., 2025). Such biases may introduce systematic measurement error, potentially attenuating observed associations (Type II error) (Del Boca and Darkes, 2003, Tourangeau and Yan, 2007).

Moreover, participants' awareness of participating in cannabis-focused survey- and ESM-based research may have elicited measurement reactivity, whereby repeated self-monitoring alters affective states or cannabis use behavior (Ottenstein et al., 2024, Schneider et al., 2024, Stone et al., 2003). In ESM, such reactivity typically emerges through habituation or panel-conditioning effects across repeated prompts (Eisele et al., 2023, Kuppens and Myin-Germeys, 2022). Additional methodological constraints in the ESM include floor or ceiling effects in affect ratings (Simkovic and

Trauble, 2019, Liu and Wang, 2021). Taken together, these limitations reflect that social desirability-driven reporting bias, recall bias, and measurement reactivity, including habituation effects, may introduce systematic and random error, potentially biasing results toward the null hypothesis (Type II error) (Del Boca and Darkes, 2003).

Finally, in ESM it is possible to have timing inaccuracies leading to temporal misclassification of cannabis use episodes (e.g., reporting use at 22:00 when it occurred at 21:40) that could differentially affect individuals' subjective acute effects of cannabis (Metrik et al., 2011). Previously, Wycoff (2018) suggested that the temporal order of cannabis use, and affective state could be solved by using a combination of random and event-based prompts (i.e., self-initiated survey when cannabis use is about to occur). Shiffman (2009) emphasizes that this choice is mainly based on the goals of the research design.

With respect to cannabis-specific measurement, several limitations should be noted regarding the assessment of cannabis quantity, specific psychoactive compounds, and product potency. First, cannabis use was measured using two self-report measures (at baseline): a structured questionnaire and a TLFB procedure, both based on retrospective self-reporting. These instruments captured product types and patterns of use over the past 30 days, and the TLFB included self-reported estimates of grams used for each type on each day. However, the gram-level quantity estimates from the TLFB showed substantial imprecision, including implausible ranges and high variability, consistent with documented difficulties in estimating cannabis amounts (Prince et al., 2018, Robinson et al., 2014). Consequently, these quantity estimates were considered insufficiently reliable for precise quantification of cannabis exposure and were not included in the analyses across manuscripts.

Second, a related limitation concerns the lack of direct assessment of cannabinoid content. As in many other studies, objective data on THC, CBD, and THC:CBD ratios were not available. Relatedly, it was not possible to determine or measure “standard THC units”, reflecting a dose of 5 mg THC, which has been proposed to standardize dose measures across different product types in cannabis research (Freeman and Lorenzetti, 2023). Reliance on self-reported potency information reduces the precision with which pharmacological exposure can be characterized, as labelled THC/CBD

content may not accurately reflect actual exposure and may vary across products and markets (Hammond et al., 2020).

6.2 Methodological strengths across studies

This study included a naturalistic sample of individuals who use cannabis daily or near-daily and were characterized by severity levels indicating probable CUD according to CUDIT-R. This population is highly relevant due to documented health-related risks associated with heavy and problematic cannabis use, yet such individuals are often underrepresented in observational research, including ESM (Wyckoff et al., 2018). By examining this group, the three manuscripts provide unique insights into psychological, affective, health-related, and epidemiological aspects of cannabis use patterns in a subclinical population within a Danish context.

Second, the use of ESM enabled this dissertation to capture moment-to-moment dynamics of affect and cannabis use in participants' everyday lives. As noted by Trull and Ebner-Priemer (2009, 2020), ESM offers a unique opportunity to observe experiences as they naturally unfold, making it particularly well suited for identifying proximal triggers and short-term consequences of substance use (Trull and Ebner-Priemer, 2020, Trull and Ebner-Priemer, 2009). Its relevance for addiction research has also been emphasized by Shiffman (2009), who highlight ESM's capacity to assess behaviors and emotions in the context in which they occur (Shiffman, 2009b). By relying on frequent and momentary assessments, the ESM design minimizes recall bias and increases ecological validity – central advantages when studying transient states such as affect and cannabis use.

Third, a key strength of this research agenda was its two-layered data collection strategy, which enabled a more comprehensive characterization of participants than ESM alone could have provided (Hamaker, 2025). The baseline questionnaires supplied important information on stable (or relatively stable) psychological and health-related factors, while the ESM protocol captured dynamic, context-dependent processes. This combination allowed the construction of a dataset in which more enduring individual differences could be used to contextualize, qualify, or statistically adjust momentary patterns observed in daily life (Hamaker, 2025). Such structure

strengthened the analytical possibilities of the work, for example by enabling the examination of cross-level moderation (e.g., whether personality traits shape short-term bidirectional associations). As a result, the study offers a deeper and more integrated understanding of cannabis use among individuals with high frequency of use.

Another methodological strength relates to the mode of ESM delivery. The ESM protocol was implemented using REDCap, a widely used and secure platform for administering personalized surveys (Harris et al., 2009, Harris et al., 2019). This choice prioritized participant privacy, autonomy, and data security, while reducing participant burden by avoiding the need to download an additional smartphone application. To support these aims, ESM survey links were delivered via SMS using an integration between REDCap and a Danish messaging gateway (Harris et al., 2009, Harris et al., 2019). This approach likely enhanced accessibility and perceived safety within the Danish context, where SMS-based communication is widely used. Although REDCap-based SMS delivery affords less experimental control over user interaction compared to app-based platforms, it represents a participant-centered and ethically sensitive design choice aligned with real-world feasibility in intensive longitudinal research.

6.3 Future directions

While the dissertation can help fill some of the gaps identified in Chapter 2, this work – and the existing literature - also points to important areas for future research.

First, future studies should examine temporal dynamics and bidirectional links between cannabis use, affect, sleep, and mental health. Although this dissertation used both cross-sectional and ESM designs, sleep and mental health were examined only at the between-person level. Intensive within-person designs that repeatedly assess sleep quality, affect, and cannabis use could clarify whether sleep disturbances and psychological distress function not only as correlates, but also as proximal predictors of subsequent use or as consequences of use. In other words, such designs are needed to directly test bidirectional processes related to cannabis

use and sleep quality, as well as cannabis use and mental health such as anxiety/depressive symptoms.

Second, future work should more explicitly model heterogeneity in affect–use dynamics. Although the present study did not estimate random slopes for affect–use associations, the cross-level moderation approach adopted here provides a theoretically targeted test of interindividual differences. By examining baseline emotion regulation difficulties as a moderator of within-person affect–use associations, this approach aligns with reinforcement-based models that emphasize vulnerability-specific affective functions of substance use. Future studies combining random slope estimation with theoretically informed moderators may further clarify whether heterogeneity in affective reinforcement reflects continuous variation, distinct subgroups, or dynamic shifts over time.

Third, future research should focus on transitions in cannabis use behavior. The present studies focused on individuals with daily or near-daily use and severity levels indicating probable CUD, a group of high public health relevance. Examining earlier stages of use, escalation periods, and attempts to reduce or stop use would help clarify how emotion regulation difficulties and affective reinforcement processes change across stages of cannabis involvement. This would help determine whether the mechanisms identified here represent risk factors for escalation, maintenance processes, or barriers to change.

Fourth, future work would benefit from integrating multimodal and more objective assessments. Combining ESM self-report data with objective measures of sleep, activity, and cannabis exposure (e.g., actigraphy, wearable devices, or biological markers, and measures of potency such as standard THC units) could improve the precision of distinguishing affective and pharmacological processes. This integration may be particularly valuable in naturalistic settings where self-reported product characteristics and timing of use remain imperfect.

Finally, future research should examine the clinical and preventive implications of these findings. The observed heterogeneity in affective reinforcement processes suggests that interventions targeting emotion regulation and affective coping may be

particularly relevant for individuals with near-daily cannabis use and indications of CUD. Rather than assuming uniform mechanisms of use maintenance, treatment approaches may benefit from tailoring intervention components to individuals' affective profiles, for example by strengthening adaptive emotion regulation strategies among those who show relief-oriented use patterns.

In this context, Just-in-Time Adaptive Interventions (JITAs) represent a promising strategy for translating momentary findings into applied settings. By leveraging real-time data on affective states, timing of use, and contextual factors, JITAs could be designed to interrupt short-term reinforcement processes as they unfold in daily life, for example by delivering regulatory strategies or alternative coping prompts at moments of elevated risk. Such approaches may be particularly well suited to address the rapid, short-lived affective feedback loops identified in this dissertation.

An important next step for future work is to evaluate whether improvements in emotion regulation capacity and sleep quality lead to subsequent reductions in cannabis use severity and related harms. Testing these pathways longitudinally and within-person would help establish whether these factors function not only as correlates, but also as modifiable intervention targets capable of altering the course of high-frequency and hazardous cannabis use.

6.4 General conclusions

Across the three manuscripts, this dissertation advances the understanding of cannabis use among individuals with daily or near-daily use by integrating stable emotional vulnerabilities, momentary affective mechanisms, and broader health-related correlates. Rather than treating cannabis use as a static behavior or focusing on isolated outcomes, the present work conceptualizes cannabis use as a process unfolding across time scales and levels of analysis.

Manuscript 1 examined emotion regulation difficulties as a vulnerability factor, demonstrating that individuals with greater difficulties, particularly in the context of coping-motivated use, exhibit more severe patterns of cannabis involvement. These findings situate emotion regulation difficulties as a stable, person-level characteristic

that shapes how cannabis is used and why it becomes problematic. Manuscript 2 focused on momentary affective processes in daily life, showing that cannabis use is embedded within real-time affective dynamics characterized primarily by positive reinforcement processes, with negative reinforcement emerging in a vulnerability-dependent manner. This manuscript clarifies how positive affective states both precede and follow cannabis use and highlights the importance of within-person mechanisms that are not captured by between-person designs alone. Manuscript 3 examined health-related correlates of cannabis use severity, demonstrating consistent associations with poorer sleep quality and elevated symptoms of anxiety and depression. These findings position sleep and mental health as part of the broader cost associated with sustained and severe cannabis use.

Taken together, these findings refine existing affective and reinforcement-based models of cannabis use by demonstrating that reinforcement processes are not uniform across individuals. Instead, affective reinforcement appears to be conditional upon underlying emotional vulnerabilities, such as difficulties in emotion regulation. This conditional framework helps reconcile inconsistencies in the existing literature, particularly regarding the role of negative affect and negative reinforcement in daily-life studies of cannabis use.

Methodologically, the dissertation illustrates the value of combining cross-sectional and intensive longitudinal approaches to study cannabis use. By integrating baseline assessments with experience sampling data, the present work captures both stable individual differences and momentary processes as they unfold in everyday life, which allows for a more precise examination of these phenomena.

From a clinical and preventive perspective, the findings suggest that interventions targeting emotion regulation capacities and affective coping may be particularly relevant for individuals with high-frequency cannabis use and indications of CUD. Moreover, the identification of short-term affective reinforcement processes highlights potential opportunities for momentary or adaptive interventions aimed at disrupting habitual use patterns.

Future research should extend this framework by examining bidirectional associations between cannabis use, affect, sleep, and mental health at the within-person level, and by testing whether changes in emotion regulation and sleep quality are associated with subsequent reductions in cannabis use severity.

In sum, this dissertation contributes to a more nuanced understanding of cannabis use by demonstrating how vulnerabilities, mechanisms, and health are interconnected within a coherent multi-level framework. By situating cannabis use within dynamic affective processes and broader health contexts, the present work provides a foundation for both theoretical refinement and future research aimed at reducing the burden associated with problematic cannabis use.

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Appendix A (Manuscript 1)

Cannabis use severity and emotion regulation in people who use cannabis near-daily: the role of gender and motives

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Cannabis use severity and emotion regulation in people who use cannabis near daily: the role of gender and motives

Abstract

This naturalistic study in people who use cannabis near daily examined gender differences in emotion regulation difficulties, associations between cannabis use severity and emotion regulation difficulties, and the role of gender, and coping motives. Adults in Denmark who used cannabis near daily ($N = 192$; $M_{age} = 30.89$; 60.2% men, 33.9% women) completed questionnaires evaluating cannabis use severity (CUDIT-R), difficulties in emotion regulation (DERS-16), motives for cannabis use (MMM), substance use, and sociodemographics including gender identity. Robust regression analyses, moderation, and mediation analyses were conducted. Participants used cannabis on average 26.28 days per month; mean CUDIT-R score was 14.11 ($SD = 5.67$), indicating probable CUD. Women reported greater difficulties in emotion regulation than men. Severity of cannabis use was positively associated with difficulties in emotion regulation, with no significant moderation by gender. Coping motives partly mediated the relationship. Emotion regulation difficulties are similarly associated with cannabis use-related problems in men and women with high frequency of use, despite emotion regulation difficulties being higher in women. Coping-related motives contribute to this relationship, supporting cannabis use as an emotion regulation strategy. Future research should test these pathways in longitudinal studies and across diverse samples including gender minorities.

Keywords: Cannabis; Emotion regulation; Coping motives; Cannabis use severity; Gender; Near daily use;

Introduction

Men are nearly twice as likely to report using cannabis as women, although recent data indicate that the gender¹ difference in past-month cannabis use is narrowing in several countries (Segura et al., 2025; United Nations Office on Drugs and Crime, 2024). Cannabis use disorder (CUD) remains approximately twice as prevalent among men, who tend to initiate use earlier and report higher frequency and severity of use (Shao et al., 2023; United Nations Office on Drugs and Crime, 2025).

Biological sex differences have been observed in relation to route of administration, acute intoxication effects, and withdrawal symptoms, suggesting distinct vulnerability pathways between males and females (Gex et al., 2023; Matheson et al., 2020; Rochat et al., 2024; Sholler, Strickland, Spindle, Weerts, & Vandrey, 2021). Furthermore, women progress faster from heavy use to CUD and experience greater psychiatric comorbidity, particularly internalizing symptoms (Hesse, Vallentin-Holbech, & Thomsen, 2020; Towers, Williams, Qillawala, Rissman, & Lynch, 2023). Despite their heightened vulnerability to cannabis-related harms, women also face greater treatment barriers (European Union Drugs Agency, 2024; Hoch, Murawski, Ferri, & Feingold, 2025). Taken together, these divergent patterns raise the question of why women appear more vulnerable to cannabis-related harms despite lower levels of use.

Emotion regulation difficulties are a potential mechanism which may help explain gender differences in cannabis use sequelae. As a transdiagnostic risk factor, emotion regulation difficulties are consistently linked to a wide range of mental health problems and maladaptive coping behaviors, including hazardous substance use, substance use disorders, and affective disorders (Weiss et al., 2022b). Individuals with poorer emotion regulatory abilities may rely on substances as maladaptive strategies to manage distress (Weiss et al., 2022b). In the context of cannabis use, poorer emotion regulation has been linked to coping-motivated cannabis use, a pattern associated with greater risk for developing hazardous and problematic cannabis use (Bonn-Miller, Vujanovic, & Zvolensky, 2008). Further differentiating regulatory processes, prior work suggests that behavioral dysregulation predicts frequency and intensity of cannabis use, whereas emotion regulation difficulties predict hazardous cannabis use (Dvorak & Day, 2014). In sum, gender differences in emotion regulation and motives for use may shape not only the risk for developing problematic cannabis use, but also the clinical presentation and severity.

Previous studies examining emotion regulation difficulties among people who use cannabis have largely focused on samples with low-to-moderate frequency of cannabis use. Within these samples, findings regarding sex differences point in opposite directions. For example, in a sample with past month cannabis use (11 days/month), males reported greater levels of emotion regulation difficulties than females, and associations between emotion regulation difficulties and problematic cannabis use were stronger among males than females (Cavalli & Cservenka, 2024). In contrast, in a large university sample with similar patterns of use (9 days/month), females reported higher levels of emotion regulation difficulties than males, and these difficulties were more strongly associated with emotional distress among females and mediated the association between cannabis use and internalizing symptoms only in females (Weidberg, Gonzalez-Roz, Castano, & Secades-Villa, 2023). Consistent with these findings, females have been shown to be more likely than males to use cannabis to cope with negative emotions (Rochat et al., 2024).

These divergent patterns suggest that gender differences in cannabis-related vulnerability cannot be interpreted solely through biological sex. Biological sex and gender identity are conceptually distinct constructs but are frequently used interchangeably in cannabis use research, limiting the ability to empirically disentangle biological and social mechanisms (Matheson & Le Foll, 2023). Gender norms, roles, and relations shape emotion regulation processes, coping strategies, and the social meaning of cannabis use; however, gender identity remains an underexamined dimension in this literature (Hemsing & Greaves, 2020; Weiss et al., 2022a). Evidence from adolescent samples suggests that internalizing symptoms may partially mediate associations between gender identity and cannabis use (Buttazzoni et al., 2021), but it remains unclear whether such processes extend to adults with high frequency patterns of cannabis use. This limitation further complicates efforts to determine whether observed sex differences reflect gender-specific mechanisms or a shared vulnerability process. Accordingly, failure to account for these gendered processes risks obscuring why women experience greater cannabis-related harms despite lower levels of use.

From a developmental perspective, emotion regulation processes are shaped not only by biological factors, but also by gendered socialization processes that influence how emotions are expressed, tolerated, and managed across development (S. Nolen-Hoeksema, 2012). In this context, examining gender identity may provide additional insight into emotion regulation pathways linked to cannabis use severity. To date, no study has directly examined whether difficulties in emotion regulation are associated with severity of cannabis use among individuals with high frequency of cannabis use, nor whether such a relationship is

moderated by gender identity. Prior research suggests that difficulties in emotion regulation are associated with coping motives and indicates that women are more likely to endorse such motives; however, it remains unclear whether motives for use mediate the relationship between emotion regulation difficulties and severity of cannabis use. Addressing these gaps is critical for pinpointing psychological vulnerability pathways in individuals with heightened risk of developing cannabis-related harms (i.e. individuals with high levels of use). Additionally, elucidating potential gender-specific mechanisms holds high clinical significance for prevention and treatment, particularly given that women have been consistently shown to be more vulnerable to cannabis-related harms despite lower levels of use.

To address these gaps, the current naturalistic study included individuals who use cannabis near daily and examined: (1) whether men and women differ in emotion regulation difficulties, (2) whether emotion regulation difficulties are associated with cannabis use severity; (3) whether these associations are moderated by gender identity; and (4) whether coping motives mediate the relationship between emotion regulation difficulties and cannabis use severity. We hypothesized that higher emotion regulation difficulties would be associated with greater cannabis use severity. Given mixed findings and conceptual and methodological heterogeneity in the literature regarding gender and biological sex, we did not formulate specific directional hypotheses regarding gender. Finally, we hypothesized that coping motives would mediate the link between emotion regulation difficulties and cannabis use severity.

Methods

Participants

Participants were recruited through online and offline methods (e.g., flyers and posters). To take part in the research, the following criteria were applied: aged 18-65; understanding Danish; smoking cannabis at least three days per week during the past month; and not having used any other illicit substances in the past month. Potential participants were screened using an online questionnaire. Of the 197 eligible individuals who completed the survey, the final analysis sample comprised $n = 192$. Non-binary participants ($n = 5$) were excluded from the models testing effects of gender due to the small group size; however, they are included descriptively in Table 1 and Figure 1.

Procedures

This study was part of a larger preregistered project (<https://osf.io/c74ur/>). Ethical approval was obtained from the Research Ethics Committee at Aarhus University (BSS-2023-051; approved on August 11, 2023), and the study was registered with the Danish Data Protection Agency. All procedures complied with the Declaration of Helsinki, and participant anonymity and confidentiality were ensured. All participants were fully informed, participation was voluntary, and withdrawal was possible at any time without consequences. All participants provided informed consent. All participants resided in Denmark during the data collection. Data were collected and managed with Research Electronic Data Capture (REDCap), hosted at Aarhus University (Harris et al., 2009). After confirming eligibility, participants were invited to complete the online survey in Danish (approximately 20 min).

Measures

The online survey included questions on sociodemographic information, including gender identity (“What gender do you identify yourself as?”; 0 = women, 1 = men; 2 = non-binary, 3 = other [open ended], 4 = I don’t know, 5 = I do not want to answer), age, education, and occupation. Mental health was assessed with a validated 4-item measurement of symptoms of depression and anxiety, the Patient Health Questionnaire for Depression and Anxiety (PHQ-4; Kroenke, Spitzer, Williams, & Lowe, 2009).

To assess substance use history, participants completed questions on past month use, age of first use, the Alcohol Use Disorders Identification Test (AUDIT; Saunders, Aasland, Babor, de la Fuente, & Grant, 1993), and the Fagerström test for Nicotine Dependence (FTDN; Heatherton, Kozlowski, Frecker, & Fagerstrom, 1991) for those using alcohol and nicotine products, respectively. For descriptive purposes, participants also reported the cannabis product types and the number of days of use for each product type during the past month, from which we derived their main product preference: (1) hash, (2) flower, (3) multiple THC products, (4) THC-dominant concentrates, or (5) combined THC + CBD.

Cannabis use severity was assessed using the Cannabis Use Disorders Identification Test-Revised (CUDIT-R; Adamson et al., 2010), a validated 8-item self-report measure assessing the frequency of use, loss of control, and cannabis-related harms over the past six months. Items are rated on a 5-point Likert scale, with a total score ranging from 0 to 32, where higher scores indicate higher severity of cannabis use. Consistent with prior research, scores ≥ 8 indicate hazardous cannabis use and scores ≥ 12 indicate probable CUD.

(Adamson et al., 2010). The CUDIT-R total score was used as a continuous measure of cannabis use severity, as reflecting both the extend of cannabis involvement and the severity of cannabis-related problems, rather than frequency alone

To assess difficulties in emotion regulation, participants completed the Difficulties in Emotion Regulation Scale-16 (DERS-16; Bjureberg et al., 2016), a shortened version of the DERS-36 (Gratz & Roemer, 2004). DERS-16 is a validated measure that captures emotion regulation difficulties across five dimensions. Participants rate 16 items on a 5-point Likert scale (1 = rarely to 5 = almost always). The scores range from 16 to 80, with higher scores reflecting greater emotion regulation difficulties. The score can be interpreted as an overall sum score or across five dimensions: (1) lack of emotional clarity (two items); (2) inability to engage in goal-directed behaviors when distressed (three items); (3) difficulties controlling impulsive behaviors when distressed (three items); (4) nonacceptance of negative emotions (three items); and (5) limited access to emotion regulation strategies perceived as effective (five items). The total score and each dimension were calculated separately (Bjureberg et al., 2016).

Cannabis use motives were assessed using the Marijuana Motives Measure (MMM; Benschop et al., 2015), a validated version with 21 items that captures five motive dimensions. Participants evaluate statements about their reasons for using cannabis using a 5-point Likert scale (1 = almost never/never to 5 = almost always/always). The five dimensions include: (1) enhancement (e.g., because I like the feeling it gives me; four items); (2) coping (e.g., to forget my worries; four items); (3) conformity (e.g., for others to like me; two items); (4) expansion (e.g., to get to know myself better; five items); and (5) social (e.g., because it makes social gatherings more fun; six items). Dimension scores were calculated separately. In the present study, coping motives were included as a mediator based on previous evidence showing its role in linking emotion regulation difficulties to cannabis related outcomes (Weidberg et al., 2023).

Statistical analyses

All statistical analyses were conducted using RStudio (Version 2025.05.0+496; Posit Software, 2025). CUDIT-R total score and DERS-16 measures were non-normally distributed (Shapiro-Wilk test $p < .001$, moderate positive skewness). Therefore, bivariate associations were computed using Spearman's rank correlations. Gender differences were

explored descriptively using Wilcoxon rank-sum tests for continuous variables and chi-square or Fisher's exact test for categorical variables (cell counts were ≤ 5).

To determine the appropriate modeling framework, Gaussian and Gamma generalized linear models were compared. Gaussian models with robust standard errors performed better based on fit and interpretability ($R^2 = 0.318$; $R^2 \text{ adj} = 0.314$; AIC = 1142.72; BIC = 1152.49). Residual diagnostics did not indicate violations. Ordinary least squares (OLS) regression with HC3 robust standard errors was used to test whether overall emotion regulation difficulties (DERS-16 total) and the individual subscales were associated with cannabis use severity (CUDIT-R total). Regression analysis including all DERS-16 subscales simultaneously, was performed to examine shared versus unique variance explained in cannabis use severity. Next, a moderation analysis was performed to examine whether associations between DERS-total/subscales and cannabis use severity differed between men and women.

Finally, we tested a mediation model in which cannabis coping motives were specified as a mediator of the association between emotion regulation difficulties (DERS-16 total score) and cannabis use severity (CUDIT-R total score). Mediation was estimated using structural equation modeling in the *lavaan* package with 5,000 bootstrap resamples to derive bias-corrected 95% confidence intervals for indirect effects. Model fit indices and R^2 values were obtained for both mediator and outcome equations. Both raw and mean-centered DERS-16 scores were considered, with centering applied to improve interpretability in all models.

Given the lack of gender differences in the descriptive variables (Table 1), no covariates were added to any of the analyses. The sample was determined by feasibility, recruitment, and the available data collection period in a hard-to-reach population of adults using cannabis near daily. Because the sample size was not determined by an a priori power analysis, we conducted a post hoc power analysis after data collection using the *pwr* package. For the individual regression models testing one predictor at a time, the sample of $N = 192$ provided 80% power to detect effects of approximately $f^2 = 0.041$. For the moderation models, the power to detect interaction effects was $f^2 = 0.042$. Based on conventional benchmarks, these correspond to small effect sizes. These estimates were model-specific and did not account for the full family of tests reported in Table 2.

Results

Sample characteristics

Most of the participants were men (64.5%, $N = 127$), followed by women (33.9%, $N = 65$), and a small group identified as non-binary ($N = 5$, see Table 1 and Supplementary Table S1).

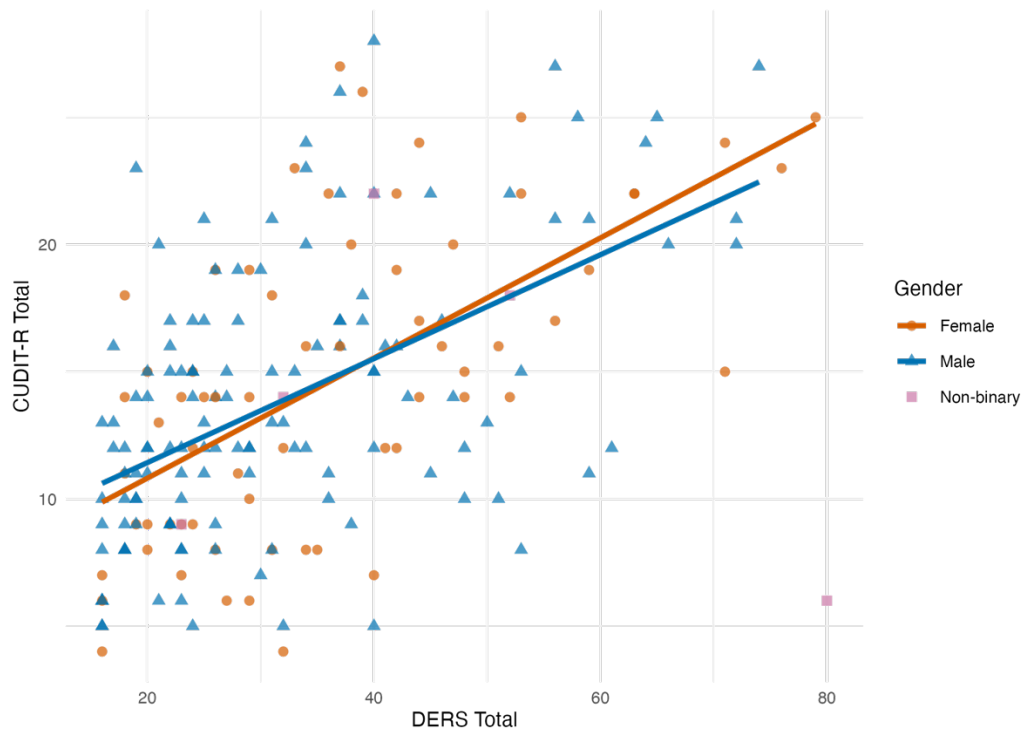
Table 1. Demographic characteristics, substance use information, and primary variables by Gender.

	Total ($N = 192$)	Women ($N = 65$)	Men ($N = 127$)	Non-binary ($N = 5$)		
Demographic Characteristics	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	<i>W</i>	<i>p</i>
<i>Age (years)</i>	30.89 (10.65)	31.91 (11.39)	30.36 (10.26)	23.60 (7.73)	4438	0.394
Education² (n)						0.336
Lower/Presecondary	55	20	35	2		
Upper secondary	54	21	33	2		
Vocational	32	6	26	0		
Higher education	39	13	26	0		
Occupation³ (n)						0.704
In education	34	13	21	1		
Employed	71	21	50	0		
Unemployed	24	6	18	3		
Retired	18	8	10	0		
Ill/Disabled	15	6	9	0		
Substance Use Characteristics						
<i>Cannabis use</i>						
Age of onset (years)	15.61(2.4)	15.84 (2.42)	15.49 (2.39)	15.40 (2.70)	3981.5	0.233
CUDIT-R	14.11 (5.67)	14.68 (6.05)	13.83 (5.46)	13.80 (6.50)	4471.5	0.345
Cannabis use (days/month)	26.28 (6.01)	26.08 (5.93)	26.39 (6.07)	21 (11.96)	4062.4	0.843
Primary Cannabis product (n)						
Flower only	26	5	21	0		
Hash only	62	28	34	1		
Multiple THC products	74	50	74	4		
THC-dominant concentrates	3	1	2	0		
Combined THC + CBD	27	7	20	0		
<i>Alcohol use</i>						
Age of onset (years)	13.65 (1.84)	13.65 (1.93)	13.66 (1.80)	12.67 (0.58)	3920.5	0.953
Past month (yes, n)	131	45	86	3		
Past month (days/month)	6.7 (7.32)	5.38 (5.8)	7.4(7.94)	4.67(2.08)	1653	0.168
AUDIT	9.56 (6.09)	9.24 (5.66)	9.73 (6.32)	9.33 (6.43)	1863.5	0.730
<i>Nicotine use</i>						
Age of onset (years)	14.45 (2.72)	14.65 (3.75)	14.35 (2.05)	11.67 (1.53)	2686.5	0.974
Past month (yes, n)	178	58	116	4		
FTND	4.20 (1.97)	4.30 (1.86)	4.15 (3.03)	4.50 (1.29)	1674.5	0.591
Scores on Primary Variables						
<i>PHQ-4</i>	3.60 (3.11)	4.09 (3.28)	3.35 (3.01)	6.20 (3.56)	4708	0.108
<i>DEERS-16</i>						

Total	33.31 (14.81)	36.35 (15.64)	31.75 (14.18)	45.40 (22.09)	4919	0.029
Clarity	3.74 (2.04)	4.02 (2.22)	3.61 (1.94)	6.20 (3.03)	4613	0.167
Goals	7.74 (3.51)	8.26 (3.57)	7.47 (3.45)	8.80 (4.09)	4667.5	0.136
Impulse	5.40 (3.10)	5.85 (3.31)	5.17 (2.98)	7.40 (5.37)	4687.5	0.112
Non-acceptance	6.56 (3.48)	7.02 (3.35)	6.32 (3.53)	9.20 (3.70)	4828	0.052
Strategies	9.87 (5.08)	11.22 (5.35)	9.18 (4.81)	13.80 (7.46)	5144	0.004
MMM						
Enhancement	15.17 (3.67)	15.14 (3.53)	15.18 (3.76)	11.00 (4.12)	4019.5	0.766
Coping	8.82 (4.20)	9.83 (4.92)	8.30 (3.70)	9.40 (4.28)	4769	0.077
Conformity	2.10 (0.51)	2.12 (0.63)	2.09 (0.44)	2.20 (0.45)	4126.5	0.997
Expansion	11.83 (5.87)	11.55 (4.09)	11.97 (6.07)	11.40 (8.17)	4089	0.916
Social	10.70 (4.4)	9.48 (4.09)	11.33 (3.01)	11.60 (6.11)	2928	0.001

The mean age was 30.89 years ($SD = 10.65$; range = 18—63). On average, participants reported using cannabis on 26.28 days in the past 30 days ($SD = 6.00$), and the mean CUDIT-R score was 14.11 ($SD = 5.67$), indicating probable CUD. See Figure 1 for gender distribution on CUDIT-R.

Figure 1. Association between cannabis use severity and emotion regulation difficulties across gender.



There were significant gender differences in overall emotion regulation difficulties, $W = 4919.0$, $p = .029$, $r = 0.15$, difficulties accessing effective emotion regulation strategies, $W = 5144.0$, $p = .004$, $r = 0.20$, and social motives for cannabis use $W = 2928$, $p < .001$, $r =$

0.23), with higher scores for women than men. On all other variables, we found no significant gender differences.

Associations between emotion regulation difficulties and cannabis use severity

Higher emotion regulation difficulties (DERS-16 total) were associated with greater cannabis use severity (CUDIT-R), $b = 0.216$, $SE = 0.023$, 95 % $CI [0.17, 0.26]$, $p < .001$. The model explained 31% of the variance in cannabis use severity ($R^2 = .314$). When examined separately, all DERS-16 subscales showed a positive association with cannabis use severity. Full model information is presented in Table 2.

Table 2. Associations between emotion regulation difficulties (DERS-16) and cannabis use severity (CUDIT-R).

Predictor (VI)	b	SE	95% CI	p	R ²	Adj. R ²	AIC	BIC	f ²
DERS-16	0.216	0.023	[0.17, 0.26]	< .001	0.318	0.314	1142.72	1152.49	0.46
Clarity	1.147	0.183	[0.79, 1.51]	< .001	0.171	0.167	1180.1	1189.8	0.21
Goals	0.877	0.099	[0.68, 1.07]	< .001	0.294	0.291	1149.2	1159.0	0.41
Impulse	0.831	0.118	[0.60, 1.06]	< .001	0.206	0.202	1171.7	1181.5	0.26
Non-acceptance	0.700	0.107	[0.49, 0.91]	< .001	0.184	0.180	1177.0	1186.8	0.23
Strategies	0.594	0.069	[0.46, 0.73]	< .001	0.283	0.279	1152.3	1162.1	0.39
Interaction term									
Gender × DERS-16	-0.032	0.048	[-0.13, 0.06]	0.507	0.319	0.308	1146.23	1162.52	0.47
Gender × Clarity	-0.273	0.377	[-1.02, 0.47]	0.470	0.174	0.161	1183.29	1199.58	0.21
Gender × Goals	0.030	0.209	[-0.38, 0.44]	0.886	0.294	0.283	1153.11	1169.40	0.42
Gender × Impulse	-0.402	0.244	[-0.88, 0.08]	0.100	0.218	0.206	1172.78	1189.07	0.28
Gender × Non-acceptance	-0.291	0.231	[-0.75, 0.17]	0.210	0.192	0.179	1179.19	1195.47	0.24
Gender × Strategies	-0.081	0.144	[-0.37, 0.20]	0.574	0.285	0.273	1155.74	1172.03	0.40

Note. Each row represents a separate OLS regression model estimated with HC3 robust standard errors. Interaction models included the main effects of gender and the corresponding emotion regulation variable, as well as their interaction term. All predictors were mean centered prior to analysis. R², R² adjusted, AIC, and BIC are reported for descriptive comparisons across models.

Abbreviations: CUDIT-R = Cannabis Use Disorders Identification Test-Revised; DERS-16 = Difficulties in Emotion Regulation Scale-16.

In the regression analysis including all DERS-16 subscales simultaneously, difficulties in maintaining goal-directed behavior under distress emerged as a unique subscale associated with cannabis use severity, whereas associations for the remaining subscales were attenuated (see Table 3). Multicollinearity was acceptable, indicating that the pattern likely refers to shared variance across all dimensions.

Table 3 Regression analysis of the joint regression model predicting cannabis use severity (CUDIT-R) across all subscales of emotion regulation difficulties

Predictor	b	SE	t	p	95% CI
(Intercept)	6.59	0.87	7.56	< .001	[4.87, 8.31]
Clarity	0.17	0.23	0.74	.462	[-0.29, 0.63]
Goals	0.51	0.15	3.50	.001	[0.22, 0.80]
Impulse	0.14	0.17	0.82	.414	[-0.19, 0.47]
Non-acceptance	-0.03	0.16	-0.21	.832	[-0.34, 0.27]
Strategies	0.24	0.14	1.72	.086	[-0.03, 0.52]

Note. *b* = unstandardized regression coefficient; SE = standard error. All predictors were entered simultaneously. Collinearity among DERS subscales was expected.

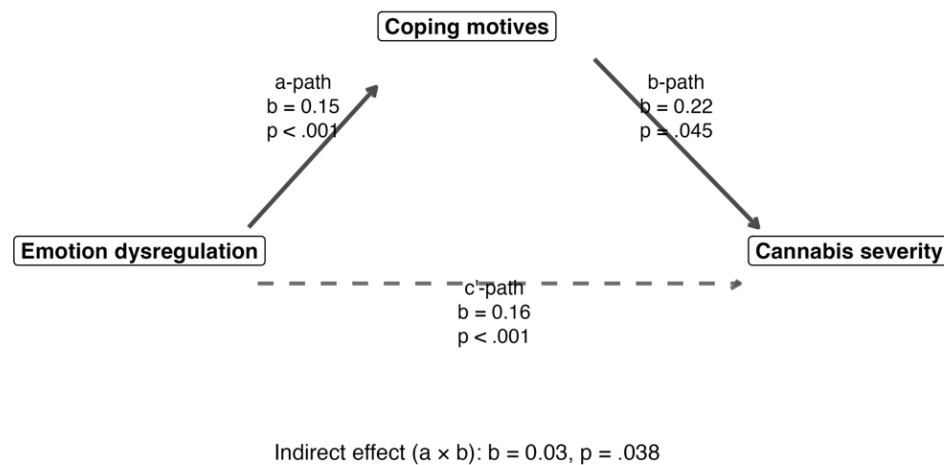
Moderation by gender

Gender did not significantly moderate the relation between the DERS-16 total/sub scores and severity of cannabis use, indicating that the strength of the relationship was similar across men and women. Details are presented in Table 2.

Mediation by coping motives

Emotion regulation difficulties were positively associated with coping motives (a-path: $B = 0.15$, $SE = 0.02$, $p < .001$, $\beta = 0.55$), and coping motives were positively associated with cannabis use severity (b-path: $B = 0.22$, $SE = 0.11$, $p = .047$, $\beta = 0.17$). The direct effect of emotion regulation difficulties on cannabis use severity remained significant when accounting for coping motives (c'-path: $B = 0.16$, $SE = 0.03$, $p < .001$, $\beta = 0.44$). The indirect effect was statistically significant ($a \times b = 0.03$, $SE = 0.02$, $p = .039$, $\beta = 0.09$), indicating that coping motives accounted for a small but significant portion of the association between emotion regulation difficulties and cannabis use severity. The total effect of emotion regulation difficulties on cannabis use severity was $B = 0.20$, $SE = 0.03$, $p < .001$, $\beta = 0.53$. Details are presented in Figure 2 and Supplementary Table S2.

Figure 2. Mediation model testing whether coping motives account for the association between emotion regulation difficulties (DERS-16 total) and cannabis use severity (CUDIT-R). Values represent unstandardized coefficients (B) with 95% confidence intervals. The indirect effect was estimated using bootstrap resampling (5,000 samples).



Discussion

This naturalistic study extends previous research by examining the relationship between emotion regulation difficulties and cannabis use severity and the role of gender in a sample with high frequency cannabis use. Additionally, it provides a novel examination of the mediating role of coping motives.

In a sample with near daily cannabis use, we found that women had higher overall emotion regulation difficulties, and higher levels of difficulties accessing effective emotion regulation strategies, compared to men. Previous studies have examined sex-based differences in emotion regulation difficulties in samples with low-to-moderate frequency of use and yielded mixed findings: Weidberg et al. (2023) found higher levels of emotion regulation difficulties in females compared to males, while Cavalli & Cservenka (2024) found the opposite pattern (Cavalli & Cservenka, 2024). To our knowledge, this is the first examination of gender differences in emotion regulation difficulties in a community sample with near daily cannabis use. The higher level of emotion regulation difficulties found in women could be an indicator of increased risk of progression to CUD however, women did not have higher levels of cannabis use severity (CUDIT-R). Alternatively, it may reflect gender-based differences in clinical presentation.

Consistent with our hypothesis, we found that higher emotion regulation difficulties were positively associated with cannabis use severity, as indexed by CUDIT-R. In a sample of undergraduate students, Buckner et al. (2024) found a similar trend: emotion regulation difficulties predicted cannabis-related problems, even after accounting for cannabis use

frequency and protective behavioral strategies, with sex showing no independent association once emotion regulation difficulties were included. Despite women reporting higher levels of emotion regulation difficulties, gender identity did not moderate the association between emotion regulation difficulties and cannabis use severity in our study. A possible interpretation is that among individuals with near daily and at higher risk for CUD, variability in gender-related psychological processes may be attenuated, such that emotion regulation difficulties confer similar risk for cannabis use severity across gender. At this stage of cannabis involvement, cannabis use may be increasingly maintained by goal-directed and affectively reinforced use patterns, which may operate similarly across gender, thereby reducing gender-specific pathways. From a gender norms perspective, this pattern suggests that while gendered socialization may shape earlier vulnerability and clinical presentation, emotion regulation difficulties may become a shared maintenance mechanism once cannabis use reaches high-frequency and hazardous levels (Greaves & Hemsing, 2020; Hemsing & Greaves, 2020). This pattern is consistent with broader evidence indicating that although women tend to report greater emotional awareness and engagement with emotions, maladaptive emotion regulation processes show comparable associations across genders, supporting the notion of a shared vulnerability process rather than gender-specific mechanisms (S. Nolen-Hoeksema, 2012; Susan Nolen-Hoeksema & Aldao, 2011). This interpretation should be considered in light of methodological heterogeneity across previous studies (e.g., in operationalization of cannabis involvement and emotion regulation) and a field under development. Nevertheless, it is consistent with previous work showing that biological sex differences are more pronounced in people with recreational or occasional use (Cavalli & Cservenka, 2024; Weidberg et al., 2023).

Beyond the primary associations, coping motives accounted for a small but statistically significant portion of the association between emotion regulation difficulties and cannabis use severity. Individuals with greater difficulties regulating emotions reported stronger coping motives, which were in turn associated with higher cannabis use severity. These results are consistent with negative reinforcement theories, which propose that substance use is maintained because it alleviates aversive emotional states, thereby making cannabis particularly reinforcing for individuals experiencing emotion regulation difficulties (Volkow et al., 2016; Zehra et al., 2019). Prospective evidence further supports the relevance of this pathway. In a three-year cohort study of young adults who used cannabis at high frequency, coping motives for cannabis use independently predicted the onset of cannabis dependence, even after accounting for cannabis exposure and stable vulnerability factors (van

der Pol et al., 2013). Together, these findings suggest that coping motives may represent one pathway through which emotion regulation difficulties reflect more severe and clinically relevant patterns of cannabis use (van der Pol et al., 2013). However, given the cross-sectional design of the present study, the mediation model should not be interpreted as establishing temporal order or a causal sequence. Rather, it represents a theory-informed test of indirect association consistent with the hypothesis that coping motives may link emotion regulation difficulties to cannabis use severity. Alternative directional pathways are also plausible and should be evaluated in longitudinal studies. A longitudinal design would be better suited to determine whether emotion regulation difficulties prospectively predict increases in cannabis use severity, whether coping motives function as an intervening mechanism over time, and whether reciprocal associations are present. Importantly, the indirect effect observed in the present study was statistically significant but modest, while the direct association remained substantial, suggesting that coping motives explain part, but not all, of the association between emotion regulation difficulties and cannabis use severity. This pattern underscores the role of emotion regulation difficulties as a broad vulnerability factor for problematic cannabis use, possibly operating through multiple reinforcing processes instead of a single pathway.

All DERS-16 subscales were associated with cannabis use severity; however, these associations largely reflected shared variance across emotion regulation dimensions, as difficulties maintaining goal-directed behavior under distress emerged as the only dimension that explained unique variance in cannabis use severity. Prior studies reporting subscale-specific associations between emotion regulation difficulties and cannabis-related outcomes have primarily used the DERS-36, which allows for greater differentiation across emotion regulation subscales (Bonn-Miller, Vujanovic, Boden, & Gross, 2011; Bonn-Miller, Zvolensky, Bernstein, & Stickle, 2008; Cavalli & Cservenka, 2020, 2024). In contrast, the DERS-16 was developed as a brief measure capturing emotion regulation difficulties and is potentially less suited to isolate independent emotion regulation processes across specific domains (Bjureberg et al., 2016). Accordingly, subscale-specific interpretations should be made with caution when using this abbreviated measure.

The present study assessed gender identity, whereas prior studies focused on biological sex (Cavalli & Cservenka, 2024; Weidberg et al., 2023). In this context, the absence of moderating effects of gender, should not be interpreted as evidence that emotion regulation pathways are invariant across sex assigned at birth or gender-diverse groups in samples with high levels of use. Instead, it may indicate that gendered processes operate

upstream of heavy use, shaping risk accumulation rather than moderating severity once use becomes habitual (Greaves & Hemsing, 2020). This is particularly relevant given consistent evidence that gender minority populations report higher levels of emotion regulation difficulties, psychological distress, and cannabis use problems compared to cisgender individuals (Buckner, Zvolensky, Garey, & Businelle, 2025; Dyar, Green, Rhew, & Lee, 2025; Dyar, Hales, Rhew, & Morgan, 2025; Dyar, Lee, Rhew, & Kaysen, 2023; Matheson & Le Foll, 2023). Accordingly, future research should examine emotion regulation processes and cannabis use patterns in gender-diverse samples, including non-binary and transgender individuals, and explicitly distinguish gender identity from sex assigned at birth when examining emotion regulation pathways. Although the current study could not directly test these questions, the inclusion of gender identity allows the findings to contribute to the cumulative evidence and future meta-analyses examining emotion regulation pathways across gender-diverse identities. Furthermore, future studies employing longitudinal methods (e.g., experience sampling) are needed to explore how emotion regulation difficulties and cannabis use interact over time, including among gender minorities.

The naturalistic sample of individuals who use cannabis near daily is a strength, as this group remains underrepresented in empirical research. Studies clarifying gender differences in psychological vulnerability processes, such as emotion regulation difficulties, are critical for understanding why cannabis-related harms disproportionately affect women despite lower levels of use but have been lacking. Nevertheless, our study also has limitations that need to be noted. First, participants were restricted to adults in Denmark, who used cannabis at least three days per week, and reported no other illicit substance use in the past month. Accordingly, the findings may not generalize to less frequent cannabis users, treatment-seeking populations, polysubstance-using samples or, adolescents. In addition, because the number of non-binary participants was too small for inclusion in the gender-based models, the findings primarily reflect comparisons between men and women. Second, the cross-sectional design limits our ability to infer causality or determine directionality. Third, we did not require participants to report sex assigned at birth. Fourth, while emotion regulation difficulties may lead to higher cannabis severity, the consumption itself may also cause impairments in emotion regulation over time (Volkow et al., 2016; Zimmermann et al., 2017). Fifth, emotion regulation difficulties were assessed through self-report, which may not fully capture behavioral or physiological regulation capacities.

Conclusion

In a naturalistic study of people who use cannabis near daily, we found higher levels of emotion regulation difficulties in women compared to men. However, difficulties regulating emotions were positively associated with cannabis use severity to a similar extent in men and women, indicating that this relationship does not vary by gender identity in individuals with high frequency of use. Furthermore, coping motives partially mediated the association between emotion regulation difficulties and cannabis use severity. Together, these findings suggest that among individuals with daily or near daily cannabis use, emotion regulation difficulties may function as a shared psychological vulnerability rather than a gender-specific mechanism. This underscores the importance of affective mechanisms in understanding and addressing heavy cannabis use in public health and clinical settings and of studying the role of gender identity at different stages of cannabis use severity.

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Declaration of interest statement

The authors have no competing interests to declare.

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Data availability statement

The data will be available upon a reasonable request.

Footnote

1. Throughout this manuscript we refer to sex or gender as defined in the original studies, reflecting classification based on biological sex assigned at birth or self-identified gender.

Supplementary Table S1. Distribution of demographic characteristics, substance use information, and primary variables by Gender.

	Total	Women	Men	Non-binary
	Median (IRQ); range	Median (IRQ); range	Median (IRQ); range	Median (IRQ); range
Demographic Characteristics				
Age	28 [23–35]; 18–63	28 [23–36]; 18–63	28 [23–35]; 18–63	21 [19–23]; 18–37
Cannabis use				
Age of onset (years)	15 [14–17]; 11–28	16 [14–17]; 11–28	15 [14–17]; 11–25	15 [14–15]; 13–20
CUDIT-R	14 [10–18]; 4–28	14 [9–19]; 4–27	13 [10–17]; 5–28	14 [9–18]; 6–22
Cannabis (days/past month)	30 [25–30]; 1–30	30 [25–30]; 10–30	30 [25–30]; 6–30	24 [20–30]; 1–30
Alcohol use				
Age of onset (years)	14 [13–15]; 7–18	14 [13–15]; 7–17	14 [13–15]; 7–18	13 [12.5–13]; 12–13
Alcohol (days/past month)	3.5 [2–9]; 1–30	3 [2–8]; 1–25	4 [2–10]; 1–30	4 [3.5–5.5]; 3–7
AUDIT	8 [5–12]; 2–37	8 [5–12]; 2–25	8 [5–12]; 2–37	12 [7–13]; 2–14
Nicotine use				
Age of onset (years)	14 [13–16]; 8–35	14 [13–16]; 8–35	14 [13–16]; 10–19	12 [11–12.5]; 10–13
FTND	4 [2–6]; 1–9	4 [3–5.25]; 2–8	4 [2–6]; 1–9	4.5 [3.75–5.25]; 3–6
Primary variables				
PHQ-4	3 [1–6]; 0–12	3 [2–6]; 0–12	2 [1–5]; 0–12	6 [6–7]; 1–11
MMM				
Enhancement	16 [13–18]; 4–20	16 [13–18]; 6–20	16 [13–18]; 4–20	12 [11–14]; 4–14
Coping	8 [6–12]; 4–20	9 [6–13]; 4–20	8 [5.5–10]; 4–20	8 [7–11]; 5–16
Conformity	2 [2–2]; 2–6	2 [2–2]; 2–6	2 [2–2]; 2–5	2 [2–2]; 2–3
Expansion	10 [7–15]; 5–25	10 [7–14]; 5–25	10 [7–15.5]; 5–25	7 [7–13]; 5–25
Social	10 [7–13]; 6–26	8 [6–11]; 6–23	10 [8–14]; 6–26	9 [7–16]; 6–20
DERS-16				
Total	29 [22–41]; 16–80	33 [24–44]; 16–79	27 [21–39.5]; 16–74	40 [32–52]; 23–80
Clarity	3 [2–4]; 2–10	3 [2–4]; 2–10	3 [2–4]; 2–10	6 [5–8]; 2–10
Goals	7 [5–10]; 3–15	7 [6–11]; 3–15	7 [5–10]; 3–15	8 [7–10]; 4–15
Impulse	4 [3–6]; 3–15	5 [3–7]; 3–15	4 [3–6]; 3–15	5 [3–11]; 3–15
Non-acceptance	6 [4–8]; 3–15	6 [4–9]; 3–15	5 [4–8]; 3–15	8 [8–10]; 5–15
Strategies	8 [6–13]; 5–25	10 [7–15]; 5–25	7 [6–12]; 5–25	12 [9–17]; 6–25

Note. Values are presented as median, interquartile range (IR), and range (min-max). Total values reflect the full descriptive sample, including women, men, and non-binary participants.

Supplementary Table S2. Mediation model: Emotion regulation difficulties, coping motives, and cannabis use severity.

Effect	Estimate	Std. Error	p	Std. Estimate
a-path (DERS-16 → Coping motives)	0.15	0.02	< .001	0.55
b-path (Coping motives (MMM) → CUDIT-R)	0.22	0.11	0.047	0.17
c'-path (Direct: DERS-16 → CUDIT-R)	0.16	0.03	< .001	0.44
Indirect effect (a × b)	0.03	0.02	0.039	0.09
Total effect	0.20	0.03	< .001	0.53

Note. Estimates are unstandardized regression coefficients unless otherwise indicated. Standardized estimates are provided for effect-size comparisons. Indirect effects were tested using mediation analysis.

Abbreviations: CUDIT-R = Cannabis Use Disorders Identification Test-Revised; MMM = Marijuana Motives Measure; DERS-16 = Difficulties in Emotion Regulation Scale-16.

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Appendix B (Manuscript 2)

**Cannabis Use, Affect, and Emotion Regulation in Daily Life: A 3-Week
Experience Sampling Method Study in People Who Smoke Cannabis Almost
Daily**

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The data is available from the authors upon reasonable request.

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ABSTRACT

Background and Aims: Although many people who use cannabis report mood-enhancing effects, affective responses are thought to vary within and between individuals. This study examined the bidirectional associations between momentary affect and cannabis use and the potential moderating role of emotion regulation in this.

Design: Intensive longitudinal observational study using Experience Sampling Method (ESM)

Setting: Denmark

Participants: Adults in Denmark who use cannabis near daily ($N = 113$; $M_{age} = 31.57$ years; 60.2% male).

Method and Measurements: Participants completed a baseline survey (substance use and emotion regulation difficulties) and 21 days of experience sampling (three assessments daily; affect and use). We used multilevel models to assess: 1) the associations between positive affect (PA) and negative affect (NA) and subsequent cannabis use, 2) whether cannabis use predicted changes in affect shortly after use (at the same assessment), and 3) whether cannabis use predicted changes in affect at a later moment (at the next assessment). Next, we assessed whether emotion regulation difficulties moderated associations tested in Models 1 and 2.

Findings: Higher PA predicted a greater likelihood of cannabis use before the next assessment, and cannabis use predicted increased PA at the same assessment, but not at the next assessment. NA was not related to use or vice versa at group level. Emotion regulation difficulties moderated the association between PA and subsequent cannabis use, such that individuals with greater difficulties had a weaker positive association between positive affect and subsequent cannabis use. Furthermore,

emotion regulation difficulties moderated short-term effects of cannabis use on NA, with individuals with higher emotion regulation difficulties showing lower NA shortly after use compared to non-use moments.

Conclusions: Within individuals who use cannabis near daily, relatively higher momentary PA, but not NA, preceded and followed cannabis use. Cannabis use was associated with higher PA at the same assessment but did not predict PA hours later. Individuals with greater emotion regulation difficulties may experience short term reductions in NA after use.

Keywords: cannabis; ecological momentary assessment, experience sampling method; emotion regulation; affect; DERS; emotion dysregulation;

INTRODUCTION

Several theories argue that affect regulation, either via positive reinforcement (enhancing positive affect (PA)) or via negative reinforcement (relieving negative affect (NA)) (1), constitutes a central mechanism in the maintenance of cannabis use. Many individuals report using cannabis to cope with NA (e.g., sadness, regret, anxiety) or enhance PA (e.g., happy, satisfied, excited) (1, 2). While cannabis use to enhance PA is common among recreational cannabis users (3), coping motives are strongly associated with problematic cannabis use, psychological distress, and perceived stress (3-5). This association is thought to be stronger among individuals who report generally high NA and have greater levels of emotion regulation difficulties (1). Given the substantial heterogeneity in individual cannabis use patterns and reasons for using cannabis, understanding the daily temporal dynamics between affect and cannabis use may help distinguish affective antecedents from affective consequences of use.

Because emotion regulation processes unfold dynamically over time, understanding their role in cannabis use requires approaches such as Experience Sampling Method (ESM) that can capture within-person temporal variation in affect and behavior in a bidirectional manner. In previous ESM studies, higher momentary PA – operationalized as happy mood - predicted a greater likelihood of cannabis use later in the day (6). Consistent with this, another study found that higher PA predicted stronger intentions to use cannabis at the next assessment among university students (7). In contrast, a study of young medical outpatients found no association between PA and time periods preceding cannabis use; instead, NA was elevated across 24h before use, with this association attenuated when examining only the moment immediately prior to use (8). In contrast, an early ESM study reported that level of state anxiety neither predicted subsequent cannabis use nor changed at the next

assessment following use in university students (9). Further, another study found that higher depressed mood predicted lower cannabis use, whereas anxious mood showed no association with use in young adults (6). More recently, a study showed that lower, rather than higher, NA was associated with stronger intentions to use cannabis at the next assessment among university students (7). Importantly, these ESM studies vary markedly in key design features, including sample characteristics (e.g., age, community vs. clinical, level of cannabis use), operationalization of affect, and outcome definitions (e.g., use vs. intention to use; proximal vs. distant periods), which may explain the inconsistent results.

Beyond affective antecedents, ESM studies have examined the affective consequences of cannabis use. In a sample of university students, within-person reports of more recent cannabis use were associated with higher PA and lower NA, consistent with short-term positive reinforcement (7). Similarly, another ESM study found that PA was higher at cannabis use moments compared to non-use moments. In contrast, NA varied by social context, being higher when using alone and lower when using with others (10). Finally, in a clinical sample of adolescents recently discharged from psychiatric inpatient care, cannabis use was associated with higher PA and lower anger/irritability, but was not associated with NA (sad, guilt, hopeless, and shame) (11).

Taken together, the limited evidence gathered in recent ESM studies suggests that PA tends to predict a higher likelihood or intention of cannabis use and increases after cannabis use in student samples. In contrast, associations between NA and cannabis use are mixed in student samples and appear more conditional, especially in those with psychiatric comorbidity (11). Emotion regulation might play a crucial role in these individual differences. Emotion regulation is a multifaceted process through

which individuals influence the experience and expression of their emotions, and difficulties in this process are considered a key risk factor for developing substance use disorders (12). Supporting this, Cavalli et al. (2020) found that stressful life events and perceived stress were more strongly associated with problematic cannabis use among individuals with greater difficulties in emotion regulation (13). In a later study, they further demonstrated that greater emotion regulation difficulties was associated with more severe problematic cannabis use, with these associations differing by sex (14).

The present study aimed to examine within-person daily associations between affect (PA/NA) and cannabis use and to test whether emotion regulation difficulties moderate these associations among individuals who use cannabis near daily (see preregistration: <https://osf.io/c74ur>). We assessed whether affect (PA/NA) influenced the subsequent likelihood of using cannabis, and whether cannabis use influenced PA/NA both shortly after use and at a later moment during the day. Next, we examined the potential moderating role of trait-level emotion regulation in these associations. Although several ESM studies have reported PA-use associations in recreational users in student samples, none of these earlier ESM studies examined adults who use cannabis near daily, making it difficult to generalize PA-related findings from these studies to individuals with heavier recreational use. Near daily use represents a pattern in which cannabis is embedded in everyday emotional contexts, making affect regulation processes particularly salient. By contrast, theoretical accounts of negative reinforcement and emotion regulation provide a clear rationale for predicting NA-driven use in this population (2). Thus, our hypotheses were guided primarily by negative reinforcement theory, which offers a stronger theoretical basis for NA-driven use among near daily users than prior ESM findings currently do. We specifically expected

higher levels of NA to increase the likelihood of cannabis use, and we expected cannabis use to be associated with lower NA shortly after use. We expected these associations to be stronger in individuals with greater emotional regulation difficulties. No directional hypotheses were preregistered for PA.

METHOD

Participants

Participants were recruited through both online and offline (flyers and posters) methods. Inclusion criteria were assessed using an online screener: aged 18-65, understanding Danish, smoking cannabis at least three times per week during the last month, having a smartphone with a Danish number for SMS notifications, and not having used any other illicit substances in the last month. After completing all baseline questionnaires (N = 191), participants were invited to take part in the ESM phase the following day, of which 165 initiated the ESM. For analysis (and as pre-registered), we only included participants who reported cannabis use on at least one day in each protocol week (N = 113), ensuring within-person variability in cannabis use across the ESM period.

Procedure

The study was registered at the Danish Data Protection Agency and approved by the Research Ethics Committee at Aarhus University (approval number: BSS-2023-051). All participants were fully informed, instructed that participation was voluntary and that they could withdraw from the study at any time without consequences, given the opportunity to contact the researchers with any questions about the study, and provided online informed consent. After confirming eligibility, participants were invited

to complete the baseline questionnaire online, which included a briefing on the ESM procedure via an informative video explaining the procedure.

The day after baseline completion, participants started the 21 days of ESM with three daily assessments sent within 20 minutes of predefined anchor hours (11:00, 16:00, 21:00). Surveys were to be completed within 60 minutes, but remained open thereafter. Data were collected between January 19 and October 10, 2024, using REDCap, an online platform hosted by Aarhus University (15). Participants who completed at least 80% of the ESM surveys within 60 minutes after receiving the prompt received 300kr. No compensation was provided otherwise.

Measures

Baseline questionnaire included sociodemographic data (age, gender, education, and occupation) and substance use. Specifically, we assessed past month use of cannabis using The Timeline Follow-Back; (TLFB; 16), and past month use of alcohol and nicotine was assessed with single items. The Alcohol Use Disorders Identification Test (AUDIT; 17) was used to assess alcohol consumption and related problems during the past 6 months. The Cannabis Use Disorders Identification Test-Revised (CUDIT-R; 18) was used to assess cannabis use and related problems during the past 6 months, with scores ≥ 8 indicating hazardous consumption (18). In individuals who used nicotine daily, the Fagerström test for Nicotine Dependence (FTND; 19) was used to assess nicotine dependence.

Furthermore, participants completed the Difficulties in Emotion Regulation Scale-16 (DERS-16; 20) to assess difficulties in emotion regulation with higher scores indicating greater emotion regulation difficulties.

Momentary affect assessments (ESM) included five PA items (happy, satisfied, excited, relaxed, proud) and five NA items (anxious, regret, angry, guilt, frustration)

from the short version of the Positive and Negative Affect Schedule (PANAS). The items were rated on a scale from 1 (very little/not at all) to 5 (extremely much) in response to the question “Right now, I feel...[affect]?”. The PA and NA items were chosen to capture a broad range of valence and arousal properties, following and expanding on previous studies (2, 7). Sum scores were created for the PA and NA sub-scales separately, with total scores ranging from 5 to 25 and higher scores indicating more intense affect.

Momentary cannabis use assessments included “Have you used cannabis today?” (1 = yes; 0 = no). If the participants replied “yes”, they were asked about what time they used cannabis the last time. Based on this information, we created a momentary binary indicator of cannabis use (Ct_0 ; 1 = yes, 0 = no).

Time until next use was defined as the time difference between the end of the current assessment and the next observed cannabis use event for that participant, based on corrected cannabis use timestamps.

Time since last use was defined as the time difference between the end of the current assessment and the most recent prior cannabis use event for that participant, also based on corrected cannabis use timestamps. To reduce skewness, timing variables were log-transformed, and the final model versions were person-mean centered to isolate within-person variation.

ANALYTIC METHOD

Preprocessing

Data was exported from REDCap, followed by a series of coherence and ordering checks on the timestamps (start, end, and last cannabis use hour) to identify and correct any temporal inconsistencies (21). When a participant completed an evening survey the following morning, or reported cannabis use in the morning survey,

referring to consumption that occurred the previous evening, the use timestamp was adjusted to the corresponding date (i.e., previous day) to reflect the timing of the event. No major temporal inconsistencies were found. To preserve power, surveys submitted after the 60-minute window were not automatically excluded. However, entries were cross-checked with earlier and later responses to evaluate and delete entries with temporal inconsistencies.

Using the *esmtools* package (21) in R Studio (Version: 2025.09.2+418; 22), time-series features were created using the lagged/lead function for PA and NA, and timing in minutes indicators for cannabis use (e.g., time since last use, time until next use).

Statistical modeling

To examine bidirectional relationships between cannabis use and affect, we used multilevel linear modeling (MLMs) with random intercepts per participant to account for within-person repeated daily assessments. Momentary affect and timing variables were modeled at the within-person level, whereas baseline cannabis use frequency and emotion regulation difficulties were modeled at the between-person level. Models were estimated separately for PA and NA. All analyses were preregistered (<https://osf.io/c74ur>). In addition to preregistered analyses, two explorative analyses were conducted (Model 3 and Model 5).

Within-person predictors included affect (PA, NA), time until next use, and time since last use (derived from ESM survey timestamps). Between-person predictors included cannabis use days in the last month (baseline TLFB) and emotion regulation difficulties at baseline. Within-person variables were person-mean-centered to estimate within-person associations, whereas between-person variables were grand-mean-centered (23). Cannabis use days in the last month (baseline) were left-skewed,

but as reversing and transforming the scale did not improve the fit, we maintained the non-transformed grand-mean centered version for the final models. All time variables (e.g., minutes until next use, and minutes since last use) were log-transformed after centering to reduce skewness. The final models did not explicitly include time-of-day, prompt number, or study day covariates. Autoregressive residual structure did not improve change the results or improve the fit. However, lagged affect terms were included in the affect models to partially account for temporal dependence across assessments.

Model diagnostics indicated no major violations of assumptions, and missing data were assumed to be missing at random. Model fit was summarized using AIC, BIC, and marginal/conditional R^2 . Unadjusted p-values were reported, given the exploratory nature of the study. No formal a priori sample size calculation was conducted. The target sample size was determined based on feasibility and prior ESM studies with a similar design (7). Analyses were conducted in R Studio (Version 2025.09.2+418) (22) using the *lme4* package (24).

Model specifications

Model 1 examined whether momentary affect at t_0 (PA t_0 or NA t_0) predicted the likelihood of cannabis use at the subsequent assessment (t_1 , 1 = use, 0 = no use). We used logistic multilevel models, including days of cannabis use in the last month (between-person) and time until next use (within-person) as covariates. See Table 1 for more details.

Table 1. Bidirectional structure of hypothesized effects and moderation. All models were controlled for number of cannabis use days in the previous 31 days and time until/since use.

Description	Model	Temporal	Relation
Preceding	1	$t_0 \rightarrow t_1$	Affect $\rightarrow$ future cannabis use
Short-term	2	$t_0 \rightarrow t_0$	Cannabis use $\rightarrow$ short-term affect
Delayed	3	$t_0 \rightarrow t_1$	Cannabis use $\rightarrow$ subsequent affect
Moderation 1	4	Baseline $\times t_0 \rightarrow t_1$	Emotion regulation difficulties moderate affect $\rightarrow$ future cannabis use
Moderation 2	5	Baseline $\times t_0 \rightarrow t_0$	Emotion regulation difficulties moderate use $\rightarrow$ short-term affect

Model 2 examined whether cannabis use since the previous assessment (i.e., measured at the same assessment, t_0 ; 1 = use, 0 = no use) was associated with momentary affect at t_0 (t_0 ; PA t_0 or NA t_0). We used linear multilevel models, including pre-use affect, measured at the previous assessment (t_{-1}), days of cannabis use in the last month (between-person), and time since last use (within-person) as covariates.

Capturing potentially delayed effects, Model 3 examined whether cannabis use since the previous assessment (t_0 ; 1 = use, 0 = no use) was associated with momentary affect at the next assessment (t_1 ; PA t_1 or NA t_1). We used linear multilevel models, including pre-use affect measured at the previous assessment (t_0), days of cannabis use in the last month (between-person), and time since last use (within-person) as covariates.

In Models 2 and 3, cannabis use at t_0 was included as a binary indicator of whether cannabis use had occurred since the previous assessment, whereas time since last use was included as a continuous timing covariate indicating the recency of the most recent use event. Including both variables allowed us to distinguish the occurrence of cannabis use from how proximal that use was to the assessment.

Model 4 examined whether individual differences in baseline emotion regulation difficulties moderated the association between momentary affect and subsequent cannabis use. To test this, we repeated the analysis described as Model 1, adding the interaction between momentary affect at t_0 (PA t_0 or NA t_0) and baseline emotion regulation difficulties (between-person; PA/NA $\times$ Emotion Regulation) as the predictor of interest.

Model 5 tested whether individual differences in baseline emotion regulation difficulties moderated the association between cannabis use and affect shortly after use. To test this, we repeated the analysis described as Model 2, adding the interaction between cannabis use (t_0 , 1 = yes, 0 = no use) and baseline emotion regulation difficulties (between-person; PA/NA $\times$ Emotion Regulation) as the predictor of interest.

RESULTS

Sample characteristics (baseline)

A total of 113 people who use cannabis near daily completed the survey (Table 2). Most of the participants (60.2 %, $N = 68$) were male, and the mean age was 31.6 (SD = 10.4; range, 18-63). On average, at baseline, participants used cannabis on 27.06 days (SD = 6.72) in the last month, and the average CUDIT-R score was 14.36 (SD = 5.57), probable CUD (18). The majority used alcohol (68.5%) and nicotine (75%) in the last month (Table 2).

Table 2. Characteristics of the sample (Mean [SD] and Median [IQR]).

Variables	Mean (S)	Median (IQR)
Age	31.57 (10.28)	29 (23 – 36)
Gender, n		
Male	68	
Female	43	
Non-Binary	2	
Education, n		
Lower / Pre-secondary	27	
Upper-secondary	29	
Vocational studies	25	
Higher Education	26	
Other / None / Missing	4	
Occupation, n		
Employed	40	
In education	21	
Unemployed	19	
Temporarily / chronic ill or disabled	11	
Retired	11	
Other / None / Missing	11	
Cannabis use		
CUDIT-R	14.36 (5.57)	14 (11 – 18)
Frequency of use (31 days) [†]	27.06 (6.72)	31 (26 – 31)
Age of onset	15.68 (2.71)	15 (14 – 17)
Alcohol use		
Alcohol use (previous 30 days; Ref: Yes), n	76	
AUDIT	9.63 (6.41)	8 (6 – 11)
Frequency of use (previous 30 days)	6.92 (7.65)	4 (2 – 10)
Age of onset	13.61 (1.74)	13 (13 – 15)
Nicotine use		
Nicotine use (previous 30 days; Ref: Yes), n	85	
FTND	4.12 (1.89)	4 (2 – 6)
Age of onset	14.37 (2.18)	14 (13 – 16)
Emotion regulation		
DERS-16 Total	33.43 (14.69)	29 (22 – 40)
Affect (ESM)		
PA (person mean across ESM)	14.94 (4.33)	15 (12 – 18)
NA (person mean across ESM)	6.44 (2.03)	6 (5 – 7)

Note: Values represent Mean (Standard Deviation) and Median (Interquartile Range) for continuous variables and n for categorical variables. PA and NA represent person-level means across the ESM period. N = 113 participants. [†] 2 Participants did not provide days of use in the past month. CUDIT-R = Cannabis Use Disorders Identification Test-Revised; AUDIT = Alcohol Use Disorders Identification Test; FTND = Fagerström test for Nicotine Dependence; DERS-16 = Difficulties in Emotion Regulation Scale-16; PA = Positive Affect; NA = Negative Affect; ESM = Experience Sampling Methods.

Descriptive summary of momentary data

The analyzed data included 7119 valid observations provided by 113 participants. On average, participants completed 85.72% of the ESM surveys (SD = 15.69; median = 92.1%), with completion rates varying only slightly across survey types: morning, 87% (SD = 16.35); afternoon, 85.54% (SD = 16.54); and evening, 84.61% (SD = 18.16). The response rates ranged from 36.5% to 100% supporting within-person analyses. Of the 6,103 completed ESM assessments, 83.7% were submitted within the 60-minute response window, with a median response delay of 10.8 minutes. Across the three daily prompts, within-window completion rates were 80.1%, 86.0%, and 85.0%, respectively. Responses submitted after 60 minutes were retained unless they generated temporal inconsistencies in the cannabis use timing variables.

Across all ESM surveys, participants reported between 7 and 63 cannabis use events ($M = 27.1$, $SD = 14.8$; median = 24), indicating between-person variability in use frequency. Overall, cannabis use was reported in 43% of all events, corresponding to an average of 27 events of use during the 21-day assessment period ($SD = 14.76$). The median time since last cannabis use was 5.8 hours [IQR = 1.02–21.03], whereas the median time until next cannabis use was 11.06 hours [IQR = 4.23–23.08]. Mean PA was 14.92 (SD between-person = 4.33, SD within-person = 2.90, $ICC = 0.691$), and mean NA was 6.46 (SD between-person = 2.03, SD within-person = 1.90, $ICC = 0.534$).

Associations between affect and subsequent cannabis use

Higher momentary $PA(t_0)$ was associated with a greater likelihood of cannabis use at the subsequent assessment (t_1) ($b = 0.05$, $p = .018$; Table 3). In contrast,

momentary NA (t_0) did not influence the likelihood of cannabis use at the subsequent assessment (t_1).

Table 3. Logistic mixed model predicting future cannabis use (t_1) based on affect (t_0) and covariates (Model 1).

	b	SE	OR	p	
Model 1					
Positive Affect					
(Intercept)	-0.36	0.30	0.7 [0.39, 1.26]	0.235	$R^2_m = 0.62$
PA t_0 WP	0.05	0.02	1.05 [1.01, 1.09]	0.018	$R^2_c = 0.9$
Minutes Until Next Use WP	-3.95	0.14	0.02 [0.01, 0.03]	< .001	AIC = 2359.1
Use Days BP	0.23	0.05	1.26 [1.15, 1.38]	< .001	BIC = 2391.6
Negative Affect					
(Intercept)	-0.35	0.30	0.7 [0.39, 1.27]	0.242	$R^2_m = 0.62$
NA t_0 WP	0.01	0.03	1.01 [0.95, 1.07]	0.808	$R^2_c = 0.9$
Minutes Until Next Use WP	-3.93	0.14	0.02 [0.02, 0.03]	< .001	AIC = 2364.6
Use Days BP	0.23	0.05	1.26 [1.15, 1.38]	< .001	BIC = 2397.2

Note. Logistic mixed-effects models included random intercepts by participant. WP = Within-person; BP = Between-person; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion; CI = Confidence Interval; SE = Standard Error; OR = Odds Ratio; R^2_m = Marginal R^2 ; R^2_c = Conditional R^2 ; PA = Positive Affect; NA = Negative Affect. All p values are two-tailed. $N = 4978$ assessments; $N = 113$ participants.

Associations between cannabis use and affect shortly after use

PA(t_0) was significantly higher following cannabis use (t_0 ; use since previous assessment) compared to when people indicated not having used cannabis ($b = 0.59$, $p < .001$; Table 4). In contrast, NA(t_0) did not differ following periods with and without cannabis use(t_0).

Table 4. Linear mixed model predicting short-term affect (t_0) based on cannabis use (t_0) (Model 2) and Linear mixed model predicting subsequent cannabis use (t_1) based on affect (t_0) and covariates (Model 3).

	b	SE	CI	p	
Model 2					
Positive Affect					
(Intercept)	14.67	0.43	[13.83, 15.51]	< .001	$R^2_m = 0.04$ $R^2_c = 0.74$ AIC = 25354.4 BIC = 25400.3
C t_0 WP	0.59	0.16	[0.27, 0.91]	< .001	
PA t_{-1} WP	0.30	0.03	[0.25, 0.35]	< .001	
Minutes Since Last Use WP	-0.06	0.04	[-0.14, 0.01]	0.119	
Use Days BP	-0.07	0.06	[-0.19, 0.06]	0.291	
Negative Affect					
(Intercept)	6.47	0.20	[6.07, 6.86]	< .001	$R^2_m = 0.06$ $R^2_c = 0.58$ AIC = 21189.0 BIC = 21234.9
C t_0 WP	-0.06	0.11	[-0.28, 0.17]	0.613	
NA t_{-1} WP	0.30	0.04	[0.22, 0.38]	< .001	
Minutes Since Last Use WP	0.01	0.03	[-0.04, 0.06]	0.738	
Use Days BP	0.06	0.02	[0.02, 0.1]	0.005	
Model 3					
Positive affect					
(Intercept)	14.97	0.43	[14.13, 15.81]	< .001	$R^2_m = 0.03$ $R^2_c = 0.74$ AIC = 25099.3 BIC = 25145.0
Cannabis use t_0	-0.03	0.17	[-0.36, 0.3]	0.858	
PA t_0 WP	0.31	0.03	[0.25, 0.37]	< .001	
Minutes Since Last Use WP	0.08	0.04	[0, 0.16]	0.061	
Use Days BP	-0.06	0.06	[-0.18, 0.06]	0.380	
Negative Affect					
(Intercept)	6.53	0.21	[6.12, 6.94]	< .001	$R^2_m = 0.06$ $R^2_c = 0.58$ AIC = 20959.1 BIC = 21004.9
Cannabis use t_0	-0.18	0.10	[-0.38, 0.02]	0.076	
NA t_0 WP	0.30	0.04	[0.22, 0.38]	< .001	
Minutes Since Last Use WP	-0.04	0.03	[-0.1, 0.02]	0.196	
Use Days BP	0.06	0.02	[0.02, 0.1]	0.003	

Note. Linear mixed-effects models included random intercepts for participant. WP = Within-person; BP = Between-person; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion; b = unstandardized coefficient; CI = Confidence Interval; SE = Robust Standard Error (CR2); R^2_m = Marginal R^2 ; R^2_c = Conditional R^2 ; PA = Positive Affect; NA = Negative Affect. All p values are two-tailed. $N = 5179$ assessments; $N = 113$ participants.

Associations between cannabis use and subsequent affect

PA(t_1) and NA(t_1) were not significantly higher following cannabis use (t_0 ; use since previous assessment) compared to when people indicated not having used cannabis (Table 4).

The moderating role of emotion regulation in the association between momentary affect and subsequent cannabis use

The interaction between $PA(t_0) \times$ emotion regulation difficulties was significant ($b = -0.10$, $p < .001$, see Figure 1 and Table 5), indicating that individuals with higher emotion regulation difficulties showed a weaker positive association between $PA(t_0)$ and subsequent cannabis use. In contrast, the interaction with $NA(t_0) \times$ emotion regulation difficulties was not significant, indicating that emotion regulation difficulties did not moderate NA antecedents of cannabis use.

Figure 1. Predicted momentary positive (Panel A) and negative affect (Panel B) at t_0 as a function of emotion regulation difficulties (DERS-16 total score), shown separately for cannabis use moments and non-use moments. Higher values indicate greater affect intensity and higher emotion regulation difficulties (at baseline). Lines represent model-based predictions across levels of emotion regulation difficulties.

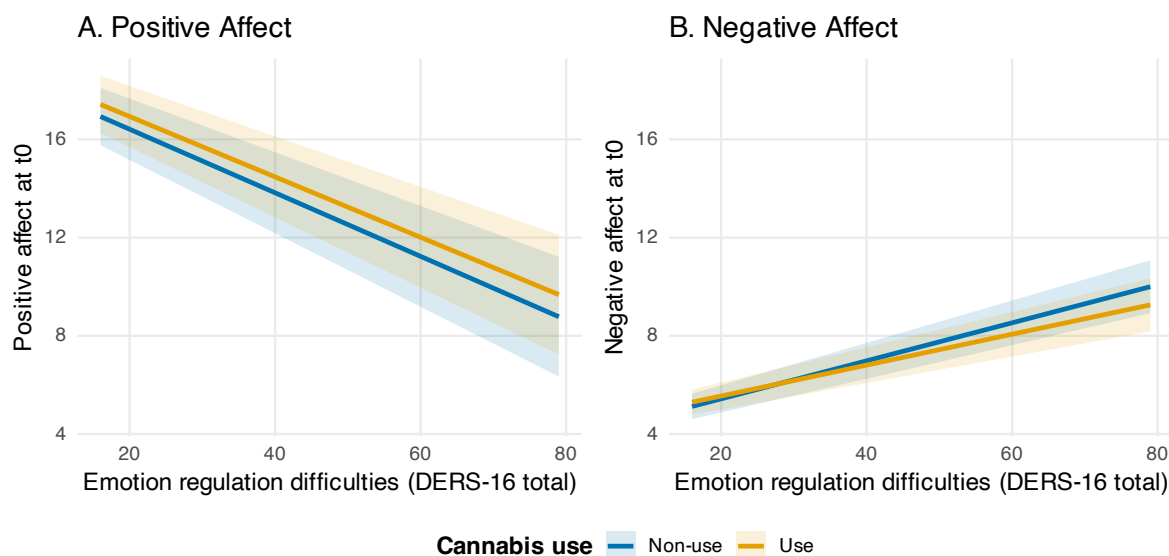


Table 5. Logistic mixed model testing whether baseline emotion regulation difficulties moderate the association between affect (t_0) and future cannabis use (t_1) (Model 4).

	b	SE	OR [CI]	p	
Model 4[†]					
Positive Affect					
(Intercept)	-0.36	0.005	0.7 [0.69, 0.70]	< .001	
PA t_0 WP	0.15	0.005	1.16 [1.15, 1.17]	< .001	$R^2_m = 0.62$
DERS-16 BP	-0.30	0.295	0.74 [0.41, 1.31]	= 0.302	$R^2_c = 0.9$
Use Days BP	1.59	0.005	4.93 [4.88, 4.98]	< .001	AIC = 2358.6
Minutes until Next Use WP	-4.20	0.005	0.02 [0.01, 0.02]	< .001	BIC = 2404.2
PA t_0 WP * DERS-16 BP	-0.10	0.005	0.91 [0.90, 0.91]	< .001	
Negative Affect					
(Intercept)	-0.34	0.30	0.71 [0.4, 1.27]	= 0.247	
NA t_0 WP	-0.03	0.06	0.97 [0.86, 1.1]	= 0.684	$R^2_m = 0.62$
DERS-16 BP	-0.29	0.30	0.74 [0.42, 1.34]	= 0.323	$R^2_c = 0.9$
Use Days BP	1.60	0.31	4.94 [2.7, 9.04]	< .001	AIC = 2365.2
Minutes Until Next Use WP	-4.18	0.14	0.02 [0.01, 0.02]	< .001	BIC = 2410.8
NA t_0 WP * DERS-16 BP	0.08	0.05	1.08 [0.98, 1.2]	= 0.126	

Note. Logistic mixed-effects models included random intercepts by participant. WP = Within-person; BP = Between-person; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion; CI = Confidence Interval; SE = Standard Error; OR = Odds Ratio; R^2_m = Marginal R^2 ; R^2_c = Conditional R^2 ; PA = Positive Affect; NA = Negative Affect; DERS-16 = Difficulties in Emotion Regulation Scale-16.

All p values are two-tailed. $N = 4978$ assessments; $N = 113$ participants.

[†] Model 4 (PA model) returned a mild convergence warning ($|grad| = 0.015 > 0.002$) and after using alternative optimizers (bobyqa with higher iterations, nloptwrap, calc.derivs = FALSE). Across all specifications, the fixed effects estimates, and significance levels were nearly identical, indicating that the convergence warning did **not** affect the results. We therefore report the bobyqa solution as the final model.

The moderating role of emotion regulation in the association between cannabis use and affect shortly after use

Emotion regulation moderated the association between cannabis use since the last assessment(t_0) and momentary NA(t_0) ($b = -0.01$, $p < .001$; see Table 6). As emotion regulation difficulties increased, the association between cannabis use and NA(t_0) became more negative, indicating that individuals with greater emotion regulation difficulties reported lower NA(t_0) at moments when they had used cannabis compared with moments of non-use. No interaction was observed for PA(t_0).

Table 6. Linear mixed model testing whether baseline emotion regulation difficulties moderate the association between cannabis use (C_{t_0}) and affect (t_0) shortly after use (Model 5).

Model 5	b	SE	CI	p	
Positive Affect					
(Intercept)	14.67	0.38	[13.91, 15.43]	< .001	
C_{t_0}	0.60	0.14	[0.34, 0.87]	< .001	
DERS-16 _{BP}	-0.13	0.03	[-0.18, -0.08]	< .001	$R^2_m = 0.17$
PA prev _{WP}	0.30	0.01	[0.28, 0.33]	< .001	$R^2_c = 0.74$
Use Days _{BP}	-0.03	0.06	[-0.14, 0.09]	0.615	AIC = 25335.6,
Minutes Since Last Use _{WP}	-0.06	0.03	[-0.12, 0.01]	0.085	BIC = 25394.5.
$C_{t_0} * \text{DERS-16}_{BP}$	0.01	0.01	[0, 0.02]	0.250	
Negative Affect					
(Intercept)	6.47	0.17	[6.13, 6.81]	< .001	
C_{t_0}	-0.08	0.09	[-0.26, 0.1]	0.379	
DERS-16 _{BP}	0.08	0.01	[0.05, 0.1]	< .001	$R^2_m = 0.2$
NA prev _{WP}	0.30	0.01	[0.27, 0.33]	< .001	$R^2_c = 0.58$
Use Days _{BP}	0.04	0.03	[-0.01, 0.09]	0.154	AIC = 21145.5
Minutes Since Last Use _{WP}	0.00	0.02	[-0.04, 0.04]	0.990	BIC = 21204.5
$C_{t_0} * \text{DERS-16}_{BP}$	-0.01	0.00	[-0.02, -0.01]	< .001	

Note. Linear mixed-effects models included random intercepts for participant. WP = Within-person; BP = Between-person; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion; b = unstandardized coefficient; CI = Confidence Interval; SE = Robust Standard Error (CR2); R^2_m = Marginal R^2 ; R^2_c = Conditional R^2 ; PA = Positive Affect; NA = Negative Affect; DERS-16 = Difficulties in Emotion Regulation Scale-16. All p values are two-tailed. Model summary: N = 5179 assessments; N = 113 participants.

DISCUSSION

This study examined within-person, bidirectional associations between affect and cannabis use among individuals who use cannabis near daily. Overall, the findings provided stronger support for positive reinforcement than for negative reinforcement. Contrary to our expectations, NA did not significantly affect the likelihood of subsequent cannabis use. However, higher PA was linked to increased likelihood of subsequent cannabis use, and PA levels were higher immediately after use compared to moments at which no recent use was reported. Furthermore, baseline emotion regulation difficulties weakened the association between PA and increased likelihood of subsequent cannabis use, but it did not relate to PA following cannabis use. In contrast, looking at NA, individuals with greater baseline emotion regulation difficulties reported lower NA after use moments relative to non-use moments.

Affect and subsequent cannabis use

Contrary to our hypotheses, higher NA was not associated with a greater likelihood of subsequent cannabis use in this community sample of near daily users. This pattern is consistent with earlier ESM research in young adult samples showing null or inverse associations between NA and cannabis use (6, 7). Furthermore, we observed unexpectedly low overall levels of NA and small moment-to-moment variability in NA (Figure 1), which is in line with research suggesting relatively limited emotional variability among individuals who regularly smoke cannabis (25). Additionally, our assessment frequency (3 times per day) could have resulted in us missing the more transient elevated NA levels commonly expected just before use (e.g., craving). Low levels and reduced moment-to-moment variability in NA can constrain the ability to detect NA-use coupling in ESM data, particularly when NA elevations are brief or unfold over pre-use windows rather than at the immediate preceding assessment, as observed in previous ESM studies (e.g., 8, 26).

In our study of near daily users, higher PA was linked to an increased likelihood of subsequent cannabis use. This finding is consistent with previous ESM studies in non-clinical samples (6, 7, 26). At the same time, this pattern does not generalize uniformly across populations: among young and clinically engaged samples, ESM studies have reported null associations between PA and subsequent use (8). This pattern suggests that near daily cannabis use observed in community samples may be primarily pleasure-driven rather than triggered by negative reinforcement (26), similar to what has been found in student samples. Higher momentary PA before use may reflect an internally or cue-triggered hedonic motivational state (27). From a reinforcement-learning perspective, repeated pairing of cannabis use with positive states may strengthen cue-response associations, such that moments of elevated PA increase the incentive salience of cannabis-related cues and the likelihood of

subsequent use (28-30). In this sense, elevated PA may not only reflect internal hedonic states but also signal contexts in which cannabis use is available or more socially reinforced. Importantly, higher PA may also coincide with situational contexts in which cannabis use is more likely to occur, such as leisure time or social interactions, thereby amplifying the likelihood of use when individuals already feel positive. Moreover, cannabis use in PA contexts may also have social meaning, such as group identity, belonging, and shared practices, suggesting that affective and social reinforcement processes may operate jointly. These social factors remain understudied and should be examined in future cross-cultural research (8, 26).

Affect following cannabis use

Looking at affect following cannabis use, we observed higher levels of PA shortly after use, but this effect did not extend to the next assessment of PA. Based on previous findings indicating transient, reduced NA following cannabis use (26) and theories of negative reinforcement and self-medication (31), we expected lower NA following cannabis use compared with non-use moments. However, results showed that cannabis use did not affect NA levels, possibly due to the combination of limited moment-to-moment variability and overall low levels of NA in this sample, as discussed above. Importantly, increases in PA and reductions in NA reflect partially distinct affective processes, and the presence of acute hedonic effects does not necessarily imply a parallel decrease in NA. Our findings align with previous studies that suggest that individuals can experience higher PA immediately after cannabis use (7, 32), consistent with short-term pharmacological effects. Previous study, in a clinical sample, observed elevated PA and reduced anger/irritability during cannabis use moments (11). Because anger/irritability represents a specific facet of NA, such reductions may occur without producing a detectable change in a global NA scale,

which could explain the absence of NA differences in the present study, where composite NA scores were used (11).

Moderating role of Emotion Regulation on the relationship of affect before cannabis use

The significant interaction between $PA(t_0)$ and emotion regulation difficulties shows that momentary increases in PA were less predictive of subsequent cannabis use among individuals with greater difficulties regulating emotions. This pattern suggests that emotion regulation difficulties attenuate the affective salience of positive mood states as proximal cues for cannabis use. For individuals with lower emotion regulation difficulties, fluctuations in PA may serve as signals for use decisions, consistent with positive reinforcement processes. In contrast, emotion regulation difficulties did not moderate the association between $NA(t_0)$ and subsequent cannabis use, suggesting that vulnerability in emotion regulation did not impact the likelihood of using cannabis in response to momentary NA in this sample. Importantly, the null moderation effects for NA should be interpreted cautiously, as they reflect restricted NA variability and generally low NA levels. Replication is needed in samples characterized by higher self-reported NA, potentially including individuals with co-morbid psychiatric disorders, as these individuals are thought to be at higher risk of cannabis use for negative reinforcement reasons (27).

Moderating role of Emotion Regulation on the relationship of affect shortly after cannabis use (consequences)

Individuals with greater emotion regulation difficulties reported lower NA at use moments compared with non-use moments, whereas no such difference was observed among individuals with lower emotion regulation difficulties. This indicates that relief-like reductions in NA following cannabis use emerged selectively among

individuals with elevated emotion regulation difficulties, rather than as a general consequence of use. Importantly, this moderation effect was observed only for NA, and not for PA, indicating that emotion regulation difficulties shape the negative-affect consequences of cannabis use rather than its PA outcomes, which may explain why no main effect of cannabis use on NA was observed in the full sample. This pattern differs from prior ESM studies (8, 10, 11), which generally did not detect group-level NA differences at use versus non-use moments. This finding suggests that person-level differences may obscure such associations when data are averaged across participants with heterogeneous emotion regulation profiles (33-35).

Together, these findings indicate the coexistence of two reinforcement processes. First, a positive-reinforcement pathway appears to be more consistent at the within-person level in this sample, as reflected in PA involvement. Second, our findings suggest that relief-like reductions in NA following use are not a general characteristic of near daily cannabis use, but instead emerge selectively among individuals who have elevated emotion regulation difficulties. This person-level heterogeneity aligns with reinforcement-based models positing that two affective functions across individuals maintain cannabis use, as discussed in previous studies (1).

Clinical and applied implications

Across the full sample, cannabis use was consistently linked to PA, suggesting positive reinforcement represents a common affective pathway in individuals who use cannabis near daily. Within this broader pattern, individuals with lower emotion regulation difficulties showed a predominantly PA-driven use profile, consistent with greater psychological flexibility and less reliance on substance use as an emotion regulation strategy as described in process-based models (36). In contrast, individuals

with higher emotion regulation difficulties showed selective NA patterns, consistent with transdiagnostic mechanisms such as experiential avoidance, affective rigidity, and reduced self-regulatory capacity (37, 38). This distinction provides a process-level explanation for why relief-based use does not emerge uniformly across participants despite being reported in cross-sectional studies and highlights important clinical implications for tailoring interventions to individuals' affective profiles.

From a public health perspective, these micro-level affective dynamics highlight how vulnerability processes can accumulate and stabilize over time, contributing to the persistence of cannabis use at the population level. These findings illustrate how short-term effects and feedback loops between affect and behavior can evolve into a self-reinforcing pattern. Prevention and intervention efforts may therefore benefit from addressing the quantity of use, but also the temporal and emotional patterns that sustain it.

Limitations and future directions

Several limitations should be acknowledged. First, social norms and legal factors around cannabis in Denmark may influence reporting patterns, potentially leading to underreporting of NA due to stigma, social desirability, or concerns about legal or regulatory implications, before or after cannabis use. Future ESM designs that incorporate passive or multimodal affective data (e.g., wearable physiology) may help overcome these constraints. Second, the sample comprised of individuals who use cannabis from the Danish community, which may limit the generalizability of findings to people who use cannabis less frequently and to other cultural contexts. Third, the absence of motives-specific, event-level measurement of potency, cannabis strains, and routes of administration during ESM assessments may have introduced heterogeneity in affective responses and reduced the precision of estimates linking

cannabis use, intoxication level, and affective responses. Fourth, because of the intensity of self-monitoring, we cannot rule out behavioral reactivity, whereby the ESM protocol itself may have influenced participants' affect or cannabis use patterns (39). Additional methodological constraints include potential habituation to repeated self-reports over 21 days, floor/ceiling effects in affect scales, timing inaccuracies in self-reported cannabis use, and software constraints limiting survey closure. Finally, the absence of momentary assessments of motives for use limits the ability to determine whether specific episodes were driven by coping, enhancement, habit, or contextual factors.

CONCLUSION

This study shows that cannabis use in daily life is closely tied to PA, both as a precursor and a short-term outcome, among individuals who use cannabis near daily. In contrast, NA played a minimal role, suggesting that negative reinforcement is not the primary momentary driver of use in this population. Emotion regulation difficulties shaped these dynamics, attenuating the association between PA and subsequent use while strengthening the association between cannabis use and lower NA at cannabis use moments relative to non-use moments. Together, these findings indicate that cannabis may serve different affective functions across individuals. Some use may be pleasure-oriented, whereas among individuals with greater emotion regulation difficulties, cannabis may function as a selective, and momentary relief strategy.

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Appendix C (Manuscript 3)

Beyond cannabis use severity: associations of cannabis product type with sleep quality and mental health

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ABSTRACT

Rationale: People who use cannabis often report using it to improve sleep and/or manage mental health symptoms. Cannabis products differ in their typical cannabinoid composition, particularly in their relative THC and CBD content, which may impact health outcomes differently.

Objectives: This naturalistic study investigated associations between cannabis use severity, sleep quality, and symptoms of anxiety/depression among people who use cannabis near daily and examined the role of cannabis product types.

Methods: Danish adults who reported using cannabis on 26.15 ± 6.23 days/month completed an online survey (197 participants; 64.5% men, $M_{\text{age}} 30.7 \pm 10.63$ years) measuring past-month use of cannabis (including product types), cannabis use and related problems (CUDIT-R), sleep quality (B-PSQI) and symptoms of anxiety/depression (PHQ-4). OLS regression models were used to examine associations and interaction terms were used to test moderation.

Results: Higher cannabis use severity was associated with poorer sleep quality and higher levels of anxiety/depression symptoms, which was not moderated by cannabis product type. Product type explained additional variance in sleep quality, but not in anxiety/depression symptoms. Individuals who reported using hash only or multiple THC-dominant product types exhibited poorer sleep quality compared with those who used flower only, independent of cannabis use severity.

Conclusions: Cannabis use severity was linked with sleep and mental health problems, and product type did not alter the strength of these associations. Use of hash or multiple THC-dominant product types was associated with more sleep problems than use of flower, adding to the literature showing that products with greater THC concentration are more harmful. Longitudinal studies are needed to establish causal relationships.

Key-words: Cannabis; Sleep; Anxiety; Depression; Hash; THC; CBD.

1. Introduction

Sleep problems constitute a major public health concern due to their strong associations with adverse physical and mental health outcomes (Roth et al. 2006; Lim et al. 2023;). Cannabis use is widely reported as a sleep aid, and improving sleep is among the most endorsed motives for its use (Winiger et al. 2021; Gex et al. 2024). However, sleep disturbances have been linked to cannabis use both as antecedents and as potential consequences, indicating bidirectional and context-dependent associations between sleep problems and cannabis-related behaviors (Babson et al. 2013; Babson and Bonn-Miller 2014; Wong et al. 2019). Moreover, while some studies report reductions in sleep-onset latency (i.e., time it takes to fall asleep) and improvements in subjective sleep quality among individuals who use cannabis, other studies report greater sleep fragmentation (i.e., frequent and short awakenings), reduced sleep duration, and suppressed rapid eye movement (REM) sleep, particularly among individuals with more frequent and problematic use patterns (Babson et al. 2013; Conroy et al. 2016; Edwards and Filbey 2021; Winiger et al. 2021; Walsh et al. 2025). Together, these findings underscore the need to better understand when, for whom, and under which conditions cannabis use is associated with sleep-related benefits or harms.

In parallel with its use for sleep, cannabis use is also reported as a strategy for regulating negative affective states and feelings, including feelings of anxiety and depression (Gex et al. 2024). Empirical findings on links between cannabis and mental health are heterogeneous. While some studies report associations between cannabis use and short-term reductions in anxiety or depression symptoms in specific contexts (Wycoff et al. 2018), other studies have found greater psychological distress and elevated anxiety and/or depression symptoms among cannabis users, particularly among individuals with more frequent or problematic patterns of use (Danielsson et al. 2016; Struble et al. 2019; Xue et al. 2021; Steeger et al. 2025;). These results mirror the heterogeneity observed for sleep outcomes, suggesting that cannabis-related mental health effects are similarly context- and pattern-dependent.

One important dimension that may help explain the heterogeneity in cannabis-sleep and cannabis-mental health associations is the severity of cannabis use-related problems, including the development of Cannabis Use Disorder (CUD) symptoms. These include impaired control, tolerance, withdrawal, and psychosocial problems (Bonnet and Preuss 2017). Higher severity of cannabis use-related problems has been associated with greater psychological distress, functional impairment, and concurrent psychiatric disorders (Hasin et al. 2016; Volkow et al. 2016; Hasin and Walsh 2020; Hjorthoj et al. 2023). In parallel, regular exposure to cannabinoids has been linked to withdrawal-related sleep complaints, and greater sleep fragmentation (Edwards and Filbey 2021). Thus, these findings suggest that both the amount of use and the severity of use-related problems reflects the broader behavioral and clinical context that may shape when, for whom, and under which conditions cannabis use is associated with sleep-related and mental health-related outcomes, rather than exerting uniform effects across people who use cannabis for either sleep or mental health outcomes.

Beyond this, product-related differences represent an additional source of variation in cannabis-sleep and cannabis-mental health associations. Product-related differences, e.g., in terms of cannabinoid content, may be

particularly relevant for sleep-related outcomes, given the evidence that cannabinoids can differently influence sleep continuity, architecture, and circadian regulation (Edwards and Filbey 2021). Similarly, there is evidence that different cannabis profiles vary in their associations with anxiety and depression symptoms (Freeman and Winstock 2015). Accordingly, product-specific cannabinoid profiles may contribute to differences in sleep and mental health outcomes independently from the severity of cannabis use-related problems. In naturalistic research, cannabis product types used in a recreational/non-medicinal context do not precisely reflect individual cannabinoid dose or potency; instead, they may serve as indicators of systematic differences in product characteristics, routes of administration, and user experiences (Freeman and Winstock 2015). At the population level, cannabis resin (such as hash) is typically characterized by higher average Delta9-tetrahydrocannabinol (THC) concentrations compared to flower products, and its use has been associated with greater cognitive impairment, anxiety, dependence liability, and sleep disruption (Freeman et al. 2021; Petrilli et al. 2023). Moreover, products high in THC can interfere with circadian regulation, including reducing total sleep time and suppressing rapid eye movement (REM) sleep (Edwards and Filbey 2021; Bell et al. 2022; Walsh et al. 2025). In contrast, flower products, which typically contain lower levels of THC and more variable levels of cannabidiol (CBD) have been reported to help with sleep onset and relaxation (Kuhathasan et al. 2022). Despite these distinctions, most studies do not differentiate between cannabis products, thereby limiting knowledge on potential product-specific inference for both sleep quality and mental health outcomes (Rømer Thomsen et al. 2017).

The predominant type of cannabis products used varies in European countries, but in most countries flower products are the most used cannabis product (EUDA 2024b). However, in the Scandinavian country of Denmark, hash is by far the most used cannabis product (Danish Health Authority 2023; Callesen et al. 2024; EUDA 2024b; Frederiksen et al. 2025). Examining product-specific associations in Denmark provides a unique opportunity because flower-based products dominate the existing literature on cannabis and health outcomes. Similar to other European countries and trends globally (Freeman et al. 2021), a marked increase in THC concentration in cannabis resin (including hash) has been reported in Denmark since 2000 (Rømer Thomsen et al. 2019), with Denmark reporting the highest THC concentration in seized cannabis resin across Europe (Freeman et al. 2019).

Present study

The goal of the present cross-sectional study was to investigate associations between cannabis use severity, sleep quality, and symptoms of anxiety/depression and the role of cannabis product type in this, in a naturalistic sample of people who use cannabis near daily. We hypothesized that (1) greater cannabis use severity would be associated with poorer sleep quality and higher levels of anxiety/depression symptoms; (2) use of hash products would be associated with poorer sleep quality and higher levels of anxiety/depression relative to use of flower products only; and (3) product type would moderate the associations between cannabis use severity, sleep quality, and symptoms of anxiety/depression, with stronger severity–outcome links among people who use hash only compared to those who use flower products only..

2. Methods

2.1 Participants

Participants were recruited using multiple online and offline strategies designed to reach adults with frequent cannabis use. These included flyers and posters distributed at The Copenhagen Cannabis Exhibition in 2023, at Aarhus University campuses and in cannabis-related community settings, online advertisements posted directly in cannabis-related forums (with moderator approval), email invitations to individuals who had previously agreed to be contacted in future research about cannabis, and snowball dissemination through cannabis community networks in Denmark. A study website provided information about eligibility, procedures, and contact details. To take part in the research, the following criteria were applied: aged 18-65; able to understand Danish; having smoked cannabis at least three days per week during the past 30 days (regardless of the frequency of other consumption methods); and not having used any other illicit substances during the past 30 days. Potential participants were screened using an online questionnaire. The full sample included 197 participants. The sample size for the present study was determined by the number of participants that enrolled in the parent experience sampling method (ESM) study (Donate et al. 2026).

2.2 Procedure

This cross-sectional study is part of a larger study preregistered on the Open Science Framework (OSF; <https://osf.io/c74ur/>), although the hypotheses and analyses of the present manuscript were not preregistered as a standalone study on the OSF. All procedures were performed in line with principles of the declaration of Helsinki. Ethical approval was obtained from the Research Ethics Committee at Aarhus University (BSS-2023-051) and the study was registered at the Danish Data Protection Agency. All participants were fully informed, were instructed that participation was voluntary, and that they could withdraw from the study at any time without consequences. Informed consent was obtained online from all participants. All participants were living in Denmark during the data collection period, which ran from January to October 2024. Data were collected and managed with Research Electronic Data Capture (REDCap), hosted at Aarhus University (Harris et al. 2009). After confirming their eligibility statement, participants were invited to complete the online survey in Danish ($\approx$ 20 min).

Measures

The online survey included questions on sociodemographic characteristics: gender, age, education, and occupation.

To assess substance use history, participants completed questions on past 30-day use, age of onset, number of days used in the past 30 days. Participants who reported alcohol and/or nicotine use also completed the Alcohol Use Disorders Identification Test (AUDIT; Saunders et al. 1993), and the Fagerström test for Nicotine Dependence (FTND; Heatherton et al. 1991) for those using alcohol and nicotine, respectively.

Cannabis product type was assessed using a multi-select questionnaire in which participants indicated all cannabis product types/forms used in the past 30 days. Response options were tailored to the Danish population and included hash, pot (i.e., flower), skunk (i.e., a high-potency cannabis flower product), legally available low-THC products (i.e., $< 0.2\%$ THC), prescribed cannabis (e.g., Bediol, Bedrocan), other smoked or inhaled forms (e.g., vaping), and unknown type. The category “other smoked or inhaled forms” captured products such as THC

vape liquids, concentrate oils, wax, tincture, dabs, and/or other inhaled cannabis forms reported in free-text responses. Because these response options were not mutually exclusive and reflected both cannabis product type, legal status, and mode of administration, participants could endorse all applicable options. Lastly, the category “unknown type” referred to cannabis products that participants were unable to clearly identify or classify based on the available response options. Based on these responses, participants were subsequently classified into five mutually exclusive cannabis product user groups: (1) flower only (pot and skunk), (2) hash only, (3) multiple THC-dominant product types (use of two or more THC-dominant products without CBD), (4) THC-dominant concentrates (exclusive use of high-THC concentrates or extracts), and (5) combined THC + CBD products (co-use of at least one THC-dominant and one CBD-dominant product). For each endorsed product type, participants also reported the number of days used in the past 30 days. Supplementary Table S1 describes how the original questionnaire response options were recoded into the five derived product-use categories.

To assess severity of cannabis use in the past six months, participants completed the Cannabis Use Disorders Identification Test-Revised (CUDIT-R; Adamson et al. 2010). The CUDIT-R is a validated, dimensional measure of cannabis use severity that captures frequency of use and cannabis use-related problems on a continuum, including subclinical levels of CUD. The total scores range from 0 to 32, and higher scores indicate higher cannabis use severity. Established classification suggests that scores below 8 indicate hazardous use, whereas score of 13 or more indicate probable CUD (Adamson et al. 2010).

To assess symptoms of anxiety and depression over the past two weeks, participants completed the validated Patient Health Questionnaire (PHQ-4; Kroenke et al. 2009), which includes two items for each, scored from 0 to 3. The total scores range from 0 to 12. Higher scores indicate more severe symptoms of anxiety and depression. Established severity thresholds classify scores as normal (0-2), mild (3-5), moderate (6-8), and severe (9-12) (Kroenke et al. 2009).

To assess sleep quality, participants completed the brief Pittsburgh Sleep Quality Index (B-PSQI; Sancho-Domingo et al. 2021): A brief, validated measure assessing the quality of sleep over the past month. Responses include self-reported times/durations or frequency rated on a 4-point Likert scale. Items are grouped and scored into five sleep components: (1) subjective sleep quality, (2) sleep latency, (3) sleep duration, (4) sleep efficiency, and (5) night disturbances. The sum of component scores (each coded 0—3) yields a total global score (B-PSQI) ranging from 0 to 15, where higher scores indicate poorer sleep quality. A score of 5 has been proposed as the optimal cutoff for poor sleep quality in this measure (Sancho-Domingo et al. 2021).

3. Statistical analysis

All statistical analyses were conducted using RStudio (Posit Software, 2025). Descriptive statistics were computed for cannabis use severity (CUDIT-R), sleep quality (B-PSQI), anxiety/depression symptoms (PHQ-4), and cannabis product categories: (1) flower only (pot and skunk), (2) hash only, (3) multiple THC-dominant product types, (4) THC-dominant concentrates, and (5) combined THC + CBD products. For analytic purposes, hash was treated as a separate category from THC-dominant concentrates, which referred to high-potency cannabis extracts such as dabs, wax, shatter, THC vape liquids and concentrate oils. Means, standard deviations (SDs), frequencies, and percentages were used to describe the sample. Normality of continuous variables was assessed using visual

inspection of histograms and Q-Q plots. Because most continuous variables had symmetric distributions, Pearson's correlations were used.

We fitted ordinary least squares (OLS) regression models with HC3 robust standard errors to test whether cannabis use severity (CUDIT-R) was associated with (a) sleep quality (B-PSQI) and (b) anxiety/depression symptoms (PHQ-4). Each outcome was analyzed in a separate model. Continuous predictors were mean-centered. Effect sizes are presented as unstandardized regression coefficients with 95% robust confidence intervals, and model fit is reported using R^2 and adjusted R^2 .

To examine the role of product type in the associations between cannabis use severity, sleep quality, and symptoms of anxiety/depression, we first estimated unadjusted linear models with cannabis product type entered as a categorical predictor. Next, we tested whether product type explained additional variance beyond cannabis use severity by refitting the models with cannabis product type entered as a categorical predictor and dummy-coded using flower only as the reference category (ANCOVA). Finally, to test moderation, interaction terms between cannabis use severity and cannabis product type were added to the model. When significant, interactions were probed using simple slopes and pairwise comparisons of slopes. For descriptive purposes, adjusted marginal means for each product type were also compared using main-effects models. Changes in explained variance across nested models were additionally examined using model comparison indices. In supplementary analyses, models were additionally adjusted for age, gender, past-30-day nicotine use, and past-30-day alcohol use. We also repeated the analyses after excluding the THC-dominant concentrates group because of its very small sample size. Model assumptions (linearity, heteroscedasticity, residual distribution, and multicollinearity) were checked using visual diagnostics and robust tests. As all models were estimated with HC3 standard errors, no additional resampling-based procedures (e.g., bootstrapping) were applied. Two-sided p-values < .05 were considered statistically significant. Analyses were conducted using all available data and the following packages: *tidyverse* (Wickham et al. 2019), *sandwich* (Zeileis et al. 2020), *lmtest* (Zeileis et al. 2002), and *emmeans* (Lenth et al. 2026).

We conducted a post hoc power analysis to estimate the smallest effects detectable with 80% power at $\alpha = .05$. With an analytic sample size of 193 for sleep models and 194 for anxiety/depression models, the study was powered to detect effects in the small-to-moderate range.

3. Results

3.1 Sample characteristics

A total of 197 individuals completed the survey. Most of the participants were men ($n = 127$, 64.5%), followed by women ($n = 65$, 33%), and a small group identified as non-binary ($n = 5$, 2.5%). The M_{age} was 30.7 years ($SD = 10.63$; range = 18-63). The majority of the participants completed lower/pre-secondary education ($n = 57$, 28.9%) or upper secondary education ($n = 56$, 28.4%).

On average, participants reported cannabis use on 26.15 days in the past 30 days ($SD = 6.23$), and the mean CUDIT-R score was 14.26 ($SD = 5.99$), indicating probable CUD (Adamson et al. 2010). The majority of

participants reported use of alcohol (n = 134) and nicotine (n = 141) in the past 30 days. Sample characteristics, substance use, mental health, and cannabis types are presented in Table 1.

Table 1. Sample characteristics, substance use, and cannabis product types

Variable	Mean (SD)	Median (IQR)
<i>Age</i>	30.7 (10.6)	28 (23 – 35)
<i>Gender, n</i>		
Men	127	
Women	65	
Non-binary	5	
<i>Education, n</i>		
Lower/pre-secondary education	57	
Upper secondary education	56	
Vocational education	32	
Higher education ^a	39	
<i>Occupation</i>		
Employed	71	
In education	35	
Unemployed	27	
Temporary/chronic ill	15	
Retired	18	
<i>Cannabis</i>		
Use days past month	26.15 (6.23)	30 (25 – 30)
Age of cannabis onset	15.6 (2.4)	15 (14 – 17)
CUDIT-R	14.1 (5.7)	14 (10 – 18)
<i>Alcohol</i>		
Age of alcohol onset	13.6 (1.8)	14 (13 – 15)
Alcohol use days past month ^b	6.66 (7.25)	3.5 (1 – 8)
AUDIT	9.6 (6.1)	8 (5 – 12)
<i>Nicotine</i>		
Age of nicotine onset	14.4 (2.7)	14 (13 – 16)
Nicotine use days past month ^c	25.81 (8.01)	30 (25 – 30)
FTDN	4.2 (1.9)	4 (2 – 6)
<i>Sleep and mental health</i>		
B-PSQI	3.5 (2.5)	3 (2 – 5)
PHQ-4	3.7 (3.1)	3 (1 – 6)
Cannabis products (non-mutually exclusive) [§], n (%)		
Hash only	161 (81.7)	
THC-dominant concentrates	14 (7.1)	
THC flower	126 (64)	
CBD flower	26 (13.2)	
CBD non-flower	2 (1)	
Derived cannabis product categories (mutually exclusive) [†], n (%)		
1. Flower only	26 (13.2)	
2. Hash only	63 (32)	
3. THC-dominant concentrates	3 (1.5)	
4. [‡] Multiple THC-dominant products	78 (39.6)	
5. [¥] Combined THC + CBD products	27 (13.7)	

Note.

Values are presented as mean (SD) and/or median (Q1 – Q3).

^a Thirteen participants reported other/none in education categories.

^b Among participants who reported alcohol use in the past 30 days (n = 134), all provided valid information on the number of days of alcohol and were included in the descriptive of alcohol use days.

^c Among participants who reported nicotine use in the past 30 days (n = 141), 138 provided valid information on the number of days of use and were included in the descriptive of nicotine use days.

[§] Participants could endorse more than one cannabis product; therefore, percentages do not sum to 100%.

† Derived cannabis product categories groups were mutually exclusive and based on patterns of product use: flower only, hash only, THC-dominant concentrates, multiple THC-dominant products, and combined THC + CBD products.

‡ Multiple THC-dominant products refer to the use of more than one THC-dominant product (e.g., flower, hash, concentrates).

‡ Combined THC + CBD products refer to the use of at least one THC-dominant and one CBD-dominant product.

See Supplementary Table S1 for details on how original product responses were recorded into derived categories.

Abbreviations.

PHQ-4 = Patient Health Questionnaire-4;

B-PSQI= Pittsburgh Sleep Quality Index – brief version;

CUDIT-R = Cannabis Use Disorder Identification Test-Revised;

AUDIT = Alcohol Use Disorders Identification Test;

FTND = Fagerström Test for Nicotine Dependence;

SD = Standard Deviation;

IQR = Interquartile Range.

In a non-mutually exclusive selection of cannabis products used in the past month, hash was the product endorsed by most ($n = 161$), followed by THC flower products ($n = 126$). In the derived cannabis product categorization, which is mutually exclusive, most people endorsed using multiple THC-dominant product types ($n = 78$) and hash only ($n = 63$). Descriptively, across derived cannabis product categories, participants who reported using flower only had lower scores in CUDIT-R, B-PSQI, and PHQ-4, whereas participants who reported using hash (only) showed relatively higher scores in CUDIT-R. Participants who reported THC-dominant concentrates showed the highest scores in sleep problems (B-PSQI) and anxiety/depression symptoms (PHQ-4), followed by multiple THC-dominant product types (see Figure 1).

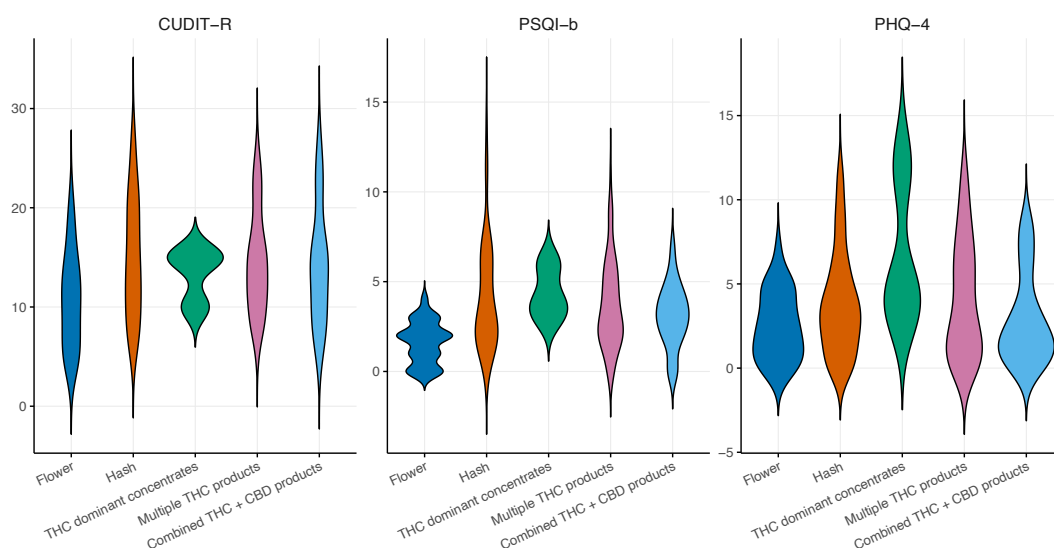


Figure 1 Violin displaying the distribution, mean, and 95% confidence intervals of cannabis use severity (CUDIT-R), sleep quality (PSQI-b) and anxiety/depression (PHQ-4) across dominant product types (flower only, hash only, multiple THC products, THC concentrates, and combined THC + CBD products).

Detailed descriptive statistics stratified by derived cannabis product type groups are presented in Table

2.

Table 2. Descriptive statistics of cannabis use severity (CUDIT-R), sleep quality (B-PSQI), and mental health (PHQ-4) by derived cannabis product categories

Derived cannabis product Categories [†]	CUDIT-R		B-PSQI		PHQ-4	
	Mean (SD)	Median (IQR)	Mean (SD)	Median (IQR)	Mean (SD)	Median (IQR)
1. Flower only	10.7 (4.9)	11 (6.2 – 14.0)	1.6 (1.2)	2.0 (1.0 – 2.0)	2.5 (2.0)	2.0 (1.0 – 4.0)
2. Hash only	15.0 (6.1)	14 (9.5 – 20.0)	4.1 (3.0)	3.0 (2.0 – 6.0)	3.8 (3.1)	3.0 (2.0 – 5.5)
3. THC-dominant concentrates	13.3 (2.9)	15 (12.5 – 15.0)	4.3 (1.5)	4.0 (3.5 – 5.0)	6.7 (4.6)	4.0 (4.0 – 8.0)
4. [‡] Multiple THC-dominant products	14.5 (5.2)	14 (11.0 – 17.0)	3.7 (2.4)	3.0 (2.0 – 5.0)	4.1 (3.5)	3.0 (1.0 – 6.8)
5. [¥] Combined THC + CBD products	14.3 (6.0)	14 (9.5 – 16.5)	3.1 (1.7)	3.0 (2.0 – 4.0)	2.9 (2.7)	2.0 (1.0 – 4.0)
6 Total sample	14.1 (5.7)	14.0 (10.0 – 18.0)	3.5 (2.5)	3.0 (2.0 – 5.0)	3.7 (3.1)	3.0 (1.0 – 6.0)

Note.

Sample characteristics and clinical measures by derived cannabis product categories, reported as mean (SD) and median (IQR). Values are descriptive only, and no statistical comparisons were conducted.

[†] Derived cannabis product groups were mutually exclusive and based on patterns of product use.

[‡] Multiple THC-dominant products refer to the use of more than one THC-dominant product (e.g., flower, hash, concentrates).

[¥] Combined THC + CBD products refer to the use of at least one THC-dominant and one CBD-dominant product.

Abbreviations.

PHQ-4 = Patient Health Questionnaire-4; B-PSQI= Pittsburgh Sleep Quality Index – brief version; CUDIT-R = Cannabis Use Disorder Identification Test-Revised; IQR = Interquartile Range.

3.2 Associations between cannabis severity and sleep quality, and role of cannabis type

In the linear regression model, greater cannabis use severity was associated with poorer sleep quality (B-PSQI), $b = 0.13$, $SE = 0.03$, 95% CI [0.07, 0.19], $p < .001$ (see Figure 2). The model explained 8.1% of the variance in sleep quality ($R^2 = .081$; see Table 3, Panel A1). Before adjustment for cannabis use severity, cannabis product type was associated with differences in sleep quality (Table 3, Panel A2), indicating higher mean B-PSQI scores among individuals using hash only ($b = 2.49$, $p < .001$), multiple THC-dominant product types ($b = 2.08$, $p < .001$), and combined THC + CBD products ($b = 1.49$, $p = .025$), compared with individuals using flower products only.

In the additive linear regression model, greater cannabis use severity was independently associated with poorer sleep quality (B-PSQI), $b = 0.10$, $SE = 0.03$, 95% CI [0.04, 0.16], $p < .001$. After adding cannabis product type, the model fit for sleep quality was significantly improved, explaining an additional 6.8% of the variance in sleep quality, $\Delta R^2 = .068$, $F(4,190) = 3.79$, $p = .005$ (Table 3, Panel A3). Compared with flower-only use, hash-only use ($b = 2.07$, $p < .001$) and multiple THC-dominant product use ($b = 1.71$, $p = .002$) were associated with poorer sleep quality.

In supplementary covariate-adjusted analyses, the association between cannabis use severity and poorer sleep quality remained statistically significant ($b = 0.12$, $p < .001$ in the severity-only model; $b = 0.09$, $p = .004$ in the additive model). Hash-only use ($b = 1.89$, $p = .001$) and multiple THC-dominant product use ($b = 1.55$, $p = .006$) also remained associated with poorer sleep quality relative to flower-only use, whereas the combined THC + CBD category was no longer statistically significant. The adjusted additive model explained more variance than the unadjusted model ($R^2 = .203$ vs. $R^2 = .149$). When the very small high-THC concentrate group was excluded, the overall pattern remained similar, supporting the robustness of the sleep findings.

Table 3. Main and additive effects of cannabis use severity and (A) sleep quality and (B) anxiety and depression symptoms

Predictor	b	SE	95% CI	p
Panel A1. Sleep quality (B-PSQI) — Severity-only model				
Intercept	3.48	0.17	[3.14, 3.83]	< .001
Cannabis use severity	0.13	0.03	[0.07, 0.19]	< .001
<i>Model fit. AIC = 906.8; BIC = 916.6; $R^2 = 0.081$; R^2 adj. = 0.076; $N = 196$</i>				
Panel A2. Main effect of product type				
Intercept	1.62	0.47	[0.68, 2.55]	< .001
Hash only	2.50	0.56	[1.39, 3.61]	< .001
High-THC concentrates	2.72	1.47	[-0.18, 5.62]	0.066
Multiple THC-dominant products	2.09	0.55	[1.01, 3.17]	< .001
Combined THC + CBD products	1.50	0.66	[0.19, 2.80]	0.025
<i>Model fit. AIC = 908.4; BIC = 928.1; $R^2 = 0.101$; R^2 adj. = 0.082; $N = 196$</i>				
Panel A3. Additive model (ANCOVA)				
Intercept	1.95	0.47	[1.02, 2.88]	< .001
Cannabis use severity	0.10	0.03	[0.04, 0.16]	0.001
Hash only	2.07	0.56	[0.96, 3.18]	< .001
High-THC concentrates	2.46	1.44	[-0.38, 5.29]	0.089
Multiple THC-dominant products	1.71	0.55	[0.63, 2.79]	0.002
Combined THC + CBD products	1.14	0.66	[-0.15, 2.44]	0.083
<i>Model fit. AIC = 899.7; BIC = 922.7; $R^2 = 0.149$; R^2 adj. = 0.126; $N = 196$; $\Delta R^2 = 0.068$; ΔR^2 adj. = 0.050; F-change(4,190) = 3.79; $p = 0.005$</i>				
Panel B1. Anxiety/depression (PHQ-4) — Severity-only model				
Intercept	3.66	0.20	[3.28, 4.05]	< .001
Cannabis use severity	0.27	0.03	[0.21, 0.34]	< .001
<i>Model fit. AIC = 959.9; BIC = 969.8; $R^2 = 0.245$; R^2 adj. = 0.241; $N = 197$</i>				
Panel B2. Main effect of product type				
Intercept	2.50	0.61	[1.30, 3.70]	< .001
Hash only	1.34	0.72	[-0.08, 2.77]	0.065
High-THC concentrates	4.17	1.89	[0.44, 7.89]	0.029
Multiple THC-dominant products	1.56	0.70	[0.18, 2.95]	0.027
Combined THC + CBD products	0.39	0.85	[-1.29, 2.07]	0.648
<i>Model fit. AIC = 1011.5; BIC = 1031.2; $R^2 = 0.048$; R^2 adj. = 0.028; $N = 197$</i>				
Panel B3. Additive model (ANCOVA)				
Intercept	3.42	0.54	[2.34, 4.49]	< .001
Cannabis use severity	0.27	0.04	[0.20, 0.34]	< .001
Hash only	0.18	0.65	[-1.10, 1.46]	0.780
High-THC concentrates	3.46	1.66	[0.19, 6.73]	0.038
Multiple THC-dominant products	0.54	0.63	[-0.70, 1.78]	0.388
Combined THC + CBD products	-0.57	0.76	[-2.06, 0.92]	0.452
<i>Model fit. AIC = 960.0; BIC = 983.0; $R^2 = 0.275$; R^2 adj. = 0.256; $N = 197$; $\Delta R^2 = 0.030$; ΔR^2 adj. = 0.015; F-change(4,191) = 1.96; $p = 0.102$</i>				

Note.

Unstandardized coefficients (b) are reported with HC3 robust standard errors and 95% confidence intervals. The reference category for cannabis product was flower only. CUDIT-R was centered before entering the model.

‡ Multiple THC-dominant products refer to the use of more than one THC-dominant product (e.g., flower, hash, concentrates).

¥ Combined THC + CBD products refer to the use of at least one THC-dominant and one CBD-dominant product.

Predictor	b	SE	95% CI	p
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Abbreviations. B-PSQI = Pittsburgh Sleep Quality Index – brief version. PHQ-4 = Patient Health Questionnaire-4; CUDIT-R = Cannabis Use Disorder Identification Test-Revised.

Cannabis product type did not significantly moderate the association between cannabis use severity and sleep quality in either the primary or covariate-adjusted models, indicating that the association between severity and poorer sleep quality was similar across product categories (see Table 4, Panel A).

Table 4. Moderation of the association between cannabis use severity and sleep quality (A) and anxiety/depression symptoms (B) by cannabis product type in the full sample.

Predictor	b	SE	95% CI	p
Panel A. Sleep quality (B-PSQI) — Interaction model				
Main effects (from interaction model)				
Intercept	1.89	0.57	[0.76, 3.02]	0.001
Cannabis use severity	0.08	0.10	[-0.11, 0.28]	0.404
Hash only	2.15	0.65	[0.87, 3.43]	0.001
Multiple THC-dominant products	1.76	0.63	[0.51, 3.01]	0.006
Combined THC + CBD products	1.20	0.73	[-0.24, 2.65]	0.102
Interaction terms				
Severity × Hash	-0.00	0.11	[-0.22, 0.21]	0.973
Severity × Multiple THC-dominant products	0.04	0.11	[-0.18, 0.26]	0.695
Severity × Combined THC + CBD products	0.02	0.13	[-0.23, 0.26]	0.893
<i>Model fit. AIC = 892.0; BIC = 921.4; R² = 0.149; R² adj. = 0.117; N = 193; ΔR² = 0.002; ΔR² adj. = -0.012; F-change(3,185) = 0.16; p = 0.926</i>				
Panel B. Anxiety/depression (PHQ-4) — Interaction model				
Main effects (from interaction model)				
Intercept	3.17	0.65	[1.88, 4.45]	< .001
Cannabis use severity	0.20	0.11	[-0.02, 0.42]	0.078
Hash only	0.43	0.74	[-1.02, 1.89]	0.557
Multiple THC-dominant products	0.79	0.72	[-0.63, 2.21]	0.276
Combined THC + CBD products	-0.32	0.83	[-1.97, 1.32]	0.701
Interaction terms				
Severity × Hash	0.07	0.13	[-0.18, 0.32]	0.570
Severity × Multiple THC-dominant products	0.09	0.13	[-0.16, 0.34]	0.468
Severity × Combined THC + CBD products	0.08	0.14	[-0.20, 0.36]	0.588
<i>Model fit. AIC = 946.9; BIC = 976.3; R² = 0.269; R² adj. = 0.242; N = 194; ΔR² = 0.002; ΔR² adj. = -0.010; F-change(3,186) = 0.18; p = 0.912</i>				

Note.

Unstandardized coefficients (b) are reported with HC3 robust standard errors and 95% confidence intervals. The reference category for cannabis product was flower only. CUDIT-R was centered before entering the model.

‡ Multiple THC-dominant products refer to the use of more than one THC-dominant product (e.g., flower, hash, concentrates).

¥ Combined THC + CBD products refer to the use of at least one THC-dominant and one CBD-dominant product.

Abbreviations. B-PSQI = Pittsburgh Sleep Quality Index – brief version. PHQ-4 = Patient Health Questionnaire-4; CUDIT-R = Cannabis Use Disorder Identification Test-Revised.

3.3. Associations between cannabis use severity and anxiety/depression symptoms, and role of cannabis type

In the linear regression model, greater cannabis use severity was associated with higher levels of anxiety/depression symptoms, $b = 0.27$, $SE = 0.034$, 95% CI [0.21, 0.34], $p < .001$ (see Figure 2). This model explained 24.5% of the variance in anxiety/depression symptoms ($R^2 = .245$; see Table 3, Panel B1).

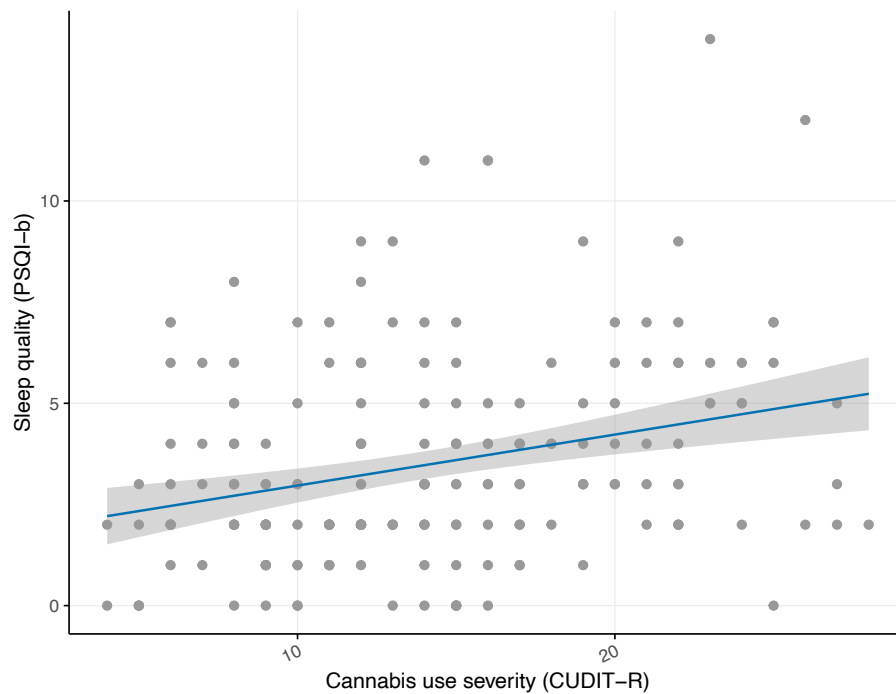


Figure 2 Scatterplot showing the association between cannabis use severity (CUDIT-R) and sleep quality (PSQI-b).

Before adjustment for cannabis use severity, cannabis product type was associated with differences in anxiety/depression symptoms (Table 3, Panel B2). Compared with individuals using flower products only, higher average PHQ-4 scores were observed among those using THC-dominant concentrates ($b = 4.17$, $p = .029$) and multiple THC-dominant products ($b = 1.59$, $p = .025$), whereas differences for hash only users did not reach statistical significance ($b = 1.34$, $p = .065$). Although THC-dominant concentrates showed a similar direction of association, estimates for this group should be interpreted with caution due to the very small number of participants in this category (see Figure 3).

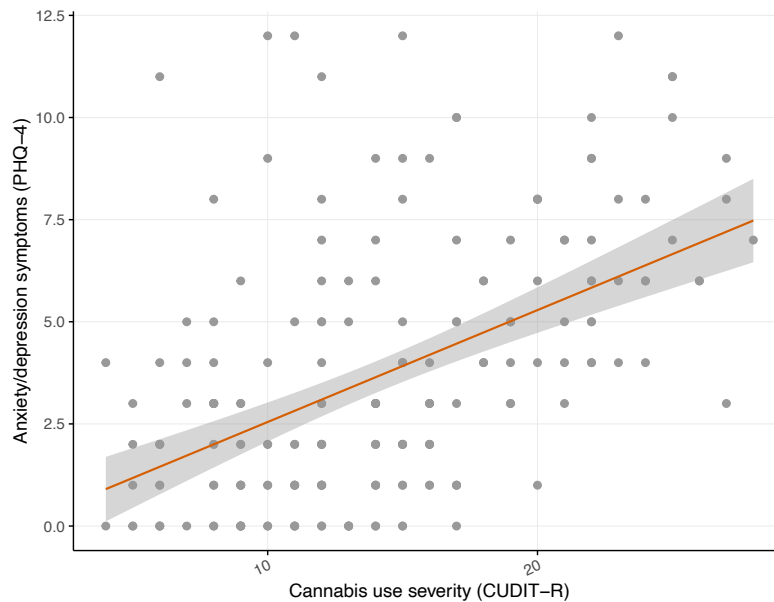


Figure 3 Scatterplot showing the association between cannabis use severity (CUDIT-R) and symptoms of anxiety/depression (PHQ-4).

In the additive linear regression model, greater cannabis use severity remained associated with anxiety/depression symptoms (PHQ-4), $b = 0.27$, $SE = 0.04$, 95% CI [0.21, 0.34], $p < .001$. The inclusion of cannabis product type did not significantly improve the model fit, $F(4, 190) = 1.99$, $p = .098$ (see Table 3, Panel B3). When the THC-dominant concentrate group was excluded, the overall pattern remained comparable (see Supplementary Table S3).

In supplementary covariate-adjusted analyses, the association between cannabis use severity and higher anxiety/depression symptoms remained robust ($b = 0.26$, $p < .001$ in both the severity-only and additive models). The adjusted additive model did not show clear evidence that product type explained additional variance beyond cannabis use severity, age, gender, nicotine use, and alcohol use ($\Delta R^2 = .029$, $p = .107$). In this adjusted model, only the high-THC concentrate group differed significantly from the flower-only group ($b = 3.45$, $p = .037$), but this estimate should be interpreted cautiously given the very small size of that subgroup. After excluding the concentrate group, the pattern remained comparable, with no consistent evidence that product type contributed additional explanatory value for anxiety/depression symptoms beyond cannabis use severity and covariates.

Cannabis product type also did not significantly moderate the association between cannabis use severity and anxiety/depression symptoms in either the primary or covariate-adjusted models, indicating that the association between severity and anxiety/depression symptoms did not differ across product categories (see Table 4, Panel B).

4. Discussion

This cross-sectional naturalistic study of 197 adults with near daily cannabis use extends previous research by examining associations between cannabis use severity, sleep quality, and anxiety/depression symptoms within the same population, while also considering the role of cannabis product type. Our findings showed that greater

cannabis use severity was robustly associated with poorer sleep quality and higher levels of anxiety/depression symptoms. Cannabis product type was associated with sleep quality, such that the use of hash only or multiple THC-dominant product types was associated with poorer sleep quality than the use of flower only. However, no moderation was observed in the association of cannabis use severity and sleep quality. In contrast, product type was not associated with cannabis use severity and anxiety/depression, nor did it moderate the association between cannabis use severity and symptoms of anxiety/depression.

Consistent with our hypothesis, greater cannabis use severity was associated with poorer sleep quality. This finding aligns with prior evidence indicating that cumulative levels of cannabis exposure and cannabis problems, rather than isolated or short-term use, are more strongly linked to disruptions in sleep quality. For example, greater daily cannabis dosage has been associated with reduced sleep efficiency and increased nocturnal awakenings, despite shorter sleep onset latency (Bell et al. 2022). Polysomnography studies indicate that heavy cannabis use is accompanied by marked impairments in sleep continuity and sleep architecture, particularly during early abstinence, including reduced sleep time, sleep efficiency, and altered REM and slow-wave sleep, independent of depressive or withdrawal symptoms (Bolla et al. 2008). Similarly, findings from treatment-seeking samples indicate that severe cannabis use is accompanied by sleep disturbances in both subjective and objective measures (Pacek et al. 2017). These findings suggest that poorer sleep quality among individuals with greater cannabis use severity may reflect cumulative rather than acute sedative effects of cannabinoid exposure, which has been hypothesized to involve tolerance to initial sedative effects, and long-term disruption of sleep architecture (Edwards and Filbey 2021).

Consistent with our hypothesis, greater cannabis use severity was associated with higher levels of symptoms of anxiety/depression. This pattern is in line with epidemiological and clinical evidence showing consistent associations between severity of cannabis use and severity of internalizing symptoms and broader psychiatric burden (Volkow et al. 2016; Hasin and Walsh 2020; Sorkhou et al. 2024). Importantly, converging evidence indicates that this association is likely bidirectional. On one hand, internalizing symptoms and related vulnerability processes, such as difficulties in emotion regulation and reliance on coping-motivated substance use, have been shown to increase the risk of developing CUD and cannabis-related problems (Moitra et al. 2015; Weidberg et al. 2023). On the other hand, chronic and severe cannabis use might exacerbate symptoms of anxiety and depression over time, contributing to greater psychological distress and psychiatric burden (Cerdeira et al. 2020; Hasin and Walsh 2020). From this perspective, internalizing symptoms may index a broader transdiagnostic vulnerability profile characterized by negative affect, emotion regulation difficulties, and reliance on external emotion regulation strategies to manage distress, such as cannabis use. For example, Babson et al. (2013) showed that depression was more strongly associated with problematic cannabis use among individuals with poorer sleep quality; suggesting that affective and sleep-related vulnerabilities may jointly amplify cannabis-related harm. Likewise, studies of distress tolerance and coping motives indicate that difficulties in tolerating negative affectivity partially explain the link between depression, anxiety and cannabis-related problems (Bujarski et al. 2012).

The inclusion of product type significantly improved model fit beyond cannabis use severity, but product type did not moderate the association between cannabis use severity and sleep quality. In line with our hypothesis,

individuals who reported using only hash products exhibited poorer sleep quality compared to those who reported using only flower, even after accounting for cannabis use severity and covariates. A similar pattern was observed among individuals who reported using multiple THC-containing products. These differences may reflect variation in cannabinoid concentration, route of administration, rather than differences in the severity-sleep association itself. Indeed, although sleep quality differed across product types, the association between greater cannabis use severity and poorer sleep quality was comparable across product categories. This pattern suggests that product type contributes additively to sleep outcomes rather than functioning as a conditional moderator of severity-sleep associations. This interpretation is consistent with laboratory studies indicating that THC acutely reduces sleep onset latency, while repeated high-dose exposure leads to tolerance and limited benefits for sustained sleep quality (Gorelick et al. 2013). Relatedly, daily diary studies further suggest that short-term subjective sleep effects may vary depending on cannabinoid composition and context of use, with better perceived sleep quality among individuals using products with higher CBD-to-THC ratios (Bidwell et al. 2024). However, these within-person and short-term effects are not directly comparable to the cumulative and between-person associations examined in the present cross-sectional study.

In contrast to the sleep outcomes, and contrary to our hypothesis, cannabis product type did not explain additional variance in anxiety/depression symptoms beyond cannabis use severity, nor did it moderate the association between cannabis use severity and anxiety/depression symptoms. Although experimental studies have suggested that acute administration of THC can induce anxiety symptoms in healthy volunteers under controlled conditions (Bhattacharyya et al. 2017), observational evidence comparing high-potency versus low-potency cannabis use has found little and inconsistent support for an association with depression, and mixed findings for anxiety (Petrilli et al. 2022). Together with our findings, this suggests that between-person differences in anxiety and depression symptoms among people who use cannabis near daily are more linked to overall use severity and related vulnerability processes than to product potency or cannabis product composition per se. This interpretation is consistent with evidence showing that severity indicators are more robustly associated with dependence and psychiatric burden than product-specific characteristics (Edwards and Filbey 2021; Petrilli et al. 2022).

Taken together, these findings suggest that sleep and mental health outcomes may relate to cannabis use through partially distinct pathways. While sleep quality appears sensitive to both overall use severity as well as to specific product characteristics, anxiety and depressive symptoms are more closely linked to overall use severity (Babson and Bonn-Miller 2014; Edwards and Filbey 2021). These findings underscore the importance of distinguishing between outcomes that are more directly impacted by proximal pharmacological properties of cannabis use and those that reflect broader patterns of dysregulated use and psychological vulnerability. Although sleep quality and anxiety/depression symptoms are known to be bidirectionally related, they were examined as separate outcomes in the present study, because our primary aim was to test whether cannabis use severity and product type showed differential associations across sleep and mental health domains. Future studies could extend this work by modeling the interrelationship between sleep disturbances and internalizing symptoms together, which may offer a more integrated understanding of how these vulnerabilities co-occur and potentially reinforce one another.

Importantly, these findings should be interpreted in the context of the Danish illegal cannabis market. In contrast to many other European countries, hash is the most popular product in Denmark (Rømer Thomsen et al. 2019), a pattern that was also evident in the present study. Moreover, THC concentrations in seized cannabis resin (such as hash) have increased 3-fold from a mean of 8.3% in 2000 to 25.3% in 2017, reaching some of the highest levels reported in Europe (Freeman et al. 2019; Rømer Thomsen et al. 2019). Similar increases have been documented across Europe and North America (Freeman et al. 2019; Freeman et al. 2021; EUDA 2024a; United Nations Office on Drugs and Crime 2024). In such high-potency markets, resin-based products (like hash) and multi-product use may confer heightened risk for sleep-related problems, potentially reflecting differences in cumulative THC exposure and use patterns rather than isolated pharmacological effects. This context may help contextualize why people who use hash and multiple THC-dominant product types showed poorer sleep quality than those who used flower only. The predominance of use of resin-based products like hash may place the Danish population of cannabis users at greater risk for the development of cannabis use-related problems. However, cross-national studies incorporating objective measures of cannabinoid exposure extending beyond the border of Denmark are needed to further investigate this.

This study has important strengths, including the use of a naturalistic sample of people who use cannabis near daily and the detailed assessment of real-world cannabis product types as used in the illegal market in Denmark (e.g., hash, flower, concentrates and mixed products), rather than relying on coarse or subjective indicators of “high” versus “low” potency. This approach captures meaningful variation in cannabinoid profiles, routes of administration, and use patterns, which increases ecological validity and clinical relevance. At the same time, this study has limitations, including a cross-sectional design that limits inferences about causality; the lack of THC:CBD ratio data; and the fact that our sleep quality measurement was based on a self-reported questionnaire, which may be influenced by recall bias, expectancy effects, and subjective interpretation of sleep problems. These limitations highlight the need for further research in this area. To allow more precise separation of pharmacological and CUD-related effects, future studies should incorporate between- and within-person designs using experience sampling methods (ESM) or other longitudinal methods that integrate objective sleep measures (e.g., polysomnography, actigraphy, wearable-based measures) with quantitative and qualitative assessments of cannabinoid exposure (e.g., THC and CBD concentrations or THC:CBD ratios), to disentangle pharmacological, behavioral, and vulnerability-related mechanisms linking cannabis use to sleep and mental health.

5. Conclusion

In this cross-sectional study of adults with near daily cannabis use, greater cannabis use severity was associated with poorer sleep quality and higher levels of anxiety/depression, whereas cannabis product type was associated with sleep quality but not with mental health outcomes. Specifically, use of hash and multiple THC-containing products was linked to poorer sleep quality compared to flower use, independent of cannabis use severity. Together these findings indicate that sleep and mental health may be related to cannabis use through partially distinct pathways, with sleep outcomes reflecting both product-specific characteristics and CUD-related vulnerability processes, and mental health outcomes reflecting primarily CUD-related vulnerability processes.

These results highlight the importance of distinguishing between cannabis product types and use-related problems when evaluating health outcomes in the non-medical cannabis population.

6. Declarations

Disclosure of potential conflicts of interest

The authors have no competing interests to declare that are relevant to the content of this article.

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Data availability

The data is available from the authors upon reasonable request.

Authors contributions

A.P.G.D. contributed to the conceptualization and methodology of the study; curated the data; conducted the formal analyses; carried out the investigation; managed the project; acquired funding; prepared the visualizations; and wrote the original draft of the manuscript. J.C. contributed to the conceptualization and methodology of the study and provided supervision and critically reviewed and edited the manuscript. K.R.T. contributed to the conceptualization and methodology of the study and provided supervision and critically reviewed and edited the manuscript. All authors reviewed and approved the final manuscript.

Compliance with Ethical Standards

Ethics approval

All procedures were performed in line with principles of the declaration of Helsinki. Ethical approval was obtained from the Research Ethics Committee at Aarhus University (BSS-2023-051) and the study was registered at the Danish Data Protection Agency. This study is part of a larger study preregistered on the Open Science Framework (OSF; <https://osf.io/c74ur/>).

Informed consent

All participants were fully informed, were instructed that participation was voluntary, and that they could withdraw from the study at any time without consequences, and all provided informed consent (online).

Clinical trial number

Not applicable.

Supplementary Table S1. Recording of cannabis product responses into derived product categories.

Final category	Meaning	Inclusion criteria	Examples
Hash only	Exclusive use of hash-type resin products	One THC-dominant product class reported, with hash/resin keywords only; no flower, concentrates, or CBD products	hash, hasj, resin, rosin
Flower only	Exclusive use of THC-dominant cannabis flower products	One THC-dominant product class reported, with flower keywords only; no hash, concentrates, or CBD products	pot, skunk, weed, Bedrocan (THC varieties)
THC-dominant concentrates	Exclusive use of high-THC concentrates or extracts	One THC-dominant product class reported, with concentrate keywords only; no flower, hash, or CBD products	dabs, wax, shatter, THC vape liquid
Multiple THC-dominant products	Use of two or more THC-dominant product classes without CBD	Two or more THC-dominant product classes reported and no CBD product reported	hash + flower; flower + concentrate; hash + concentrate
Combined THC + CBD products	Co-use of at least one THC-dominant and one CBD-dominant product	At least one THC-dominant product and at least one CBD-dominant product reported	flower + CBD flower; hash + CBD; flower + CBD oil

Supplementary Table S2. Covariate-adjusted main and additive effects of cannabis use severity and cannabis product type on sleep quality and anxiety/depression symptoms in the full sample.

Predictor	b	SE	95% CI	p
Panel A1. Sleep quality (B-PSQI) — Severity-only model				
Intercept	4.43	0.95	[2.57, 6.30]	< .001
Cannabis use severity	0.12	0.03	[0.05, 0.18]	< .001
Age	-0.00	0.02	[-0.04, 0.03]	0.815
Male	-1.35	0.37	[-2.07, -0.63]	< .001
Non-binary	0.20	1.41	[-2.58, 2.99]	0.886
Nicotine use (Yes)	-0.19	0.63	[-1.44, 1.06]	0.763
Alcohol use (Yes)	0.33	0.38	[-0.42, 1.09]	0.382
<i>Model fit. AIC = 885.6; BIC = 911.7; R² = 0.151; R² adj. = 0.124; N = 192</i>				
Panel A2. Main effect of product type				
Intercept	2.79	1.05	[0.73, 4.85]	0.008
Hash only	2.29	0.57	[1.16, 3.42]	< .001
High-THC concentrates	2.39	1.46	[-0.49, 5.27]	0.104
Multiple THC-dominant products	1.87	0.55	[0.78, 2.96]	< .001
Combined THC + CBD products	1.43	0.66	[0.12, 2.73]	0.032
Age	-0.02	0.02	[-0.05, 0.02]	0.311
Male	-1.22	0.37	[-1.95, -0.49]	0.001
Non-binary	-0.16	1.41	[-2.95, 2.62]	0.908
Nicotine use (Yes)	0.03	0.63	[-1.22, 1.28]	0.965
Alcohol use (Yes)	0.43	0.38	[-0.33, 1.18]	0.265
<i>Model fit. AIC = 888.2; BIC = 924.1; R² = 0.166; R² adj. = 0.125; N = 192</i>				
Panel A3. Additive model (ANCOVA)				
Intercept	3.08	1.03	[1.05, 5.11]	0.003
Cannabis use severity	0.09	0.03	[0.03, 0.15]	0.004
Hash only	1.89	0.58	[0.75, 3.03]	0.001
High-THC concentrates	2.17	1.44	[-0.67, 5.00]	0.133
Multiple THC-dominant products	1.55	0.55	[0.46, 2.64]	0.006
Combined THC + CBD products	1.14	0.66	[-0.16, 2.43]	0.085
Age	-0.01	0.02	[-0.04, 0.02]	0.581
Male	-1.17	0.36	[-1.89, -0.45]	0.002
Non-binary	0.03	1.39	[-2.71, 2.77]	0.983
Nicotine use (Yes)	-0.19	0.63	[-1.43, 1.04]	0.760
Alcohol use (Yes)	0.33	0.38	[-0.41, 1.07]	0.385
<i>Model fit. AIC = 881.6; BIC = 920.7; R² = 0.203; R² adj. = 0.159; N = 192; $\Delta R^2 = 0.052$; ΔR^2 adj. = 0.035; F-change(4,181) = 2.92; $p = 0.022$</i>				
Panel B1. Anxiety/depression (PHQ-4) — Severity-only model				
Intercept	6.06	1.07	[3.94, 8.18]	< .001
Cannabis use severity	0.26	0.04	[0.19, 0.33]	< .001
Age	-0.05	0.02	[-0.09, -0.01]	0.008
Male	-0.58	0.41	[-1.40, 0.24]	0.165
Non-binary	2.23	1.60	[-0.93, 5.40]	0.165
Nicotine use (Yes)	-0.07	0.72	[-1.49, 1.34]	0.917
Alcohol use (Yes)	-0.59	0.43	[-1.44, 0.26]	0.174
<i>Model fit. AIC = 939.0; BIC = 965.1; R² = 0.295; R² adj. = 0.273; N = 193</i>				
Panel B2. Main effect of product type				
Intercept	4.99	1.33	[2.36, 7.61]	< .001
Hash only	1.26	0.73	[-0.18, 2.70]	0.086
High-THC concentrates	4.10	1.86	[0.43, 7.78]	0.029
Multiple THC-dominant products	1.18	0.70	[-0.21, 2.56]	0.095
Combined THC + CBD products	0.08	0.84	[-1.59, 1.74]	0.929
Age	-0.07	0.02	[-0.12, -0.03]	< .001
Male	-0.69	0.47	[-1.62, 0.24]	0.147
Non-binary	1.60	1.80	[-1.95, 5.15]	0.376

Predictor	b	SE	95% CI	p
Nicotine use (Yes)	0.79	0.81	[-0.80, 2.39]	0.326
Alcohol use (Yes)	-0.43	0.49	[-1.39, 0.53]	0.380
<i>Model fit. AIC = 986.1; BIC = 1022.0; R² = 0.128; R² adj. = 0.085; N = 193</i>				
Panel B3. Additive model (ANCOVA)				
Intercept	5.81	1.18	[3.48, 8.14]	< .001
Cannabis use severity	0.26	0.04	[0.19, 0.33]	< .001
Hash only	0.11	0.66	[-1.20, 1.42]	0.872
High-THC concentrates	3.45	1.65	[0.21, 6.70]	0.037
Multiple THC-dominant products	0.27	0.63	[-0.98, 1.51]	0.675
Combined THC + CBD products	-0.77	0.75	[-2.26, 0.71]	0.306
Age	-0.05	0.02	[-0.09, -0.01]	0.008
Male	-0.54	0.42	[-1.36, 0.28]	0.197
Non-binary	2.15	1.59	[-0.99, 5.29]	0.178
Nicotine use (Yes)	0.16	0.72	[-1.25, 1.58]	0.820
Alcohol use (Yes)	-0.71	0.43	[-1.56, 0.14]	0.100
<i>Model fit. AIC = 939.0; BIC = 978.2; R² = 0.324; R² adj. = 0.287; N = 193; $\Delta R^2 = 0.029$; ΔR^2 adj. = 0.014; F-change(4,182) = 1.93; $p = 0.107$</i>				

Note.

Unstandardized coefficients (b) are reported with HC3 robust standard errors and 95% confidence intervals. The reference category for cannabis product was flower only. Models were adjusted for age, gender, recent nicotine use, and recent alcohol use. CUDIT-R was centered before entering the model.

‡ Multiple THC-dominant products refer to the use of more than one THC-dominant product (e.g., flower, hash, concentrates).

¥ Combined THC + CBD products refer to the use of at least one THC-dominant and one CBD-dominant product.

Abbreviations. B-PSQI = Pittsburgh Sleep Quality Index – brief version. PHQ-4 = Patient Health Questionnaire-4; CUDIT-R = Cannabis Use Disorder Identification Test-Revised.

Supplementary Table S3. Covariate-adjusted main and additive effects of cannabis use severity and cannabis product type on sleep quality and anxiety/depression symptoms after excluding the small high-THC concentrates subgroup.

Predictor	b	SE	95% CI	p
Panel A1. Sleep quality (B-PSQI) — Severity-only model				
Intercept	4.44	0.97	[2.52, 6.36]	< .001
Cannabis use severity	0.12	0.03	[0.05, 0.18]	< .001
Age	-0.00	0.02	[-0.04, 0.03]	0.800
Male	-1.34	0.37	[-2.07, -0.60]	< .001
Non-binary	0.23	1.42	[-2.58, 3.04]	0.871
Nicotine use (Yes)	-0.20	0.66	[-1.50, 1.10]	0.762
Alcohol use (Yes)	0.31	0.38	[-0.45, 1.07]	0.416
<i>Model fit. AIC = 874.5; BIC = 900.4; R² = 0.149; R² adj. = 0.121; N = 189</i>				
Panel A2. Main effect of product type				
Intercept	2.89	1.06	[0.79, 4.99]	0.007
Hash only	2.30	0.58	[1.17, 3.44]	< .001
Multiple THC-dominant products	1.87	0.55	[0.78, 2.97]	< .001
Combined THC + CBD products	1.44	0.66	[0.12, 2.75]	0.032
Age	-0.02	0.02	[-0.05, 0.02]	0.282
Male	-1.19	0.38	[-1.93, -0.45]	0.002
Non-binary	-0.15	1.42	[-2.95, 2.66]	0.919
Nicotine use (Yes)	-0.06	0.65	[-1.35, 1.23]	0.926
Alcohol use (Yes)	0.42	0.38	[-0.33, 1.18]	0.272
<i>Model fit. AIC = 875.2; BIC = 907.7; R² = 0.164; R² adj. = 0.126; N = 189</i>				
Panel A3. Additive model (ANCOVA)				
Intercept	3.15	1.05	[1.09, 5.22]	0.003
Cannabis use severity	0.09	0.03	[0.03, 0.15]	0.005
Hash only	1.91	0.58	[0.76, 3.05]	0.001
Multiple THC-dominant products	1.55	0.56	[0.46, 2.65]	0.006
Combined THC + CBD products	1.14	0.66	[-0.16, 2.45]	0.085
Age	-0.01	0.02	[-0.04, 0.02]	0.542
Male	-1.15	0.37	[-1.87, -0.42]	0.002
Non-binary	0.04	1.39	[-2.71, 2.79]	0.975
Nicotine use (Yes)	-0.26	0.65	[-1.54, 1.01]	0.686
Alcohol use (Yes)	0.32	0.38	[-0.42, 1.07]	0.392
<i>Model fit. AIC = 868.9; BIC = 904.5; R² = 0.200; R² adj. = 0.160; N = 189; $\Delta R^2 = 0.051$; ΔR^2 adj. = 0.039; F-change(3,179) = 3.79; p = 0.011</i>				
Panel B1. Anxiety/depression (PHQ-4) — Severity-only model				
Intercept	6.07	1.07	[3.94, 8.19]	< .001
Cannabis use severity	0.26	0.03	[0.19, 0.33]	< .001
Age	-0.05	0.02	[-0.09, -0.02]	0.005
Male	-0.46	0.41	[-1.27, 0.35]	0.260
Non-binary	2.38	1.57	[-0.73, 5.48]	0.133
Nicotine use (Yes)	-0.07	0.73	[-1.51, 1.36]	0.919
Alcohol use (Yes)	-0.68	0.43	[-1.52, 0.16]	0.113
<i>Model fit. AIC = 917.2; BIC = 943.2; R² = 0.308; R² adj. = 0.285; N = 190</i>				
Panel B2. Main effect of product type				
Intercept	5.24	1.34	[2.60, 7.87]	< .001
Hash only	1.32	0.72	[-0.11, 2.75]	0.071
Multiple THC-dominant products	1.20	0.70	[-0.18, 2.57]	0.087
Combined THC + CBD products	0.09	0.84	[-1.56, 1.74]	0.911
Age	-0.08	0.02	[-0.12, -0.04]	< .001
Male	-0.55	0.47	[-1.48, 0.38]	0.248
Non-binary	1.68	1.79	[-1.84, 5.20]	0.348
Nicotine use (Yes)	0.56	0.82	[-1.06, 2.19]	0.495
Alcohol use (Yes)	-0.44	0.48	[-1.39, 0.51]	0.357

Predictor	b	SE	95% CI	p
<i>Model fit. AIC = 966.9; BIC = 999.4; R² = 0.119; R² adj. = 0.080; N = 190</i>				
Panel B3. Additive model (ANCOVA)				
Intercept	6.00	1.18	[3.67, 8.34]	< .001
Cannabis use severity	0.26	0.04	[0.19, 0.33]	< .001
Hash only	0.17	0.66	[-1.13, 1.46]	0.802
Multiple THC-dominant products	0.29	0.63	[-0.95, 1.53]	0.645
Combined THC + CBD products	-0.75	0.75	[-2.22, 0.72]	0.315
Age	-0.06	0.02	[-0.09, -0.02]	0.004
Male	-0.41	0.42	[-1.23, 0.41]	0.324
Non-binary	2.22	1.58	[-0.89, 5.33]	0.160
Nicotine use (Yes)	-0.02	0.73	[-1.46, 1.42]	0.981
Alcohol use (Yes)	-0.72	0.43	[-1.56, 0.12]	0.091
<i>Model fit. AIC = 919.9; BIC = 955.6; R² = 0.320; R² adj. = 0.286; N = 190; $\Delta R^2 = 0.012$; ΔR^2 adj. = 0.001; F-change(3,180) = 1.06; $p = 0.369$</i>				

Note.

Unstandardized coefficients (b) are reported with HC3 robust standard errors and 95% confidence intervals. The reference category for cannabis product was flower only. Models were adjusted for age, gender, recent nicotine use, and recent alcohol use. CUDIT-R was centered before entering the model.

‡ Multiple THC-dominant products refer to the use of more than one THC-dominant product (e.g., flower, hash, concentrates).

¥ Combined THC + CBD products refer to the use of at least one THC-dominant and one CBD-dominant product.

Abbreviations. B-PSQI = Pittsburgh Sleep Quality Index – brief version. PHQ-4 = Patient Health Questionnaire-4; CUDIT-R = Cannabis Use Disorder Identification Test-Revised.

Supplementary Table S4. Covariate-adjusted moderation of the association between cannabis use severity and sleep quality (A) and anxiety/depression symptoms (B) by cannabis product type in the full sample.

Predictor	b	SE	95% CI	p
Panel A. Sleep quality (B-PSQI) — Interaction model				
Main effects (from interaction model)				
Intercept	3.15	1.12	[0.93, 5.37]	0.006
Cannabis use severity	0.09	0.10	[-0.11, 0.28]	0.388
Hash only	1.94	0.67	[0.61, 3.26]	0.004
Multiple THC-dominant products	1.56	0.65	[0.29, 2.84]	0.016
Combined THC + CBD products	1.15	0.74	[-0.31, 2.61]	0.123
Age	-0.01	0.02	[-0.04, 0.02]	0.523
Male	-1.14	0.38	[-1.88, -0.40]	0.003
Non-binary	-0.03	1.42	[-2.83, 2.76]	0.981
Nicotine use (Yes)	-0.24	0.65	[-1.53, 1.04]	0.709
Alcohol use (Yes)	0.31	0.39	[-0.46, 1.07]	0.430
Interaction terms				
Severity × Hash	-0.01	0.11	[-0.23, 0.21]	0.907
Severity × Multiple THC-dominant products	0.02	0.11	[-0.20, 0.24]	0.891
Severity × Combined THC + CBD products	0.02	0.13	[-0.23, 0.27]	0.872
<i>Model fit. AIC = 874.7; BIC = 920.0; R² = 0.201; R² adj. = 0.146; N = 189; ΔR² = 0.001; ΔR² adj. = -0.013; F-change(3,176) = 0.07; p = 0.977</i>				
Panel B. Anxiety/depression (PHQ-4) — Interaction model				
Main effects (from interaction model)				
Intercept	5.87	1.27	[3.37, 8.38]	< .001
Cannabis use severity	0.22	0.11	[0.00, 0.44]	0.047
Hash only	0.31	0.76	[-1.18, 1.80]	0.684
Multiple THC-dominant products	0.40	0.73	[-1.03, 1.84]	0.579
Combined THC + CBD products	-0.63	0.84	[-2.29, 1.02]	0.450
Age	-0.06	0.02	[-0.09, -0.02]	0.004
Male	-0.39	0.42	[-1.22, 0.45]	0.360
Non-binary	2.13	1.60	[-1.03, 5.29]	0.185
Nicotine use (Yes)	-0.01	0.74	[-1.46, 1.45]	0.993
Alcohol use (Yes)	-0.73	0.44	[-1.60, 0.14]	0.098
Interaction terms				
Severity × Hash	0.02	0.13	[-0.23, 0.27]	0.879
Severity × Multiple THC-dominant products	0.06	0.13	[-0.19, 0.31]	0.645
Severity × Combined THC + CBD products	0.04	0.14	[-0.24, 0.32]	0.762
<i>Model fit. AIC = 925.5; BIC = 971.0; R² = 0.321; R² adj. = 0.275; N = 190; ΔR² = 0.001; ΔR² adj. = -0.011; F-change(3,177) = 0.11; p = 0.953</i>				

Note.

Unstandardized coefficients (b) are reported with HC3 robust standard errors and 95% confidence intervals. The reference category for cannabis product was flower only. Models were adjusted for age, gender, recent nicotine use, and recent alcohol use. CUDIT-R was centered before entering the model.

‡ Multiple THC-dominant products refer to the use of more than one THC-dominant product (e.g., flower, hash, concentrates).

¥ Combined THC + CBD products refer to the use of at least one THC-dominant and one CBD-dominant product.

Abbreviations. B-PSQI = Pittsburgh Sleep Quality Index – brief version. PHQ-4 = Patient Health Questionnaire-4; CUDIT-R = Cannabis Use Disorder Identification Test-Revised.

Supplementary Table S5. Covariate-adjusted moderation of the association between cannabis use severity and sleep quality (A) and anxiety/depression symptoms (B) by cannabis product type after excluding the small high-THC concentrates subgroup.

Predictor	b	SE	95% CI	p
Panel A. Sleep quality (B-PSQI) — Interaction model				
Main effects (from interaction model)				
Intercept	3.15	1.12	[0.93, 5.37]	0.006
Cannabis use severity	0.09	0.10	[-0.11, 0.28]	0.388
Hash only	1.94	0.67	[0.61, 3.26]	0.004
Multiple THC-dominant products	1.56	0.65	[0.29, 2.84]	0.016
Combined THC + CBD products	1.15	0.74	[-0.31, 2.61]	0.123
Age	-0.01	0.02	[-0.04, 0.02]	0.523
Male	-1.14	0.38	[-1.88, -0.40]	0.003
Non-binary	-0.03	1.42	[-2.83, 2.76]	0.981
Nicotine use (Yes)	-0.24	0.65	[-1.53, 1.04]	0.709
Alcohol use (Yes)	0.31	0.39	[-0.46, 1.07]	0.430
Interaction terms				
Severity × Hash	-0.01	0.11	[-0.23, 0.21]	0.907
Severity × Multiple THC-dominant products	0.02	0.11	[-0.20, 0.24]	0.891
Severity × Combined THC + CBD products	0.02	0.13	[-0.23, 0.27]	0.872
<i>Model fit. AIC = 874.7; BIC = 920.0; R² = 0.201; R² adj. = 0.146; N = 189; ΔR² = 0.001; ΔR² adj. = -0.013; F-change(3,176) = 0.07; p = 0.977</i>				
Panel B. Anxiety/depression (PHQ-4) — Interaction model				
Main effects (from interaction model)				
Intercept	5.87	1.27	[3.37, 8.38]	< .001
Cannabis use severity	0.22	0.11	[0.00, 0.44]	0.047
Hash only	0.31	0.76	[-1.18, 1.80]	0.684
Multiple THC-dominant products	0.40	0.73	[-1.03, 1.84]	0.579
Combined THC + CBD products	-0.63	0.84	[-2.29, 1.02]	0.450
Age	-0.06	0.02	[-0.09, -0.02]	0.004
Male	-0.39	0.42	[-1.22, 0.45]	0.360
Non-binary	2.13	1.60	[-1.03, 5.29]	0.185
Nicotine use (Yes)	-0.01	0.74	[-1.46, 1.45]	0.993
Alcohol use (Yes)	-0.73	0.44	[-1.60, 0.14]	0.098
Interaction terms				
Severity × Hash	0.02	0.13	[-0.23, 0.27]	0.879
Severity × Multiple THC-dominant products	0.06	0.13	[-0.19, 0.31]	0.645
Severity × Combined THC + CBD products	0.04	0.14	[-0.24, 0.32]	0.762
<i>Model fit. AIC = 925.5; BIC = 971.0; R² = 0.321; R² adj. = 0.275; N = 190; ΔR² = 0.001; ΔR² adj. = -0.011; F-change(3,177) = 0.11; p = 0.953</i>				

Note.

Unstandardized coefficients (b) are reported with HC3 robust standard errors and 95% confidence intervals. The reference category for cannabis product was flower only. Models were adjusted for age, gender, recent nicotine use, and recent alcohol use. CUDIT-R was centered before entering the model.

‡ Multiple THC-dominant products refer to the use of more than one THC-dominant product (e.g., flower, hash, concentrates).

¥ Combined THC + CBD products refer to the use of at least one THC-dominant and one CBD-dominant product.

Abbreviations. B-PSQI = Pittsburgh Sleep Quality Index – brief version. PHQ-4 = Patient Health Questionnaire-4; CUDIT-R = Cannabis Use Disorder Identification Test-Revised.

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Appendix D (Co-authorship declaration)

Declaration of co-authorship¹

Date: 27-01-2026

This declaration concerns the following article/manuscript:

Title:	Cannabis Use, Affect and Emotion Regulation in Daily Life: A 3-Week Experience Sampling Method Study In People who Smoke Cannabis Almost Daily
Authors:	Ana Paula Gonçalves Donate, MS, Janna Cousijn, PhD, Emese Kroon, PhD, Kristine Rømer Thomsen, PhD

The article/manuscript is:

- ☐ Published, state full reference:
- ☐ Accepted, state journal:
- ☐ Invited for revision, state journal:
- ☒ Submitted
- ☐ In preparation

Date of the current version of the manuscript, if not published or accepted: 17-12-2025

Please fill out Table 1 regarding contribution to the manuscript for all authors. The respective author has contributed to the elements:

- A. Research idea: Identifying, developing, specifying, and formulating the overarching research question and aim.
- B. Theory: Organizing theoretical perspectives, developing arguments and hypotheses, specifying theoretical model.
- C. Research design: Developing and planning design for test or exploration of the research question.
- D. Data collection: Preparing and organizing data collection, data collection, preparing data for analysis and storage.
- E. Data analysis: Application of empirical techniques to analyze or synthesize study data including providing support for interpretations such as visualizations etc.
- F. Writing: Drafting and revising manuscript presenting the research idea and results

of this article/manuscript as follows:

- 4 Has essentially delivered this part.
- 3 Major contribution
- 2 Equal contribution
- 1 Minor contribution
- 0 Did not contribute to this part.
- N/A Not relevant or not applicable

¹ Attribution of authorship should be based on criteria a-d adopted from the [Vancouver guidelines](#) (see also [rules and guidelines from Aarhus University](#)) and all individuals who meet these criteria should be recognized as authors. The co-author has contributed:

- a) to the conception or design of the work, or the acquisition, analysis, or interpretation of data for the work, *and*
- b) drafting the work or revising it critically for important intellectual content, *and*
- c) to the final approval of the version to be published, *and*
- d) agrees to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Table 1. Individual contributions and signature of each co-author¹

Author	Extent of contribution (4-0) per element (A.-F.)						Signature of the author ²
	A. Research Idea	B. Theory	C. Research Design	D. Data Collection	E. Data Analysis	F. Writing	
Ana Paula Gonçalves Donate	3	4	3	4	3	3	<i>Donate</i>
Janna Cousijn	1	0	1	0	1	1	<i>Janna</i>
Emese Kroon	0	0	1	0	1	1	<i>Emese</i>
Kristine Rømer Thomsen	1	0	1	0	1	1	<i>Kristine Rømer Thomsen</i>

¹More rows can be added for additional authors.

²All authors must confirm the declaration either by signature or email.

If relevant, you may add more information on the work and collaboration such as open science practices or more detailed specifications of authors' contributions here:

This study is preregistered at Open Science Framework (OSF; <https://osf.io/c74ur/>). Ethical approval was obtained from the Research Ethics Committee at Aarhus University (BSS-2023-051), and the study was registered at the Danish Data Protection Agency.

Declaration of co-authorship¹

Date: 27-01-2026

This declaration concerns the following article/manuscript:

Title:	Cannabis use severity and emotion regulation in people who use cannabis near daily: the role of gender and motives
Authors:	Ana Paula Gonçalves Donate, MS, Janna Cousijn, PhD, Kristine Rømer Thomsen, PhD

The article/manuscript is:

- ☐ Published, state full reference:
- ☐ Accepted, state journal:
- ☐ Invited for revision, state journal:
- ☒ Submitted
- ☐ In preparation

Date of the current version of the manuscript, if not published or accepted: 17-12-2025

Please fill out Table 1 regarding contribution to the manuscript for all authors. The respective author has contributed to the elements:

- A. Research idea: Identifying, developing, specifying, and formulating the overarching research question and aim.
- B. Theory: Organizing theoretical perspectives, developing arguments and hypotheses, specifying theoretical model.
- C. Research design: Developing and planning design for test or exploration of the research question.
- D. Data collection: Preparing and organizing data collection, data collection, preparing data for analysis and storage.
- E. Data analysis: Application of empirical techniques to analyze or synthesize study data including providing support for interpretations such as visualizations etc.
- F. Writing: Drafting and revising manuscript presenting the research idea and results

of this article/manuscript as follows:

- 4 Has essentially delivered this part.
- 3 Major contribution
- 2 Equal contribution
- 1 Minor contribution
- 0 Did not contribute to this part.

¹ Attribution of authorship should be based on criteria a-d adopted from the [Vancouver guidelines](#) (see also [rules and guidelines from Aarhus University](#)) and all individuals who meet these criteria should be recognized as authors. The co-author has contributed:

- a) to the conception or design of the work, or the acquisition, analysis, or interpretation of data for the work, *and*
- b) drafting the work or revising it critically for important intellectual content, *and*
- c) to the final approval of the version to be published, *and*
- d) agrees to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Author	Extent of contribution (4-0) per element (A.-F.)						Signature of the author ²
	A. Research Idea	B. Theory	C. Research Design	D. Data Collection	E. Data Analysis	F. Writing	
Ana Paula Gonçalves Donate	3	4	3	4	4	3	
Janna Cousijn	1	0	1	0	0	1	
Kristine Rømer Thomsen	1	0	1	0	0	1	

¹More rows can be added for additional authors.

²All authors must confirm the declaration either by signature or email.

If relevant, you may add more information on the work and collaboration such as open science practices or more detailed specifications of authors' contributions here:

This cross-sectional study is part of a larger study preregistered at Open Science Framework (OSF; <https://osf.io/c74ur/>). Ethical approval was obtained from the Research Ethics Committee at Aarhus University (BSS-2023-051), and the study was registered at the Danish Data Protection Agency.

Declaration of co-authorship¹

Date: 27-01-2026

This declaration concerns the following article/manuscript:

Title:	Beyond cannabis use severity: associations of cannabis product type with sleep quality and mental health.
Authors:	Ana Paula Gonçalves Donate, MS, Janna Cousijn, PhD, Rømer Thomsen, PhD

The article/manuscript is:

- ☐ Published, state full reference:
- ☐ Accepted, state journal:
- ☐ Invited for revision, state journal:
- ☒ Submitted
- ☐ In preparation

Date of the current version of the manuscript, if not published or accepted: 17-12-2025

Please fill out Table 1 regarding contribution to the manuscript for all authors. The respective author has contributed to the elements:

- A. Research idea: Identifying, developing, specifying, and formulating the overarching research question and aim.
- B. Theory: Organizing theoretical perspectives, developing arguments and hypotheses, specifying theoretical model.
- C. Research design: Developing and planning design for test or exploration of the research question.
- D. Data collection: Preparing and organizing data collection, data collection, preparing data for analysis and storage.
- E. Data analysis: Application of empirical techniques to analyze or synthesize study data including providing support for interpretations such as visualizations etc.
- F. Writing: Drafting and revising manuscript presenting the research idea and results

of this article/manuscript as follows:

- 4 Has essentially delivered this part.
- 3 Major contribution
- 2 Equal contribution
- 1 Minor contribution
- 0 Did not contribute to this part.
- N/A Not relevant or not applicable

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- a) to the conception or design of the work, or the acquisition, analysis, or interpretation of data for the work, *and*
- b) drafting the work or revising it critically for important intellectual content, *and*
- c) to the final approval of the version to be published, *and*
- d) agrees to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Table 1. Individual contributions and signature of each co-author¹

Author	Extent of contribution (4-0) per element (A.-F.)						Signature of the author ²
	A. Research Idea	B. Theory	C. Research Design	D. Data Collection	E. Data Analysis	F. Writing	
Ana Paula Gonçalves Donate	3	4	3	4	4	3	
Janna Cousijn	1	0	1	0	0	1	
Kristine Rømer Thomsen	1	0	1	0	0	1	

¹More rows can be added for additional authors.

²All authors must confirm the declaration either by signature or email.

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